paeds path

Every topic, as a tree.

165 topics. Each tree: topic (dark bar), key point (green), then the same six branches every time, ending in your own notes lines. Italic line = memory hook. Use your PDF reader's outline/bookmarks to jump to any topic; zoom in freely, it's all vector.

Definition · Causes · Pathophysiology · Clinical picture · Diagnosis · Treatment

Section I:

Symptoms, differentials & acute care

(55 topics)

I-1 The unconscious child

The unconscious child is an airway problem before a neurology problem: ABCDE and a bedside glucose first, then decide structural (focal, asymmetric) versus metabolic/toxic (global, symmetric) - and never LP into raised ICP or shock.

- **Definition**
  - Unconscious child: an airway problem before a neurology problem.
  - *A before brain: airway first, always.*
- **Causes**
  - AEIOU-TIPS is the mnemonic list of causes of coma.
  - Alcohol, epilepsy, insulin, opiates, uraemia, trauma, infection, psychiatric, shock, space-occupying lesion.
  - *AEIOU-TIPS: vowels then TIPS.*
- **Patho-physiology**
  - Structural causes: focal and asymmetric; metabolic/toxic: global and symmetric.
  - CPP = MAP - ICP; raised ICP → Cushing triad (hypertension, bradycardia, irregular breathing).
  - Uncal herniation: CN III compression → dilated pupil; tonsillar → cardiopulmonary arrest.
  - *Cushing triad: high BP, slow pulse, odd breathing.*
- **Clinical picture**
  - Opioids: respiratory and CNS depression, pinpoint pupils.
  - Decorticate: flexed arms; decerebrate: extended arms.
  - Expose: purpura, odd bruises (abuse), needle marks.
  - *Decorticate = flexed to the 'cor' (heart).*
- **Diagnosis**
  - Never skip a bedside glucose.
  - Head CT if GCS ≤ 14, skull fracture or focal signs.
  - Never do a lumbar puncture with raised ICP or shock.
  - *'Don't Ever Forget Glucose' (DEFG).*
- **Treatment**
  - ABCDE; GCS of 8 or less → intubation; naloxone for pinpoint pupils.
  - Meningitis possible: ceftriaxone immediately, don't wait for lumbar puncture.
  - Raised ICP: elevate head, hyperventilate, mannitol.
  - *GCS 8 → intubate.*

I-2 Causes and treatment of convulsion in childhood

A seizure lasting ≥5 minutes is status epilepticus: check glucose at 0 min and climb the 5-5-5 ladder (benzo, benzo, second-line, RSI at 25 min); in neonates every seizure is pathological until a cause is found.

- **Definition**
  - Status epilepticus: seizure lasting ≥ 5 min → treat on the 5-5-5 ladder.
  - *5-5-5: benzo, benzo, second line.*
- **Causes**
  - A neonatal seizure is always pathological: HIE, hypoglycaemia, low Ca/Na, sepsis, haemorrhage.
  - Toddler: febrile seizure commonest (3-5% of children), meningitis, NAI, West syndrome.
  - Older: epilepsy, trauma, tumour, AVM.
  - *Neonatal seizure = always look for a cause.*
- **Patho-physiology**
  - Excitatory synapses mature before inhibitory ones → low seizure threshold.
  - Rapid temperature rise or metabolic insult → threshold crossed.
  - *Accelerator before brakes.*
- **Clinical picture**
  - Absence: blank stare, provoked by hyperventilation.
  - Focal impaired awareness: staring, head tilting, mouth chewing, confusion.
  - West syndrome: bilateral jerking, hypsarrhythmia.
  - *Absence: blow on a windmill to provoke it.*
- **Diagnosis**
  - Bedside glucose at 0 min; check Ca, Na; screen for infection.
  - First seizure: EEG and CT head without contrast (emergency) or MRI.
  - *Glucose at zero.*
- **Treatment**
  - 5 min first benzodiazepine; 10 min second benzodiazepine; 15 min levetiracetam, phenytoin or valproate; 25 min RSI.
  - Neonates: phenobarbital, pyridoxine if resistant; West: ACTH.
  - *ACTH for West; phenobarbital for neonates.*

I-3 Headache in childhood

Most childhood headache is primary (migraine/tension); your job is to find the few secondary ones with three questions - acute or chronic, progressive or not, red flags - and any red flag means urgent MRI.

- **Definition**
  - Primary (migraine, tension-type - benign) vs secondary headache.
  - *Primary = benign; secondary = search.*
- **Causes**
  - Primary: migraine and tension-type = overwhelming majority.
  - Childhood CNS tumors are usually infratentorial: pilocytic astrocytoma, medulloblastoma, ependymoma.
  - Also idiopathic intracranial hypertension, medication overuse headache.
  - *Kids' tumours live downstairs (infratentorial).*
- **Patho-physiology**
  - Raised ICP → worse on waking, coughing, lying flat; vomiting without nausea.
  - Raised ICP → new CN VI palsy = false localising sign.
  - *Six is stretched first.*
- **Clinical picture**
  - Migraine: throbbing, nausea, photophobia; 4 to 72 hours; in children often bilateral, as short as 2 hours.
  - Tension: bilateral, constant, no aura; chronic = 15 days/month for more than 3 months.
  - *Kids' migraine: shorter and both sides.*
- **Diagnosis**
  - Three questions: acute or chronic, progressive or not, red flags.
  - Red flags: awakening from sleep, coordination difficulties, focal signs, age under 5 years.
  - Any red flag → urgent neuroimaging, preferably MRI.
  - *Red flag → MRI.*
- **Treatment**
  - Acute migraine: NSAIDs or acetaminophen first-line; also triptans.
  - Idiopathic intracranial hypertension: acetazolamide (± furosemide).
  - Headache diary for triggers; prophylaxis if frequent.
  - *Acetazolamide dries the CSF tap.*

I-4 Febrile infant

A febrile infant under 3 months cannot localise infection and looks deceptively well: under 28 days it is a full septic screen with LP and IV ampicillin plus gentamicin/cefotaxime for everyone; remember UTI is the commonest serious bacterial infection.

- **Definition**
  - Infant with fever ≥ 38 °C; under 3 months they cannot localise infection.
  - *Under 3 months: fever = full respect.*
- **Causes**
  - Neonatal sepsis: group B streptococcus, E. coli and Listeria; consider HSV.
  - Commonest serious bacterial infection: urinary tract infection.
  - *GBS, E. coli, Listeria: the birth canal trio.*
- **Patho-physiology**
  - Immature immunity + no vaccine cover + no localising signs → appearance unreliable.
  - *Looks well ≠ is well.*
- **Clinical picture**
  - Meningitis = poor feeding, 'not right'; no neck stiffness.
  - Non-blanching rash → meningococcal sepsis; fever >5 days → think Kawasaki.
  - *Fever > 5 days: think Kawasaki.*
- **Diagnosis**
  - Under 28 days: full septic screen incl. lumbar puncture for all.
  - 1-3 months: FBC, CRP, culture, urine; LP if unwell.
  - Over 3 months: traffic-light system; urine in everyone: clean catch, not a bag.
  - *Under 28 days: everyone gets the LP.*
- **Treatment**
  - Under 28 days: admit + IV ampicillin (amoxicillin) + gentamicin/cefotaxime; ± aciclovir.
  - Listeria is resistant to cephalosporins; non-blanching rash → antibiotics first.
  - Well over 3 months with viral source → home with safety-netting.
  - *Listeria laughs at cephalosporins.*

I-5 Rash and fever in children (incubation periods as well!)

Fever plus rash: first ask 'does it blanch?' - a non-blanching rash is meningococcal sepsis until proven otherwise - then pattern-match by morphology, site of onset, timing relative to fever and incubation period.

- **Definition**
  - Fever + rash: first ask does it blanch?, then pattern-match.
  - *Glass first, guess second.*
- **Causes**
  - First measles, second scarlet fever, third rubella, fifth parvovirus, sixth roseola.
  - Chickenpox: varicella-zoster virus (VZV), airborne.
  - Emergencies: meningococcaemia, Kawasaki, SJS/TEN, toxic shock.
  - *1 measles, 2 scarlet, 3 rubella, 5 parvo, 6 roseola.*
- **Patho-physiology**
  - Non-blanching + fever → meningococcal sepsis until proven otherwise.
  - Petechiae above the nipple line after coughing or vomiting: raised venous pressure.
  - Chickenpox lesions are in different stages.
  - *Chickenpox: 'stars in all stages'.*
- **Clinical picture**
  - Measles rash starts behind the ears; Koplik spots; death from pneumonia.
  - Scarlet fever: sandpaper rash, strawberry tongue; parvovirus: slapped cheek.
  - Rubella: suboccipital nodes. Coxsackie A: hand, foot and mouth disease.
  - *Slapped cheek = parvo; sandpaper = scarlet.*
- **Diagnosis**
  - Glass test; then morphology, site of onset, timing vs fever.
  - Incubation: varicella 10-21 days; rubella 14-21 days; measles 10-14 d; scarlet 2-5 d.
  - *Scarlet is quickest: 2-5 days.*
- **Treatment**
  - Suspected meningococcal sepsis → benzylpenicillin or ceftriaxone immediately.
  - Severe chickenpox: IV acyclovir; newborn of infected mother: VZIG.
  - Kawasaki → IVIG within 10 days.
  - *Kawasaki: IVIG inside 10 days.*

I-6 Fever of Unknown Origin (FUO) and Kawasaki syndrome

FUO is usually an uncommon presentation of a common infection, found by repeating the history and examination daily; Kawasaki is the FUO-mimic you must not miss - treat with IVIG 2 g/kg plus aspirin within 10 days to protect the coronaries.

- **Definition**
  - FUO: fever >38 °C ≥ 8 days, no cause after initial evaluation.
  - Kawasaki: fever ≥ 5 days plus 4 of 5 principal features.
  - *Kawasaki: fever 5 + 4 of 5.*
- **Causes**
  - FUO is most often infection (~50%), then inflammatory disease (~20%), then malignancy (~10%).
  - Kawasaki classically affects Asian children.
  - *FUO: infection 50, inflammation 20, malignancy 10.*
- **Patho-physiology**
  - Kawasaki hits the coronary arteries → aneurysm → rupture or myocardial infarction.
  - sJIA with falling ESR, cytopenias, high ferritin → macrophage activation syndrome.
  - *Falling ESR in sJIA = danger, not recovery.*
- **Clinical picture**
  - Kawasaki conjunctivitis: bilateral, non-purulent, limbal sparing.
  - Thrombocytosis in the second week; high CRP/ESR, sterile pyuria.
  - sJIA: daily spiking fevers, salmon-pink macular rash.
  - *Kawasaki eyes: red but no pus.*
- **Diagnosis**
  - FUO: repeated daily history and examination is the highest-yield investigation.
  - Suspected Kawasaki: CRP or ESR; echocardiogram at diagnosis, then 1-2 and 4-6 weeks.
  - *FUO: look again tomorrow.*
- **Treatment**
  - IVIG 2 g/kg plus aspirin within 10 days; aspirin exception despite Reye syndrome.
  - No defervescence → second IVIG; refractory → steroids/infliximab.
  - Incomplete Kawasaki: under 1 year, highest aneurysm risk; delay vaccines 8 months.
  - *IVIG 2 g/kg + aspirin, within 10 days.*

I-7 Differential diagnosis and treatment of neonatal respiratory distress

Neonatal respiratory distress is pattern recognition on gestation and delivery: preterm = RDS, term C-section = TTN, post-term with meconium = MAS/PPHN - but always think sepsis, cardiac, CDH, choanal atresia and TE fistula; CPAP for all.

- **Definition**
  - Newborn with tachypnoea >60, grunting, flaring, recession, cyanosis.
  - *Over 60 = too fast for a newborn.*
- **Causes**
  - RDS: prematurity, maternal diabetes, caesarean; TTN: term caesarean; MAS: term and post-term.
  - Sepsis; cyanotic CHD, diaphragmatic hernia, choanal atresia, TE fistula.
  - *Pre-RDS, section-TTN, post-MAS.*
- **Patho-physiology**
  - Prematurity → inadequate surfactant → alveoli fail to open → poor oxygenation.
  - TTN: delayed clearance of fetal lung fluid.
  - PPHN: pulmonary vascular resistance fails to fall; meconium inactivates surfactant.
  - *Surfactant keeps the balloons open.*
- **Clinical picture**
  - TTN: term C-section baby with grunting, settles in 24-72 hours.
  - CDH: scaphoid abdomen; choanal atresia: pink when crying.
  - TE fistula: coughing, choking, cyanosis.
  - *Pink when crying = choanal atresia.*
- **Diagnosis**
  - RDS CXR: ground-glass with air bronchograms; lecithin:sphingomyelin ratio ≥ 2.
  - No response to 100% oxygen → duct-dependent CHD; PPHN: echocardiography.
  - NG tube coiling in esophagus → esophageal atresia.
  - *Coiled tube = blind oesophagus.*
- **Treatment**
  - CPAP for all; RDS: maternal antenatal glucocorticoids, surfactant when FiO_2 exceeds 0.30.
  - CDH: never bag-mask; intubate with NG tube on suction.
  - Sepsis: ampicillin + gentamicin (cefotaxime if meningitis); PPHN: nitric oxide.
  - *CDH: no bag-mask, tube + suction.*

I-8 Neonatal jaundice

Neonatal jaundice is sorted by onset and by splitting the bilirubin: jaundice in the first 24 hours is always pathological (haemolysis until proven otherwise), and prolonged conjugated jaundice with pale stools is biliary atresia until a Kasai is done before 60 days.

- **Definition**
  - Classify by onset and conjugated vs unconjugated; UGT conjugates bilirubin.
  - *UGT = the liver's bilirubin glue.*
- **Causes**
  - Jaundice in the first 24 hours is always pathological: Rh/ABO, G6PD, spherocytosis, sepsis.
  - Breast milk (β-glucuronidase); Gilbert and Crigler-Najjar: unconjugated.
  - Conjugated: biliary atresia, hepatitis, choledochal cyst, TORCH, galactosaemia.
  - *Day 1 jaundice = always pathological.*
- **Patho-physiology**
  - Phototherapy makes bilirubin water-soluble, excreted without hepatic conjugation.
  - High unconjugated bilirubin → kernicterus.
  - Rh disease: Rh- mother, Rh+ fetus; not the first pregnancy.
  - *Rh: first baby sensitises, second suffers.*
- **Clinical picture**
  - Prolonged jaundice beyond 2 weeks with pale stools and dark urine = biliary atresia.
  - Cephalohematoma is subperiosteal, does not cross sutures; encephalopathy: high-pitched cry, opisthotonus.
  - *Pale stools + dark urine = blocked bile.*
- **Diagnosis**
  - Early work-up: blood group, direct Coombs, FBC, film.
  - Split the bilirubin: conjugated is >25 µmol/L or >20% of total.
  - *Split it: >25 or >20%.*
- **Treatment**
  - Phototherapy above the line; exchange transfusion if it fails or encephalopathy.
  - Breastfeeding jaundice: increase feeds; breast milk jaundice: continue breastfeeding.
  - Anti-D at 28-32 weeks and within 72 hours; Kasai before 60 days.
  - *Kasai before 60 days.*

I-9 Neonatal cyanosis

A comfortably blue baby is cardiac, a distressed blue baby is pulmonary: use the hyperoxia test and pre/post-ductal saturations, get an echo, and if the lesion may be duct-dependent start prostaglandin E1 before the duct closes on day 2-3.

- **Definition**
  - Central cyanosis (lips, mucous membranes) = low arterial oxygen saturation.
  - *Look at the tongue, not the toes.*
- **Causes**
  - Respiratory, cardiac, PPHN (SSRIs in pregnancy), methaemoglobinaemia, polycythemia (hematocrit >65%).
  - Five Ts: truncus, transposition (day 1), tricuspid atresia, tetralogy of Fallot (commonest overall), total anomalous pulmonary venous return.
  - *Five Ts: count the fingers.*
- **Patho-physiology**
  - PPHN: pulmonary vascular resistance fails to fall → right-to-left shunt at PDA and foramen ovale.
  - Duct closes day 2-3 → collapse.
  - *Duct closes day 2-3 → crash.*
- **Clinical picture**
  - Comfortably blue = cardiac, as respiratory drive responds mainly to CO_2; distressed and blue = pulmonary.
  - Methaemoglobinaemia: chocolate-brown blood, normal PaO_2.
  - *Blue but happy = heart.*
- **Diagnosis**
  - Hyperoxia test: pulmonary cyanosis improves on 100% O_2; cardiac and PPHN do not.
  - Pre-ductal saturation on the right hand; difference over 10% suggests PPHN.
  - Echocardiography is the key investigation.
  - *Hyperoxia: lungs respond, hearts don't.*
- **Treatment**
  - Duct-dependent lesions: prostaglandin E1, avoiding excess oxygen.
  - Transposition: balloon atrial septostomy before arterial switch.
  - PPHN: inhaled nitric oxide (cGMP); refractory → ECMO; methaemoglobinaemia: methylene blue.
  - *Methaemoglobin → methylene blue.*

I-10 Hypoglycaemia and hyperglycaemia

With hypoglycaemia, take the critical sample before the dextrose and let the ketones decide: high ketones = ketotic/hormonal, low ketones = hyperinsulinism or fat-oxidation defect; in DKA give fluids first and start insulin an hour later without a bolus.

Definition
- Neonatal hypoglycaemia: blood glucose below 2.6 mmol/L.
- HHS: glucose >600 mg/dL, osmolarity >320.

Neonate low: under 2.6.

Causes
- Ketotic hypoglycaemia of childhood is commonest at 1–5 y; GSD, GH/cortisol deficiency.
- No ketones → hyperinsulinism (infant of diabetic mother) or fat-oxidation defect (MCAD).
- Hyperglycaemia → almost always type 1 DM.

No ketones: insulin or MCAD.

Patho-physiology
- Maternal diabetes → fetal hyperinsulinism → neonatal hypoglycaemia, macrosomia.
- DKA insulin bolus → rapid osmotic shifts → cerebral oedema.

No bolus in DKA: protect the brain.

Clinical picture
- Autonomic (sweating, tremor, tachycardia, hunger) → neuroglycopenic (confusion, seizures, coma).

Shaky first, confused later.

Diagnosis
- Take the critical sample (insulin, C-peptide, cortisol, GH, ketones) before giving glucose.
- Best discriminator: the ketone level at the time of the low sugar.
- DKA = glucose + ketosis + pH <7.3 or bicarbonate <18 mmol/L.

Critical sample before sugar.

Treatment
- 2 ml/kg of 10% dextrose, then infusion; glucagon if no access.
- Diazoxide for hyperinsulinism; cornstarch (GSD); avoid fasting (MCAD).
- DKA: fluids first, then insulin 0.05–0.1 U/kg/h an hour later, no bolus.

2-10: 2 ml/kg of 10%.

I-11 Differential diagnosis of lymphadenopathy

Most childhood nodes are reactive (soft, tender, <2 cm, gone in 4–6 weeks); worry about nodes over 2 cm and growing, beyond 6 weeks, supraclavicular or painless - start with an FBC and film, and always get a chest X-ray before anyone gives an anaesthetic for biopsy.

Definition
- Reactive nodes (soft, tender, <2 cm, resolve 4–6 wk) vs malignant nodes.

Reactive: soft, sore, small, short-lived.

Causes
- Localised: Staphylococcus aureus > Streptococcus pyogenes, cat scratch (Bartonella), atypical mycobacteria, TB.
- Generalised: EBV, CMV, toxoplasma, HIV, Kawasaki, JIA, SLE, leukaemia, lymphoma.

Cat scratch = Bartonella.

Patho-physiology
- EBV: T-cell hyperplasia in the node paracortex.
- Staph/Strep lymphadenitis → suppuration, abscess, fluctuance.
- Lymphoma diagnosis depends on nodal architecture; mediastinal mass threatens airway.

Lymphoma needs the whole node.

Clinical picture
- EBV: fatigue, fever, posterior cervical nodes, pharyngitis, hepatosplenomegaly.
- Worrying: over 2 cm and growing, beyond 6 weeks, supraclavicular, painless.
- Cold, violaceous, non-tender node in a toddler: atypical mycobacteria.

Supraclavicular node = sinister.

Diagnosis
- Highest-yield first test: full blood count and film; CRP, LDH, urate, EBV/CMV serology.
- Chest X-ray before biopsy to look for a mediastinal mass.

CXR before the knife.

Treatment
- Soft, mobile, non-tender nodes: observation; reactive nodes resolve in 4–6 weeks.
- Suspected lymphoma: excisional biopsy, not FNA.

Excise, don't aspirate.

I-12 Arthralgia

A child with joint pain has septic arthritis until proven otherwise - aspirate before antibiotics - then classify by age (DDH, Perthes, SUFE) and timing of pain: morning = inflammatory, after activity = mechanical, night = malignant.

Definition
- Joint pain: exclude septic arthritis first, then classify by cause and timing.

Sepsis first, sort later.

Causes
- Reactive: rheumatic fever (GAS M protein mimicry); inflammatory: JIA, SLE, HSP; malignant: leukaemia.
- Mechanical by age: DDH infancy, Perthes (4–8 years), SUFE obese adolescent.
- Septic: Staphylococcus aureus; Salmonella osteomyelitis in sickle cell.

DDH baby, Perthes child, SUFE teen.

Patho-physiology
- Osteomyelitis starts in the metaphysis (slow-flowing hairpin capillary loops).

Bugs park in the metaphyseal hairpin loops.

Clinical picture
- Septic: febrile, toxic, refuses to weight-bear; transient synovitis: afebrile, well, limps.
- Morning = inflammatory, after activity = mechanical, night = malignant.
- Jones: carditis, polyarthritis, Sydenham chorea, erythema marginatum; sJIA: quotidian fever.

Morning inflammatory, evening mechanical, night malignant.

Diagnosis
- Kocher: non-weight-bearing, fever >38.5, ESR >40, WBC >12.
- Urgent aspiration + blood cultures; MRI for osteomyelitis.
- JIA: onset before 16, lasting at least 6 weeks.

Kocher: 4 points, 4 = almost certain.

Treatment
- Aspirate the joint before antibiotics → IV antibiotics + surgical washout.
- Transient synovitis: rest and NSAIDs; JIA: methotrexate weekly.
- Rheumatic fever: penicillin prophylaxis; macrolides if anaphylaxis.

Tap before treat.

I-13 Causes and differential diagnosis of bloody stool in children

With blood per rectum, check haemodynamics and do a rectal exam first, then let age and story narrow it: neonate (swallowed blood, NEC, volvulus), toddler (fissure, intussusception, Meckel's), older child (infective colitis, IBD, HSP, HUS).

Definition
- Bloody stool: digital rectal examination and haemodynamics first, then narrow by age and story.

Finger first (DRE), then the age.

Causes
- Neonate: swallowed maternal blood, NEC, CMPA, volvulus, vitamin K deficiency.
- Toddler: fissure (commonest), intussusception, Meckel's, polyp.
- Older: infectious colitis, IBD, HSP, HUS.

Toddler bleed: fissure first.

Patho-physiology
- Meckel's bleeds because it contains ectopic gastric mucosa.
- Rule of 2s: 2% of people, 2 feet from ileocaecal valve, 2 inches, under age 2.
- E. coli O157 toxin → HUS; antibiotics increase toxin release.

Meckel's: everything is 2.

Clinical picture
- Painful, paroxysmal + sausage mass (3 mo–3 y) → intussusception.
- Painless, transfusion-requiring bleeding without diarrhoea → Meckel's.
- Bilious vomiting + blood → volvulus.

Bilious vomit = volvulus until proven otherwise.

Diagnosis
- Apt test for maternal blood.
- Meckel's → technetium-99m pertechnetate scan; intussusception → ultrasound target sign.
- HUS: haemolytic anaemia, thrombocytopenia, acute kidney injury.

Target on ultrasound = intussusception.

Treatment
- Shocked → resuscitate first: wide-bore access, fluids, cross-match.
- Intussusception: enema reduction. Volvulus: theatre. NEC: stop feeds, decompress, antibiotics.
- HUS: no antibiotics for E. coli O157.

NEC: stop, drain, treat.

I-14 Upper gastrointestinal bleeding in children (hematemesis)

In a child with haematemesis, first prove the blood is the child's own and from the GI tract, then split variceal from non-variceal - that split decides the treatment; a swallowed button battery is an emergency within hours.

Definition
- Haematemesis: confirm the blood is the child's own and GI.

Is it theirs, and is it gut?

Causes
- Non-variceal: oesophagitis, gastritis, peptic ulcer (H. pylori, NSAIDs), Mallory-Weiss, button battery.
- Variceal: portal hypertension after umbilical venous catheterisation (portal vein thrombosis), biliary atresia.
- Neonate: swallowed maternal blood, vitamin K deficiency bleeding.

UVC → portal vein clot → varices.

Patho-physiology
- Portal vein thrombosis → pre-hepatic portal hypertension → varices.
- Button battery → necrosis within hours → can erode into the aorta.

Button battery: hours, not days.

Clinical picture
- Splenomegaly, ascites, caput medusae, thrombocytopenia → variceal.
- Neonate bleeding from several sites → vitamin K deficiency.

Big spleen + bleed = varices.

Diagnosis
- Exclude impostors: epistaxis, cracked nipple (Apt test), red foods.
- Split variceal vs non-variceal - decides treatment.

Impostors: nose, nipple, food.

Treatment
- Two large cannulas, cross-match, blood; vitamin K + FFP for coagulopathy.
- Variceal: terlipressin or octreotide + prophylactic antibiotics + band ligation.
- Non-variceal: high-dose proton pump inhibitor + endoscopy; button battery → emergency removal.

Varices: terlipressin, antibiotics, bands.

I-15 Proteinuria and leukocyturia (pyuria)

For proteinuria ask 'is it real, is it big?' (first-morning urine, protein:creatinine ratio); most is transient or orthostatic, and a typical nephrotic child aged 1–10 gets steroids without biopsy. Leukocyturia is UTI until the culture is negative.

Definition
- Proteinuria: is it real, is it big? Leukocyturia = UTI until culture negative.

Pyuria = UTI until proven sterile.

Causes
- Commonest: transient proteinuria; orthostatic (upright only, overweight adolescents).
- Nephrotic age 1–10 → minimal change disease (~90%).
- Sterile pyuria: partially treated UTI, renal tuberculosis, Kawasaki disease, appendicitis.

1–10 years nephrotic = minimal change.

Patho-physiology
- Nephrotic: heavy proteinuria, oedema, albumin below 30 g/L, hyperlipidaemia.
- PSGN: low C3, normal C4, normal by 8 weeks; follows pharyngitis by 1–2 weeks, skin by 3–6 weeks.

PSGN: throat 1–2, skin 3–6 weeks.

Clinical picture
- Nephrotic complications: spontaneous bacterial peritonitis, renal vein thrombosis, PE.
- Nephritic: haematuria with red cell casts, hypertension, milder proteinuria.

Nephrotic: protein out; nephritic: blood out.

Diagnosis
- Start with urinalysis; confirm on first-morning urine; quantify with protein:creatinine ratio.
- UTI: nitrites specific, leucocyte esterase sensitive; nephritic → complement, ASOT.
- After UTI: ultrasound → MCUG (VCUG), DMSA.

DMSA for scars, MCUG for reflux.

Treatment
- Typical nephrotic 1–10 y → steroids without biopsy.
- Nephritic → urgent nephrology.

Typical nephrotic: steroids, skip biopsy.

I-16 Macroscopic and microscopic haematuria

Confirm true red cells on microscopy first, then make the one decision that drives everything: glomerular (cola urine, casts, protein, high BP → complement and renal work-up) or non-glomerular (red urine, clots → UTI, stones, trauma, Wilms).

Definition
- Haematuria: confirm red cells on microscopy via urinalysis, then glomerular vs non-glomerular.

Dipstick suspects, microscope convicts.

Causes
- Glomerular: IgA, post-streptococcal GN, HSP, Alport, thin BM, MPGN/lupus, HUS.
- Non-glomerular: UTI (commonest overall), stones, trauma, Wilms, coagulopathy, sickle cell.

Filter vs pipes: IgA, PSGN, HSP, Alport above; UTI, stones, trauma, Wilms below.

Patho-physiology
- Red cell casts prove haematuria is glomerular in origin.
- IgA 1–2 days after URTI, normal complement; post-strep 1–3 weeks, low C3.

IgA = 'Immediately' (1–2 days); post-strep = 'Patient' (1–3 weeks).

Clinical picture
- Glomerular: brown or cola-coloured, without clots, proteinuria, hypertension.
- Initial haematuria = urethral; terminal = bladder or prostate.
- Alport: sensorineural deafness and family history of renal failure.

Cola = kidney; clots = plumbing.

Diagnosis
- Dipstick positive with no red cells → myoglobinuria or haemoglobinuria; exclude beetroot, rifampicin.
- Glomerular: check complement levels; BP, protein dipstick, renal function, ultrasound.

Dipstick positive, no cells → think pigment.

Treatment
- Red cell casts → glomerular work-up; stop looking at the bladder.
- Non-glomerular → target cause: UTI, stones, trauma, tumour (Wilms spreads to lungs).

Casts → kidney, not bladder.

I-17 Causes of enuresis in childhood

Enuresis is only diagnosed from age 5; dipstick everyone and examine the spine to exclude organic and secondary causes, then manage behaviourally, using the alarm for durable cure and desmopressin (with evening fluid restriction) for quick control.

Definition
- Involuntary voiding at age ≥ 5: at least 2 times/week for ≥ 3 months; primary vs secondary.

5 years, 2 a week, 3 months: 5-2-3.

Causes
- Greatest risk factor: family history; both parents → ~70%.
- Secondary: UTI, diabetes mellitus, DI, CKD, constipation, stress/abuse, neuro, structural.

Both parents wet → 7 in 10 kids wet.

Patho-physiology
- Failure to wake, nocturnal polyuria (low ADH), small bladder capacity.
- Constipation compresses the bladder, causing detrusor overactivity.

Sleep, wee volume, small tank.

Clinical picture
- Continuous dampness in a toilet-trained girl → ectopic ureter.
- Sacral dimple/hairy patch → spinal dysraphism; polyuria, polydipsia, weight loss → DM.

Always damp girl = ureter in the wrong place.

Diagnosis
- First rule out organic causes: dipstick everyone for infection, glucose, specific gravity.
- Examine lumbosacral spine, gait, lower-limb reflexes.

Dip every wetter.

Treatment
- First-line: lifestyle modification and behavioral training (night fluid restriction, timed voiding).
- Next: enuresis alarm ± oral desmopressin; imipramine rarely used.
- Desmopressin → restrict evening fluids (hyponatraemia); the alarm is the best long-term cure.

Alarm cures, desmopressin covers (sleepovers).

I-18 Polydipsia, polyuria

Polyuria has three causes: osmotic (diabetes mellitus), diabetes insipidus and primary polydipsia. Dipstick for glucose first, then paired serum and urine osmolality, then a water deprivation test with desmopressin to separate central from nephrogenic DI.

Definition
- True polyuria: >2 l/m²/day, with polydipsia.

Over 2 litres per m² per day.

Causes
- Three causes: osmotic diuresis (diabetes mellitus), diabetes insipidus, primary polydipsia.
- Central DI: craniopharyngioma, germinoma, Langerhans cell histiocytosis, trauma.
- Nephrogenic: AVPR2, low K, high Ca, lithium.

Sugar, hormone, habit.

Patho-physiology
- Central = no ADH; nephrogenic = kidney ignores ADH.
- Primary polydipsia → low-normal serum osmolality.

Central = no signal; nephrogenic = deaf receiver.

Clinical picture
- Glycosuria, weight loss despite appetite, ketones → DM.
- DI: high serum osmolality with dilute urine.

DI: thick blood, thin urine.

Diagnosis
- First bedside test: urine dipstick for glucose; then paired serum + urine osmolality.
- Water deprivation test: stop at 3–5% body-weight loss.
- Concentrates after desmopressin → central DI → MRI of the hypothalamic-pituitary axis.

Responds to DDAVP → brain is the problem → MRI.

Treatment
- Central DI: desmopressin + assess other pituitary hormones (cortisol).
- Nephrogenic: free water, low-solute diet, thiazides, sometimes indometacin (indomethacin).

Paradox: a diuretic (thiazide) reduces urine in nephrogenic DI.

I-19 Differential diagnosis and treatment of hypertension. Pheochromocytoma.

Hypertension in a pre-adolescent child is secondary until proven otherwise, most often renal. Confirm with the right cuff on 3 occasions and check all four limbs. In phaeochromocytoma always give alpha-blockade before beta-blockade.

Definition
- BP ≥ 95th centile for age, sex, height, confirmed on 3 occasions.
- Pheochromocytoma: tumor of chromaffin cells; rule of 10s.

95th centile ×3 visits.

Causes
- Before adolescence most often kidney disease (~70–80%), then renovascular, coarctation, endocrine, drugs.
- ARPKD (infants, PKHD1); ADPKD (young adults, renin).
- Pheochromocytoma: MEN 2A/2B, VHL, NF1, SDH; familial more often bilateral.

Child BP high? Look at the kidney.

Patho-physiology
- Catecholamine-secreting tumour → episodic surges.
- Beta-blocker first → unopposed alpha → hypertensive crisis.
- Neuroblastoma (adrenal medulla): N-myc, Homer-Wright rosettes.

Alpha before beta, like the alphabet.

Clinical picture
- Episodic headache, sweating, palpitations; fasting hyperglycemia.
- Weak femorals → coarctation; hypertension + bradycardia = Cushing triad.
- MEN 2B: tall, kyphoscoliosis, lip neuromas.

Phaeo triad: headache, sweat, palpitations.

Diagnosis
- Cuff bladder covers 80% of arm; too small raises reading; BP in all four limbs.
- Pheo: metanephrines → adrenal CT/MRI → I-123 MIBG scan.
- Ultrasound for PKD; string of beads = fibromuscular dysplasia.

Biochemistry before scans.

Treatment
- Pheo: alpha-blockade first (phenoxybenzamine), then beta-blockade, then surgical excision.
- Hyperkinetic hypertension: beta-blocker; Cushing triad: do not lower BP.

Cushing triad: don't drop the BP, drop the ICP.

I-20 Differential diagnosis and treatment of cough in children

Acute cough is almost always viral; chronic cough (> 4 weeks) needs a cause. A wet cough with clubbing is suppurative lung disease, never asthma, and a sudden cough in a well toddler is a foreign body even if the X-ray is normal.

Definition
- Acute cough under 3 weeks; chronic cough lasts more than 4 weeks.

Under 3 acute; over 4 chronic.

Causes
- Acute: viral infection of the respiratory tract commonest; bronchiolitis (RSV), croup, foreign body.
- Chronic: pertussis, asthma, GERD, cystic fibrosis, protracted bacterial bronchitis, primary ciliary dyskinesia.

Acute = virus until proven otherwise.

Patho-physiology
- Bronchiolitis: small airway inflammation, usually RSV.
- Pertussis: catarrhal 1–2 weeks → paroxysmal → convalescent; coughing risks pneumothorax.
- Chronic wet cough + clubbing → suppurative lung disease, never asthma.

Clubbing is never asthma.

Clinical picture
- Bronchiolitis: wheezing/crackles, not stridor. Pertussis: whoop, post-tussive vomiting; infants apnoea and cyanosis.
- Barking → croup; nocturnal dry + wheeze → asthma; meconium ileus → CF.
- Sudden cough in a well toddler → foreign body.

Bark = croup; whoop = pertussis; sudden = foreign body.

Diagnosis
- Plot growth, check clubbing, immunisations, choking history, TB contacts.
- Suspected foreign body → rigid bronchoscopy; normal CXR doesn't exclude.

Normal CXR doesn't clear a foreign body.

Treatment
- Viral: reassure + safety-net; no over-the-counter cough medicines.
- Bronchiolitis: supportive; RSV prevention nirsevimab or palivizumab.
- Pertussis: macrolide, also prophylaxis for contacts.

Pertussis: macrolide for patient and contacts.

I-21 Sore throat (viral versus bacterial pharyngitis) and upper respiratory tract infection

Most sore throats are viral; the job is to find group A strep (abrupt fever, exudate, tender anterior nodes, no cough), treat it with 10 days of penicillin to prevent rheumatic fever, and never miss the airway emergencies.

Definition
- About 70–80% of sore throats in children are viral; the task is finding group A strep.

Hunt the strep among the viruses.

Causes
- Viruses; EBV in teenagers; strep in age 3–14.
- Emergencies: epiglottitis, quinsy, retropharyngeal abscess, diphtheria, Lemierre syndrome.

Strep 3–14.

Patho-physiology
- Untreated strep → acute rheumatic fever; antibiotics do not prevent PSGN.
- Cough, coryza or hoarseness = viral.

Antibiotics stop the heart damage, not the kidney damage.

Clinical picture
- Strep: abrupt fever, exudate, tender nodes, no cough; EBV: posterior nodes, splenomegaly.
- Quinsy: trismus, uvula deviated away; epiglottitis: drooling, muffled voice, tripod, no cough.

Quinsy pushes uvula away.

Diagnosis
- Rapid strep has low sensitivity: negative → throat culture.
- EBV: atypical lymphocytes, raised LFTs; diphtheria: gray-white pseudomembrane.

Negative rapid test → culture.

Treatment
- Strep: penicillin V for 10 days, to prevent rheumatic fever.
- EBV given amoxicillin → non-allergic maculopapular rash; no contact sport 4–6 weeks (splenic rupture).

EBV + amoxicillin = rash; spleen = no rugby.

I-22 Inspiratory stridor, croup syndrome

Inspiratory stridor means extrathoracic narrowing. In a child aged 6 months to 6 years with a barking cough it is croup: oral dexamethasone for everyone, nebulised adrenaline if severe. Grade severity by stridor at rest, recession and conscious level, and treat the toxic, drooling child as epiglottitis or tracheitis.

Definition
- Inspiratory stridor localises narrowing above the cords (extrathoracic); acute cause is usually croup.

Inspiratory = up high.

Causes
- Croup: parainfluenza virus, age 6 months to 6 years.
- Mimics: epiglottitis, bacterial tracheitis (Staph aureus), foreign body, laryngomalacia, vascular ring.

Parainfluenza = croup.

Patho-physiology
- The subglottis swells inward because it is ringed by the cricoid, the only complete cartilage ring.

Rigid ring: swelling squeezes inwards.

Clinical picture
- Croup: barking cough, stridor, not toxic; laryngomalacia: improves prone.
- Epiglottitis: drooling, toxic, no cough; tracheitis: toxic, no response to adrenaline.

Toxic + no bark = not croup.

Diagnosis
- Severity = stridor at rest, recession, conscious level, not oxygen saturation.
- Epiglottitis: do not examine the throat.

Epiglottitis: hands off the throat.

Treatment
- Every child with croup gets oral dexamethasone; severe: nebulised adrenaline, wears off in about 2 hours.
- Epiglottitis: secure the airway first; foreign body: rigid bronchoscopy.

Adrenaline lasts ~2 hours: watch for rebound.

I-23 Haemolytic anaemias in childhood

Prove haemolysis (high reticulocytes, unconjugated bilirubin and LDH, low haptoglobin), then split by the direct Coombs test: positive means immune (warm IgG or cold IgM); negative means membrane, enzyme, haemoglobin or microangiopathic causes, which the blood film usually identifies.

Definition
- Haemolysis is proven by anaemia with high reticulocytes, high unconjugated bilirubin, high LDH, low haptoglobin.

Haemolysis: retics up, hapto down.

Causes
- Positive direct Coombs = immune: warm AIHA IgG, cold AIHA IgM.
- Coombs negative: hereditary spherocytosis, G6PD, sickle cell, thalassaemia, HUS/TTP/DIC.

Warm = IgG; cold = IgM (M for 'chilly Mountain').

Patho-physiology
- Hereditary spherocytosis: membrane tethering defect, autosomal dominant.
- G6PD: triggered by infection, fava beans, drugs → bite cells, Heinz bodies.

G6PD: beans, bites, Heinz.

Clinical picture
- Jaundice, splenomegaly, dark urine; sickle cell: dactylitis, acute chest syndrome.
- Low reticulocytes in chronic haemolysis = parvovirus B19 aplastic crisis.

Low retics in a haemolyser = parvovirus.

Diagnosis
- Spherocytes, Coombs negative, positive EMA binding, raised MCHC = hereditary spherocytosis.
- Schistocytes = microangiopathic haemolysis; Hb electrophoresis for sickle and thalassaemia.

Schistocytes = shredded in small vessels.

Treatment
- Warm AIHA: stop offending drug, steroids; splenectomy last resort.
- Sickle: daily penicillin to 5 years; fever → ceftriaxone; stroke or priapism → exchange transfusion.

Sickle: penicillin till 5.

I-24 Nonhemolytic anaemias

Start every anaemia with the MCV. Microcytic anaemia in a child is iron deficiency (often from excess cow's milk) until thalassaemia trait or lead poisoning is excluded with RDW, red cell count and ferritin interpreted alongside CRP; macrocytosis or pancytopenia points to B12/folate deficiency or marrow failure.

Definition
- The first branch point in any anaemia is the MCV; the commonest in childhood is iron deficiency.

Start with the MCV.

Causes
- Microcytic: iron deficiency, thalassaemia trait, lead poisoning.
- Macrocytic: B12/folate deficiency, Diamond-Blackfan; pancytopenia: Fanconi, aplastic anaemia.

Microcytic = 'FiLTh': iron, lead, thal.

Patho-physiology
- Excess cow's milk: iron-poor, inhibits iron absorption, occult GI blood loss.
- B12/folate are needed for DNA synthesis, so cells grow large.

Too much milk, too little iron.

Clinical picture
- Lead: bluish gum-tooth line, LEAD mnemonic; B12: paresthesias, megaloblasts.
- Fanconi: short stature, café-au-lait spots, malformed thumbs.

Fanconi: thumbs, short, café-au-lait.

Diagnosis
- RDW high in iron deficiency, normal in thalassaemia trait; red cell count normal or high.
- Ferritin is an acute phase protein, so inflammation can mask deficiency.

Thal trait: many small even cells.

Treatment
- Iron deficiency: oral elemental iron, continued 3 months after Hb normalises.
- Lead 45–69: DMSA (succimer); ≥ 70 mcg/dL: IM dimercaprol + EDTA.

Iron: 3 more months to fill the tank.

I-25 Causes of failure to thrive

Failure to thrive is a pattern (weight falls across two or more centile lines) and usually a feeding problem. Watch a feed and read the growth chart (weight falls before height, head last) before ordering tests, and keep neglect on the list.

Definition
- Failure to thrive classically means crossing two or more weight centile lines downward.

Two lines down = FTT.

Causes
- Commonest cause: inadequate intake (non-organic); child neglect is the commonest maltreatment.
- Organic: cystic fibrosis, galactosaemia, congenital heart disease, enteropeptidase deficiency.

Not enough in, before rare diseases.

Patho-physiology
- Order of loss: weight → height → head circumference.
- Congenital heart disease: increased energy expenditure plus reduced intake.

Weight, height, head: W-H-H.

Clinical picture
- Short and plump with delayed bone age suggests an endocrine cause.
- Small head from early on: intrauterine, genetic or congenital infection.
- Kwashiorkor: oedema, swollen abdomen; marasmus: wasting.

Kwashiorkor swells; marasmus shrivels.

Diagnosis
- Highest-yield 'investigation': observe a feed.
- Meconium ileus → cystic fibrosis; neonatal jaundice and E. coli sepsis → galactosaemia.

Watch, don't test.

Treatment
- Treat the cause; feeding advice for intake problems.
- Suspected abuse: first contact child protective services (CPS).

Suspect abuse → call protection services.

I-26 Hepatomegaly and splenomegaly

Measure liver span rather than trusting a palpable edge, start with an FBC and film, and judge liver severity by the INR rather than the ALT. The rest of the frame is the company the organomegaly keeps: infection, haemolysis, storage disease or inflammation.

Definition
- Enlarged liver, spleen or both; a palpable liver edge is normal in infants, so measure the liver span.

Measure span, not edge.

Causes
- EBV/infectious mononucleosis, CMV, hepatitis A, malaria, visceral leishmaniasis.
- Hereditary spherocytosis, systemic JIA, glycogen storage disease type I.

Infection, blood, storage, inflammation.

Patho-physiology
- A spleen must enlarge to about 2–3 times normal size before it is palpable.
- Hypersplenism: enlarged spleen sequesters blood cells and lowers the platelet count.

Big spleen eats platelets.

Clinical picture
- Infant hepatomegaly with fasting hypoglycaemia suggests von Gierke.
- CMV newborn: blueberry muffin rash, jaundice; systemic JIA: spiking fevers, salmon-pink rash.

Blueberry muffin = congenital infection.

Diagnosis
- First test: full blood count and film.
- Synthetic function: prothrombin time (INR), not the ALT.

INR = function; ALT = damage.

Treatment
- Treat the cause.
- EBV splenomegaly: avoid contact sports for 4–6 weeks.

EBV spleen: no contact sport 4–6 weeks.

I-27 Differential diagnosis of diarrhoea

In acute diarrhoea treat the child's hydration, not the stool: ORS works because SGLT1 is intact. In chronic diarrhoea the only question that matters is whether the child is growing; toddler's diarrhoea thrives, coeliac disease and CF do not.

Definition
- In acute diarrhoea the priority is assessing hydration, not the stool.

Look at the child, not the nappy.

Causes
- Acute: mostly rotavirus and norovirus; bacterial if bloody.
- Chronic: pivotal question is whether the child is growing; toddler's diarrhoea is the thriving form.

Growing well? Probably toddler's diarrhoea.

Patho-physiology
- ORS works because the sodium-glucose cotransporter (SGLT1) stays intact.
- Antibiotics in E. coli O157 raise the risk of haemolytic uraemic syndrome.
- Osmotic diarrhoea stops with fasting; secretory continues.

Fast: osmotic stops, secretory continues.

Clinical picture
- Loose stools with recognisable food in a thriving 2-year-old = toddler's diarrhoea.
- Blood, high fever, severe pain suggest bacterial cause.

Peas and carrots in the nappy = toddler's.

Diagnosis
- Grade dehydration clinically.
- Faltering growth: coeliac screen with tTG and total IgA, sweat test, calprotectin.

tTG needs total IgA alongside.

Treatment
- ORS and continue feeding; IV only if ORS fails.
- Loperamide is not used; no routine antibiotics.

ORS first, IV if it fails.

I-28 Differential diagnosis of vomiting

Bilious vomiting means obstruction distal to the ampulla and is malrotation with volvulus until proven otherwise. Projectile vomiting in a hungry 2–8-week-old is pyloric stenosis: a hypochloraemic, hypokalaemic alkalosis that is corrected before pyloromyotomy.

Definition
- Green (bilious) vomit means obstruction distal to the ampulla of Vater.

Green means go: to theatre.

Causes
- Neonate: malrotation with volvulus, atresias, Hirschsprung, salt-wasting CAH.
- Infant: pyloric stenosis, intussusception; older: gastroenteritis, appendicitis, DKA, raised ICP.

Age sorts vomiting.

Patho-physiology
- Pyloric stenosis causes a hypochloraemic, hypokalaemic metabolic alkalosis.
- Urine is acidic despite alkalosis: paradoxical aciduria.

Pyloric: low Cl, low K, alkalosis.

Clinical picture
- Pyloric stenosis: 2-8 weeks, firstborn boy, projectile vomiting, hungry afterwards.
- Neonate with shock, hyponatraemia, hyperkalaemia = salt-wasting congenital adrenal hyperplasia.

Firstborn boy, 6 weeks, projectile.

Diagnosis
- Bilious vomiting: urgent exclusion of malrotation with volvulus.
- Pyloric stenosis imaging of choice: ultrasonography.

Bilious vomit = rule out volvulus now.

Treatment
- Correct fluids and electrolytes before surgery; alkalosis risks post-operative apnoea.
- Operation: Ramstedt pyloromyotomy; volvulus → emergency surgery.

Fix the fluids, then the pylorus.

I-29 Acute abdominal disorders in children

Appendicitis is the commonest acute abdomen, but the differential is sorted by age: NEC and volvulus in neonates, intussusception in toddlers, appendicitis at school age, torsion in adolescents. Always check the chest, urine, glucose and scrotum.

Definition
- The commonest cause of an acute abdomen in children is appendicitis.

Causes
- Sort by age: neonate NEC and volvulus; toddler intussusception; school age appendicitis.
- Adolescent: testicular or ovarian torsion, gynaecological causes; trauma: splenic rupture.

Neonate NEC, toddler telescopes, schoolkid appendix.

Patho-physiology
- Pain starts periumbilically: appendix is a midgut organ, afferents enter the cord at T10.
- Two-stage splenic rupture: subcapsular haematoma, shock hours to days later.

Appendix starts at the belly button (T10).

Clinical picture
- Neonate with bilious vomiting = malrotation with midgut volvulus until proven otherwise.
- Appendicitis: central pain moving to RLQ, anorexia, low fever.

Pain moves: umbilicus → RLQ.

Diagnosis
- Four free checks: chest, urine dipstick, glucose, scrotum in boys.
- Polytraumatised child: acute (whole-body) CT.

Chest, dip, glucose, scrotum.

Treatment
- Peritonism → surgical referral; resuscitate if unstable.
- Salvage of a torted testis falls sharply after about 6 hours.

Testis: 6 hours.

I-30 Differential diagnosis of palpable abdominal mass in childhood

With a childhood abdominal mass, first exclude benign causes (bladder, stool, hydronephrosis), then separate the well 3-year-old with a smooth flank mass that respects the midline (Wilms) from the unwell toddler whose mass crosses it (neuroblastoma). Image with ultrasound and do not keep palpating.

Definition
- Exclude benign causes, then separate Wilms tumour from neuroblastoma.

Wilms vs neuroblastoma.

Causes
- Neonate: most masses are renal (hydronephrosis, multicystic dysplastic kidney).
- Wilms: WAGR (aniridia), Beckwith-Wiedemann syndrome, hemihypertrophy; neuroblastoma: neural crest, N-myc.

WAGR: Wilms, Aniridia, GU anomalies, Retardation.

Patho-physiology
- Wilms arises from metanephric mesenchyme (blastema); it spreads mostly to the lungs.
- Do not palpate repeatedly: risk of rupture.

Wilms → lungs.

Clinical picture
- Wilms: peaks 3-4 years, well child, smooth mass, does not cross the midline, haematuria.
- Neuroblastoma: under 2 years, unwell, crosses midline, raccoon eyes, opsoclonus-myoclonus-ataxia.

Neuroblastoma crosses; Wilms stays.

Diagnosis
- Abdominal ultrasound first.
- Neuroblastoma: raised urinary VMA and HVA; small blue round cells, Homer-Wright rosettes.

VMA/HVA = neuroblastoma.

Treatment
- A suspected Wilms tumour must not be repeatedly palpated.
- Europe (SIOP): chemotherapy before surgery; US (COG): usually operate first.

SIOP: chemo first; COG: cut first.

I-31 Chronic abdominal pain

Chronic abdominal pain in a thriving child is usually functional and should be diagnosed positively, not by exclusion; hunt for red flags (pain far from the umbilicus, night waking, poor growth) and always treat constipation, the commonest organic cause.

Definition
- Most chronic abdominal pain in children is functional and should be diagnosed positively.

Functional = a positive diagnosis, not a dustbin.

Causes
- Functional: irritable bowel syndrome, functional abdominal pain.
- Organic: constipation is the commonest treatable cause; also coeliac, IBD, H. pylori, choledochal cyst, abuse.

Before anything fancy, think poo.

Patho-physiology
- The further the pain from the umbilicus, the more likely an organic cause.
- IBS is not related to structural abnormalities.

Far from the belly button = far from functional.

Clinical picture
- Functional abdominal pain never wakes the child at night.
- Key question in infancy: how the infant is growing/developing.

Wakes at night = look for a cause. Growing well = reassuring.

Diagnosis
- Best non-invasive test to separate IBD from functional pain: faecal calprotectin.
- H. pylori is urease-positive: urea breath test.

Calprotectin = inflammation in a cup; urease = breath test.

Treatment
- IBS first line: lifestyle modification, low FODMAP diet, fiber.
- Best evidence: cognitive behavioural therapy; treat constipation.

Diet first, brain best (CBT).

I-32 Differential diagnosis of oedema

Classify oedema by Starling mechanism (hydrostatic, oncotic, permeability, lymphatic); in a young child generalised periorbital morning oedema is nephrotic syndrome until proven otherwise, and a low albumin with a clean dipstick sends you to the gut or liver.

Definition
- Four Starling mechanisms: raised hydrostatic pressure, low oncotic pressure, increased capillary permeability, lymphatic obstruction.

Push up, pull down, leak, or block.

Causes
- Commonest cause of generalised oedema in a young child: nephrotic syndrome.
- Also kwashiorkor, protein-losing enteropathy, liver disease, hereditary angio-oedema.

Puffy toddler = check the urine.

Patho-physiology
- Kwashiorkor: oedema from decreased plasma oncotic pressure.
- Membranous nephropathy: GBM thickening, granular immune-complex deposits; amyloid deposits in the mesangium.

Kwashiorkor: no protein, no pull.

Clinical picture
- Oedema first appears periorbitally, in the morning.
- Recurrent non-pitting, non-itchy swelling without urticaria suggests hereditary angio-oedema.

Puffy eyes at breakfast; swelling without hives = HAE.

Diagnosis
- Nephrotic: UPCR >200 mg/mmol, low albumin.
- Low albumin with a negative urine dipstick points to the gut (protein-losing enteropathy) or liver.

Low albumin, clean dipstick: look at gut or liver.

Treatment
- Treat the mechanism.
- Nephrotic syndrome in children: mostly steroid-responsive; HAE is C1-esterase inhibitor deficiency.

Fix the force; minimal change melts with steroids.

I-33 Differential diagnosis of dehydration in childhood

Estimate the deficit (% × kg × 10), look for shock (hypotension is pre-terminal in children) and always check the sodium: doughy skin means hypernatraemia, which must be corrected no faster than 0.5 mmol/L/h to avoid cerebral oedema.

Definition
- Fluid deficit (ml) = % dehydration × weight (kg) × 10.

5% in a 10 kg child = 500 ml.

Causes
- Commonest cause: acute gastroenteritis; rotavirus is the commonest cause of severe diarrhoea.
- Cholera and Shigella (high fever, mucoid-bloody diarrhoea, seizures) also cause it.

Rotavirus = severe watery; Shigella = blood, fever, fits.

Patho-physiology
- Diarrhoea: ECF volume decreased, haematocrit and plasma protein increased.
- Rotavirus causes villous destruction and atrophy.

Lose water, thicken blood.

Clinical picture
- Doughy skin in a dehydrated infant suggests hypernatraemic dehydration.
- In children hypotension is a late, pre-terminal sign of shock.

Doughy = high sodium; hypotension = pre-terminal.

Diagnosis
- Check serum sodium; correct slowly, no faster than 0.5 mEq/L/hr.
- Hypernatraemia corrected rapidly causes cerebral oedema.

High sodium: slow down, max 0.5 per hour.

Treatment
- Without shock: ORS 50 ml/kg over 4 hours plus maintenance.
- Shock bolus: 10 ml/kg 0.9% sodium chloride; rotavirus vaccine at 2, 4, 6 months.

No shock: ORS 50/4h. Shock: 10 ml/kg saline.

I-34 Principles of fluid therapy in childhood Maintenance fluid therapy plan for a 3kg neonate and for a 20kg child.

Every fluid plan = resuscitation + deficit + maintenance + ongoing losses; use Holliday-Segar (100/50/20 ml/kg/day, i.e. 4/2/1 ml/kg/h) with isotonic fluid in children, but neonates start at 60 ml/kg/day of 10% glucose without sodium and build up over 5 days.

Definition
- Plan = resuscitation + deficit + maintenance + ongoing losses; standard maintenance is D5NS.

R-D-M-O: Resus, Deficit, Maintenance, Ongoing.

Causes
- Holliday-Segar maintenance: 100 ml/kg first 10 kg, 50 ml/kg next 10 kg, 20 ml/kg above.

100-50-20 per day; 4-2-1 per hour.

Patho-physiology
- Hypotonic saline alone can cause RBC lysis; use isotonic fluid beyond the neonatal period.
- Sodium is withheld from neonatal fluids until the postnatal diuresis begins.

Neonate: no salt till they pee it off.

Clinical picture
- Overload: oedema, hepatomegaly, crackles, weight gain.
- High ADH (meningitis, pneumonia, post-op, head injury): restrict to 50-80%.

High ADH illness = restrict to 50-80%.

Diagnosis
- Maintenance for a 20 kg child: 1500 ml/day (≈60 ml/h); hourly 4/2/1 rule.
- 3 kg neonate day 1: about 180 ml/day (7.5 ml/h).

20 kg = 1500/day = 60/h.

Treatment
- Term neonate day 1: 50-60 ml/kg/day of 10% glucose, rising to 120-150 ml/kg/day by day 5.
- Older child: 0.9% NaCl with 5% glucose; shock bolus 10 ml/kg.

Neonate starts at 60 and climbs to 150.

I-35 Disorders of sexual differentiation

Ambiguous genitalia is a medical emergency: think salt-wasting 21-hydroxylase CAH first (non-palpable gonads in a 'male' = CAH until proven otherwise), get a karyotype and 17-OHP, watch the sodium and potassium, and never assign sex without the MDT.

Definition
- Mismatch of chromosomal, gonadal or phenotypic sex; ambiguous genitalia is an emergency.

Ambiguous genitals = check the sodium today.

Causes
- Commonest cause of ambiguous genitalia in a newborn: 21-hydroxylase deficiency (CAH).
- 46,XY with female genitalia, testes and no uterus: complete androgen insensitivity.

CAH virilises girls; CAIS feminises boys.

Patho-physiology
- SRY makes the gonad a testis; AMH from Sertoli cells regresses the Müllerian ducts.
- Testosterone virilises; androgen insensitivity is an androgen receptor defect.

SRY builds, AMH erases, testosterone masculinises.

Clinical picture
- Salt-wasting CAH presents at 1-3 weeks with low sodium and high potassium.
- Androgen insensitivity: pubic hair absent; Müllerian agenesis: present.

CAH crisis at 1-3 weeks: low Na, high K.

Diagnosis
- Do karyotype; screening test for CAH: raised 17-hydroxyprogesterone.
- Non-palpable gonads in a 'male' infant = CAH until proven otherwise.

No testes felt = assume CAH girl.

Treatment
- Involve the MDT before assigning sex.
- Androgen insensitivity: bilateral gonadectomy after puberty; do not circumcise hypospadias.

Keep the foreskin for the surgeon.

I-36 Disorders of sexual development/maturation

Puberty is early before 8 (girls) / 9 (boys) and late if no breasts by 13 or testes by 14; decide central (gonadotrophin-driven, LH rises with GnRH stimulation, needs a brain MRI, especially in boys) versus peripheral (LH/FSH suppressed), and remember that small testes with virilisation means an adrenal source.

Definition
- Premature puberty: signs before 8 years in girls, 9 in boys.
- Delayed puberty: no breasts by 13, testes by 14.

Early: 8 girls/9 boys. Late: 13 breasts/14 testes.

Causes
- Central precocious puberty: girls mostly idiopathic; boys often organic (CNS lesion).
- Commonest delay: constitutional delay; Kallmann syndrome (anosmia); Turner and primary ovarian insufficiency.

Boys go early? Scan the brain.

Patho-physiology
- Central: leuprolide stimulation increases LH; peripheral: LH/FSH suppressed.
- Oestrogen closes the epiphyseal growth plate.

Central: LH rises. Peripheral: LH flat.

Clinical picture
- First sign: girls thelarche; boys testicular volume ≥4 ml.
- Boy with small testes suggests CAH/adrenal cause.

Big penis, small testes = adrenal.

Diagnosis
- Central precocious puberty needs MRI of the brain before treatment.
- Primary amenorrhea: pelvic ultrasound; Müllerian agenesis has normal secondary sex characteristics.

Central early puberty = MRI first.

Treatment
- Central precocious puberty: GnRH agonist depot (leuprolide).
- Treat the cause; premature adrenarche and newborn breast enlargement need reassurance.

Continuous GnRH = off switch.

I-37 Arrhythmias

SVT is the commonest symptomatic paediatric tachyarrhythmia: infants present with heart failure, not palpitations; treat stable SVT with vagal manoeuvres then rapid IV adenosine (never verapamil in infants) and unstable SVT with synchronised cardioversion; sudden syncope without prodrome is an arrhythmia until proven otherwise.

Definition
- Commonest symptomatic tachyarrhythmia in children: SVT; presents with abrupt onset tachycardia.

SVT switches on like a light.

Causes
- WPW: ventricular pre-excitation; long QT: Romano-Ward (AD), Jervell and Lange-Nielsen (AR, deafness).
- Congenital complete heart block: maternal anti-Ro/La antibodies; Brugada: AD.

Jervell = deaf; Ro/La = blocked baby heart.

Patho-physiology
- Long QT predisposes to torsades de pointes.
- Adenosine inhibits the AV node, causing transient heart block.

Adenosine = 6-second AV pause.

Clinical picture
- Infants with SVT (rate >220) present with heart failure, not palpitations.
- Sudden syncope without prodrome = arrhythmia.

Baby SVT = poor feeding, sweaty, failing.

Diagnosis
- WPW resting ECG: short PR, widened QRS, delta wave.
- Brugada: pseudo-RBBB with ST elevation in V1-V3.

WPW: short PR + delta. Brugada: V1-V3 ST up.

Treatment
- Stable SVT: vagal manoeuvres first, then rapid IV adenosine; verapamil contraindicated in infants.
- Unstable: synchronised cardioversion; long QT: beta-blockers; complete heart block: pacemaker.

Stable: vagal, adenosine. Unstable: shock it.

I-38 Obesity in childhood

Most childhood obesity is exogenous: these children are tall with advanced bone age, whereas a short, slowly growing obese child needs an endocrine work-up; look for insulin-resistance complications (acanthosis, PCOS, NAFLD), SCFE and OSA, and start with family-based lifestyle change.

Definition
- Excess weight judged by BMI centiles; mostly exogenous obesity.

Plot BMI on a centile chart, not adult cut-offs.

Causes
- Exogenous: child is tall with advanced bone age.
- Endocrine: child is short with slowing height velocity; commonest monogenic cause MC4R; Prader-Willi.

Tall and fat = food; short and fat = hormones.

Patho-physiology
- Prader-Willi: gene deletion, raised ghrelin.
- Insulin resistance underlies acanthosis nigricans, PCOS and fatty liver.

Prader-Willi = ghrelin up, never full.

Clinical picture
- Velvety dark skin in neck folds = acanthosis nigricans.
- Knee pain and limp in an obese teenager: SCFE; also obstructive sleep apnea.

Obese teen limping = SCFE until proven otherwise.

Diagnosis
- Genetic red flags: onset before 5 years, severe hyperphagia, developmental delay or dysmorphism.
- Metabolic syndrome: at least 3 of 5 criteria.

Under 5, starving, delayed = test genes.

Treatment
- Family-based lifestyle change first.
- GLP-1 receptor agonists (semaglutide) licensed for obesity from age 12.

Whole family changes, not just the child.

I-39 Causes of short stature

Short stature (< -2 SD) is pathological when height velocity is low and the child crosses centiles downwards; bone age separates familial (bone age = chronological) from constitutional delay (bone age delayed, normal final height), and every unexplained short girl needs a karyotype for Turner.

Definition
- Short stature = height below -2 SD.

Below -2 SD = short.

Causes
- Normal variants: familial, constitutional delay.
- Most frequent endocrine cause: hypothyroidism; also growth hormone deficiency, Turner, Prader-Willi, Fanconi anaemia.

Normal variants first; thyroid tops endocrine.

Patho-physiology
- Growth hormone and thyroid hormone increase bone growth; GH raises IGF.
- Pathological short stature = low height velocity, crossing centiles downward.

Crossing centiles down = pathology.

Clinical picture
- Familial: bone age equal to chronological age.
- Constitutional delay: bone age delayed, final height normal.

Familial: bone age = age. Constitutional: bone age behind.

Diagnosis
- Every unexplained short girl needs a karyotype to exclude Turner syndrome.
- GH deficiency needs two GH stimulation tests.

Short girl = karyotype.

Treatment
- Treat the cause: levothyroxine for hypothyroidism.
- Growth hormone for deficiency, Turner and Prader-Willi; constitutional delay: reassurance.

Thyroid gets thyroxine; constitutional gets reassurance.

I-40 Muscle weakness, paralysis and other myopathies

Localise weakness first (anterior horn cell, nerve, neuromuscular junction, muscle): tongue fasciculations and areflexia = SMA, constipation then descending paralysis = infant botulism, ascending areflexic weakness with albuminocytological dissociation = GBS, Gowers sign with huge CK = Duchenne, fatigable ptosis = myasthenia.

Definition
- Localise the weakness: horn cell, nerve, neuromuscular junction or muscle.

Horn, nerve, junction, muscle.

Causes
- SMA: deletion of SMN1; Duchenne: X-linked absence of dystrophin.
- Myasthenia: antibodies to postsynaptic ACh receptors; infant botulism: honey spores; GBS.

No honey under 1.

Patho-physiology
- Myasthenia: weakness worsens with muscle use; Lambert-Eaton (presynaptic calcium channels) improves.
- SMA severity depends on SMN2 copy number.

Myasthenia tires; Lambert-Eaton warms up.

Clinical picture
- Tongue fasciculations and areflexia in a floppy infant: SMA; infant botulism begins with constipation.
- Duchenne: Gowers sign; myasthenia: fatigable ptosis and diplopia.

Floppy + tongue fasciculations = SMA.

Diagnosis
- GBS: CSF high protein with normal cell count; serial spirometry.
- Duchenne: serum CK raised 10-100x; myotonic dystrophy: EMG then genetic testing.

GBS: protein up, cells normal.

Treatment
- GBS: IVIG or plasmapheresis; infant botulism: human-derived botulism immune globulin.
- Duchenne: steroids; myasthenic crisis: intubation, then prednisone plus IVIG.

Botulism baby = BabyBIG.

I-41 Disorders of coagulation

Pattern of bleeding localises the defect: mucocutaneous bleeding = platelets/VWF, joint and muscle bleeds = clotting factors (isolated prolonged APTT = haemophilia); in a haemophiliac any head injury means factor concentrate first, and ITP with only skin bleeding is watched, not treated.

Definition
- Mucocutaneous bleeding suggests platelets/VWF; joint and muscle bleeds suggest clotting factors.

Skin/mucosa = platelets; joints = factors.

Causes
- Haemophilia A = factor VIII; haemophilia B = factor IX.
- Commonest inherited bleeding disorder: von Willebrand disease; ITP: IgG against GPIIb/IIIa.

A = eight, B = nine.

Patho-physiology
- Desmopressin increases release of vWF from Weibel-Palade bodies.
- ITP antibodies are produced in the spleen and can cross the placenta.

DDAVP opens the vWF storehouse.

Clinical picture
- ITP: scattered petechiae, isolated thrombocytopenia, increased bleeding time.
- Severe haemophilia: spontaneous joint bleeds.

Well child + petechiae + low platelets = ITP.

Diagnosis
- Haemophilia screen: APTT prolonged, PT normal.
- Chronic ITP: low platelets for >1 year.

Haemophilia: APTT long, PT normal.

Treatment
- Haemophilia with head trauma, even without fainting: immediate factor concentrate and nearest haemophilia centre.
- Inhibitors: bypassing agent; prophylaxis emicizumab; ITP with only petechiae: observation.

Head knock in haemophilia = factor now.

I-42 Behavioural disorders (ADHD, autism spectrum disorder)

ADHD needs symptoms in two or more settings, onset before 12 and duration of at least 6 months; exclude hearing, vision, absences and OSA; methylphenidate is first line from age 5 with growth and cardiovascular monitoring. Autism is social-communication deficit plus restricted, repetitive behaviour: screen with M-CHAT-R/F, always test hearing, and treat with early intensive therapy.

Definition
- ADHD: hyperactivity, impulsivity, and/or inattention in ≥ 2 settings.
- Autism spectrum disorder: poor social interactions, repetitive / ritualized behaviors.

ADHD: 2+ settings. Autism: social + repetitive.

Causes
- ADHD: ~70-80% heritable, prematurity, fetal alcohol/tobacco.
- Autism: genetic; fragile X, Rett, fetal valproate; not MMR; more common in males.

MMR does not cause autism.

Patho-physiology
- ADHD: weak dopamine and noradrenaline signalling → impulsivity.
- Autism: altered connectivity → social cues not rewarding → routines calm.

Stimulants boost dopamine.

Clinical picture
- ADHD: onset before 12, in two or more settings, for at least 6 months.
- Autism: early childhood onset; poor eye contact, insistence on sameness, stereotypies.

ADHD: 12, 2, 6.

Diagnosis
- ADHD: clinical; exclude hearing, vision, absences, OSA.
- Autism screening at 18 months and 2 years: M-CHAT-R/F (16-30 months).
- Hearing test before diagnosing autism in non-verbal child; ADOS.

Can't talk? Test hearing first.

Treatment
- ADHD from age 5: stimulants ± CBT; methylphenidate first-line, then atomoxetine, guanfacine.
- Monitor heart rate, blood pressure, height and weight on stimulants.
- Autism: early intensive developmental therapy; risperidone for severe irritability.

Methylphenidate first; watch heart and growth.

I-43 Acid-base disorders

Read a gas in order: pH, then SMORE (pH and pCO2 moving in the Same direction = Metabolic, Opposite = RESpiratory), then the anion gap (Na - [Cl + HCO3], normal 8-12) and Winter's formula for compensation; pyloric stenosis is the classic hypochloraemic hypokalaemic alkalosis, fixed with chloride and volume before surgery.

Definition
- Disturbed pH; follows the HCO3: pCO2 ratio.
- SMORE: Same direction (pH and pCO2) → Metabolic; Opposite → RESpiratory.

SMORE: Same Metabolic, Opposite REspiratory.

Causes
- High anion gap = acid added: DKA, lactate, renal failure, toxins.
- Normal anion gap: diarrhoea, RTA (HARDASS); alkalosis: vomiting, diuretics.
- Pyloric stenosis causes hypochloraemic, hypokalaemic metabolic alkalosis.

High gap = acid in; normal gap = bicarb out.

Patho-physiology
- Metabolic acidosis: decrease in HCO3 → hyperventilation → decreased pCO2.
- Respiratory compensation is immediate; kidneys slow (3-5 days).
- Pyloric stenosis: HCl loss + aldosterone → paradoxical aciduria.

Lungs fast, kidneys slow.

Clinical picture
- Kussmaul breathing, vomiting, lethargy.
- A normal or rising pCO2 in acute severe asthma signals exhaustion.

Asthma + normal pCO2 = tiring out.

Diagnosis
- Anion gap = Na - (Cl + HCO3), normal about 8-12.
- Winter's: expected pCO2 = 1.5 × HCO3 + 8 (±2).
- Metabolic alkalosis: next step check urine chloride.

Winter's: 1.5 × HCO3 + 8 ± 2.

Treatment
- Treat the cause: fluids, insulin for DKA, dialysis; bicarbonate not routine.
- Pyloric: chloride + volume first.
- Operate once Cl^- and HCO3 normal.

Fix the salts before the knife.

I-44 Hypo- and hypernatremia

Sodium disorders are water disorders of the brain: low Na swells brain cells (headache, vomiting, seizures; give 3% saline for seizures), high Na shrinks them (doughy skin, high-pitched cry, bridging-vein bleeds); correct chronic changes slowly (≤8-10 mmol/L/24 h up, ≤0.5 mmol/L/h down) and use isotonic maintenance fluids.

Definition
- Abnormal plasma sodium → water shifts into (low) or out of (high) brain cells.

Low Na swells, high Na shrinks.

Causes
- Low: SIADH + hypotonic fluids, diarrhoea and vomiting, salt-wasting CAH, nephrotic, heart failure.
- High: gastroenteritis, poor breastfeeding in week 1, diabetes insipidus, bad formula.

Breastfed baby losing weight in week 1 = high Na.

Patho-physiology
- In hyponatraemia water moves into brain cells, causing cerebral oedema.
- High Na^+ → brain shrinks → bridging veins tear → haemorrhage.

Shrinking brain tears bridging veins.

Clinical picture
- Low: headache, vomiting, seizures; neonate low Na^+ + high K^+ → CAH.
- High: high-pitched cry, doughy skin, seizures; shock late.

Low Na + high K neonate = CAH.

Diagnosis
- Plasma and urine osmolality, urine Na, volume status.
- SIADH: low plasma osmolality, inappropriately concentrated urine, euvolaemia, urine Na >30.

SIADH: dilute blood, concentrated urine, euvolaemic.

Treatment
- Hyponatraemic seizures: bolus of 3% saline; SIADH → fluid restriction; isotonic maintenance fluids.
- Too-rapid correction of chronic hyponatraemia causes osmotic demyelination; raise ≤ 8-10 mmol/L/24 h.
- Hypernatraemia: corrected over 48 hours, no more than 0.5 mmol/L per hour; 0.9% saline for shock.

Low Na: max 8-10/day. High Na: 48 h.

I-45 Hypo- and hyperkalaemia

High potassium in a well infant is a haemolysed heel-prick until a free-flowing venous sample proves otherwise; real hyperkalaemia with ECG changes (peaked T → wide QRS → sine wave) needs calcium gluconate first to protect the heart, then shift (salbutamol, insulin-glucose) and remove. Low potassium: replace magnesium and never bolus KCl.

Definition
- Abnormal K^+; K^+ sets the resting membrane potential.

K sets the resting potential.

Causes
- High: AKI, CAH, DKA, tumour lysis, ACE inhibitors, heparin, trimethoprim.
- Commonest cause of high K in a well infant: a haemolysed sample.
- Low: diarrhoea and vomiting, pyloric stenosis, diuretics, insulin in DKA, refeeding.

Well baby with high K? Squeezed heel.

Patho-physiology
- High K^+ → inactivates fast Na^+ channels → slow conduction, fast repolarisation.
- Salbutamol and insulin lower K by stimulating the Na/K-ATPase.
- Low K may not correct until magnesium is replaced.

No Mg, no K fix.

Clinical picture
- Earliest ECG sign of hyperkalaemia: peaked T waves → wide QRS → sine wave.
- Low: weakness, ileus, flat T waves, U waves.

Peaked T first; low K gives U waves.

Diagnosis
- First exclude haemolysed heel-prick; repeat free-flowing venous sample.
- Low K: urine K (>20 = renal loss), acid-base.

Repeat before you treat (if well).

Treatment
- High with ECG changes: IV calcium gluconate 10% first (protect).
- Shift: salbutamol, insulin-glucose, sodium bicarbonate; then resin, loop diuretic, dialysis.
- Low: oral KCl; IV in fluids, never bolus; replace Mg.

Protect, shift, remove.

I-46 Causes of polyhydramnios and oligohydramnios

Amniotic fluid after 20 weeks is fetal urine in, fetal swallowing out: too much means the baby cannot swallow (atresia) or pees too much (maternal diabetes, TTTS recipient); too little means no urine, blocked urine or leak - and early oligohydramnios kills through pulmonary hypoplasia.

Definition
- Abnormal amniotic fluid: poly AFI >24-25 cm; oligo AFI under 5 cm.

Poly over 24, oligo under 5.

Causes
- Poly: oesophageal and duodenal atresia, maternal diabetes (fetal osmotic diuresis), TTTS recipient.
- Oligo: no urine (renal agenesis, ARPKD), blocked (PUV), leak (PPROM, commonest), FGR.
- ACE inhibitors / ARBs in utero may cause oligohydramnios.

Oligo: no pee, blocked pee, leaking bag. ACEi/ARB shut fetal kidneys.

Patho-physiology
- After 20 weeks fluid comes mainly from fetal urine, removed by fetal swallowing.
- Early oligohydramnios → no lung distension → lethal pulmonary hypoplasia.

Fetus pees in, drinks out. No fluid, no lungs.

Clinical picture
- Poly: preterm labour, cord prolapse, placental abruption after rapid decompression.
- Oligo: Potter sequence (pulmonary hypoplasia, twisted facies, extremity deformities, renal agenesis).
- Also growth restriction, cord compression.

Potter: Pulmonary hypoplasia, Odd face, Twisted limbs, Renal agenesis.

Diagnosis
- Ultrasound AFI; double bubble (duodenal atresia), keyhole bladder (PUV).
- Oligo before ~24 weeks → most severe lung hypoplasia.

Double bubble = duodenum; keyhole = valves.

Treatment
- Polyhydramnios newborn: NG tube before first feed to exclude oesophageal atresia.
- Severe polyhydramnios: amnioreduction or indomethacin; term, mild, asymptomatic: expectant.
- Oligohydramnios: treat cause; pulmonary hypoplasia is the killer.

Poly baby: tube before food.

I-47 Neonatal and infant resuscitation

Newborn arrest is respiratory: dry, warm, open the airway and aerate the lungs with 5 long inflation breaths - the heart rate is your monitor, and compressions (3:1) only start if HR stays under 60 after 30 s of effective ventilation.

Definition
- Newborn arrest is respiratory - aerate the lungs.
- Best indicator of success: the heart rate.

Babies arrest from lungs, not hearts: aerate first.

Causes
- Hypoxia → primary apnoea → gasping → terminal apnoea.
- Meconium aspiration: term and post-term infants.

Apnoea → gasp → final apnoea.

Patho-physiology
- First breaths clear lung fluid, create FRC → pulmonary vascular resistance falls.
- Meconium inactivates surfactant; bradycardia = hypoxia until proven otherwise.

Slow heart in a baby = no oxygen until proven otherwise.

Clinical picture
- Assess tone, breathing, heart rate; HR under 100 or not breathing → act.
- Apgar at 1 and 5 minutes: Appearance, Pulse, Grimace, Activity, Respiration.

APGAR: Appearance, Pulse, Grimace, Activity, Respiration.

Diagnosis
- Stethoscope, then ECG/pre-ductal SpO_2.
- Targets: 70-75% at 3 min, 80-85% at 5 min, 85-95% at 10 min.
- PPV with brighter left chest transillumination → left pneumothorax.

Sats climb slowly: ~70 at 3, ~80 at 5, ~90 at 10.

Treatment
- Dry, warm; 5 inflation breaths, each 2-3 seconds; term babies start in air (21%).
- Compressions 3:1 if HR under 60/min after 30 s effective ventilation.
- Adrenaline 20 µg/kg via UVC; HIE: 33-34 °C for 72 h, within 6 h.

5 long breaths in air; 3 compressions : 1 breath.

I-48 Pediatric resuscitation

Children rarely arrest from the heart: arrest is the end of progressive hypoxia or shock, so recognise respiratory failure and shock early with ABCDE - and once arrested, the rhythm is usually non-shockable (asystole/PEA) and needs CPR 15:2 plus adrenaline 10 µg/kg every 3-5 min.

Definition
- Paediatric arrest = end of progressive hypoxia or shock.
- Key to survival: elapsed time to effective resuscitation.

Kids' hearts stop last - fix air and circulation early.

Causes
- Respiratory failure or shock → tiring → hypoxia → bradycardia → asystole.
- Non-shockable rhythms (asystole, PEA) are commonest.

Hypoxia → brady → asystole: kids rarely need the defib.

Patho-physiology
- Paradoxical breathing: chest moves inward, abdomen outward during inhalation.
- PEA: organized rhythm without a palpable pulse.

Seesaw breathing: chest in, belly out.

Clinical picture
- ABCDE: stridor, recession, grunting, head bobbing, CRT, AVPU, glucose.

DEFG: Don't Ever Forget Glucose.

Diagnosis
- Pulse oximetry not informative in carbon monoxide intoxication, methaemoglobinaemia, severe anaemia, cardiac failure, cold extremities.
- SpO_2 ignores ventilation → capnography or blood gas.

Sats tell you O_2, not CO_2 - ventilation needs capnography.

Treatment
- Conscious, breathing: face mask with reservoir, high flow 10-15 l/min.
- BLS: 5 rescue breaths, then 15:2; asystole/PEA: CPR + adrenaline 10 µg/kg IV/IO every 3-5 min.
- VF / pulseless VT: unsynchronized shock 4 J/kg; amiodarone after 3rd and 5th shock.

5 breaths, then 15:2; shock 4 J/kg; amio after shocks 3 and 5.

I-49 Fever, management of fever, antipyretics, pain relief

Fever is a regulated rise in the hypothalamic set point (cytokines → PGE_2), not a disease: treat the distressed child rather than the number, and use the NICE traffic light to find the few febrile children - especially under 3 months - who are septic.

Definition
- Fever: cytokine-driven rise in hypothalamic set point.

Fever = thermostat turned up, not a broken boiler.

Causes
- Microbial products → IL-1, IL-6, TNF-α release.
- Early-onset neonatal sepsis occurs within 72 hours of birth.

Pyrogen trio: IL-1, IL-6, TNF.

Patho-physiology
- Cytokines induce PGE_2 in the hypothalamus, raising the set point → shivering, vasoconstriction.
- Antipyretics block COX; fever itself not harmful.

Block COX → no PGE_2 → thermostat resets.

Clinical picture
- NICE red: child under 3 months with temperature ≥ 38 °C, mottled, grunting, neck stiffness.
- Amber: 3-6 months with ≥ 39 °C, fever ≥ 5 days, CRT ≥ 3 s.

Under 3 months + 38 = red; 3-6 months + 39 = amber.

Diagnosis
- NICE traffic light for under-5s; red → septic work-up.
- Neonatal sepsis: blood culture, serial CRP, lumbar puncture; antibiotics within 1 hour.
- Pain scales: FLACC (preverbal), Wong-Baker (4+), numeric (8+).

FLACC for babies, faces from 4, numbers from 8.

Treatment
- Only if distressed: paracetamol 15 mg/kg every 4-6 h, ibuprofen 10 mg/kg three times daily.
- Antipyretics do not prevent febrile seizures; aspirin avoided under 16 (Reye syndrome).
- Neonatal sepsis: ampicillin + gentamicin; codeine contraindicated under 12 (CYP2D6).

Paracetamol 15, ibuprofen 10 (mg/kg).

I-50 Causes of shock in childhood

Children hold their blood pressure until 30-40% of volume is lost, so tachycardia and CRT >2 s are your early warning and hypotension means you are late - give 20 ml/kg boluses with reassessment for fluid overload after each, and remember the obstructive/duct-dependent neonate.

Definition
- Inadequate oxygen delivery; types hypovolaemic, distributive, cardiogenic, obstructive, dissociative.

Shock types: Hypo, Distributive, Cardiogenic, Obstructive, Dissociative.

Causes
- Gastroenteritis, haemorrhage, sepsis, anaphylaxis, myocarditis, SVT, tamponade.
- Neonate 1-2 weeks with weak femoral pulses: duct-dependent lesion.

Grey baby, 1-2 weeks, no femorals = duct closing.

Patho-physiology
- Children compensate with tachycardia + vasoconstriction until 30-40% volume lost.
- Hypotension is a late sign of shock; tachycardia is early.

Tachy early, hypotension late.

Clinical picture
- Tachycardia (earliest), CRT >2 s, cool mottled peripheries, oliguria, lactic acidosis.
- Bradycardia + hypotension = decompensation / pre-arrest.

Slow + low = about to go.

Diagnosis
- Cultures, gas, lactate, glucose.
- Hypotension in children 1-10 y: systolic below 70 + 2 × age mmHg.
- Reassess HR, CRT, BP, liver edge, crackles after each bolus.

Minimum systolic = 70 + 2 × age.

Treatment
- Septic shock: O2, IV/IO, fluid bolus 20 ml/kg body weight; antibiotics within 1 hour.
- Fluid-refractory: start adrenaline or noradrenaline; TSS: clindamycin suppresses toxin synthesis.
- Anaphylaxis: IM adrenaline first (150 µg under 6); duct-dependent: prostaglandin E.

Sepsis: 20 ml/kg + antibiotics in 1 hour.

I-51 Neonatal birth injuries

Birth injuries map to anatomy: scalp swellings are sorted by layer (caput crosses sutures, cephalhaematoma stops at them, subgaleal can bleed a baby into shock), and plexus palsies by root (Erb C5-6 waiter's tip, Klumpke C8-T1 claw ± Horner).

Definition
- Injuries from delivery, determined by anatomical layer and nerve root.

Layer for scalp, root for arm.

Causes
- Risk factors: difficult deliveries, shoulder dystocia, macrosomia, vacuum, forceps.
- Maternal hyperglycemia → macrosomia.

Sugar mum → big baby → stuck shoulders.

Patho-physiology
- Caput succedaneum crosses suture lines; cephalhaematoma is subperiosteal and does not.
- Subgaleal haemorrhage: injury to emissary veins, between aponeurosis and periosteum.
- Erb: lateral traction on neck during delivery.

Caput crosses, cephal can't; subgaleal swallows blood.

Clinical picture
- Cephalhaematoma: swelling limited to one cranial bone; resorption can cause jaundice.
- Subgaleal haemorrhage: hypovolaemic shock, typically after vacuum delivery.
- Erb C5-C6 waiter's tip; Klumpke C8-T1 claw hand ± Horner; clavicle: crepitus, asymmetric Moro.

Erb = waiter's tip (C5-6); Klumpke = claw (C8-T1).

Diagnosis
- Subgaleal: monitor head circumference, haemoglobin.
- Erb palsy with tachypnoea suggests phrenic nerve palsy (C3-C5).
- Clavicle fracture: check subclavian artery, brachial plexus.

C3-4-5 keeps the diaphragm alive.

Treatment
- Shoulder dystocia: McRoberts (flex hips) + suprapubic pressure.
- Commonest fracture: clavicle, heals rapidly; plexus palsy resolves within 3 months (80%).
- No biceps recovery by 3-6 months → surgery; subgaleal: volume and blood replacement.

Biceps silent at 3-6 months → surgeon.

I-52 Poisoning in childhood (toxicology)

Poisoning is accidental in toddlers and deliberate in teenagers: stabilise ABCDE and glucose, calculate mg/kg assuming the worst, and know which ingestions bypass charcoal (iron, lithium, alcohols) and which have antidotes (NAC, desferrioxamine, bicarbonate, naloxone).

Definition
- Ingestion: accidental in toddlers (1-4 y), deliberate in adolescents.

Toddler = accident; teen = intent.

Causes
- Medicines, cleaners, button batteries, paracetamol, iron, TCAs, opioids, caustics.
- Lead poisoning: homes built before 1978.

Lead paint before '78.

Patho-physiology
- Paracetamol: toxic metabolite NAPQI, detoxified by glutathione until exhausted.
- Iron: corrosive, then mitochondrial poison; TCA: Na^+ channel block → wide QRS.

NAPQI eats glutathione; TCA widens QRS.

Clinical picture
- Toxidromes: anticholinergic, cholinergic, opioid (pinpoint pupils), sympathomimetic.
- Paracetamol day 1 near asymptomatic; lead: bluish gum-tooth line.
- Aspirin: early pure respiratory alkalosis; late mixed with metabolic acidosis.

Pinpoint pupils = opioid; aspirin: alkalosis then acidosis.

Diagnosis
- ABCDE + glucose; dose in mg/kg, assume worst case.
- Paracetamol level on nomogram from 4 hours; iron X-ray + serum iron; battery CXR halo.
- Lead: capillary level raised → measure venous lead level.

Paracetamol level at 4 h, plot on nomogram.

Treatment
- Never induce vomiting; charcoal only within 1 hour, does not bind iron, lithium, alcohols.
- Antidotes: N-acetylcysteine (paracetamol); desferrioxamine (iron); bicarbonate (TCA); naloxone (opioid).
- Oesophageal battery or coin: urgent endoscopy; lead moderate DMSA, severe dimercaprol + EDTA.

Charcoal won't bind: Iron, Lithium, Alcohols (ILA).

I-53 Child abuse and shaken baby syndrome

Suspect abuse when the story does not fit the child's development or keeps changing: any bruise in a non-mobile baby, posterior rib or metaphyseal corner fractures, or the triad of subdural + retinal haemorrhages + encephalopathy - document, keep the child safe and involve child protection; welfare overrides confidentiality.

Definition
- Physical, sexual, emotional abuse or neglect; neglect is the commonest form of maltreatment.

Neglect: the commonest and quietest.

Causes
- Most significant parental risk factor: being abused themselves; abuser usually primary caregiver.
- Shaken baby: infants under 1 y, peak 2-4 months; sexual abuse peak 9-12 years.

Crying peak 2-4 months = shaking peak.

Patho-physiology
- Shaking tears the bridging veins, causing subdural haemorrhage.
- Vitreous pulls retina → retinal haemorrhages; grip → posterior rib fractures.

Shake → bridging veins tear; grip → back ribs crack.

Clinical picture
- History: delay, changing story, injury not fitting development.
- Any bruise in a non-mobile infant; scalds with sharp lines of demarcation.
- Abusive head trauma: subdural haemorrhage, retinal haemorrhages and encephalopathy.

Those who don't cruise rarely bruise.

Diagnosis
- CT then MRI, skeletal survey repeated at 11-14 days, ophthalmology, coagulation.
- Most specific: posterior rib and classic metaphyseal (corner) fractures.

Corner + back rib = abuse.

Treatment
- Suspected abuse: contact child protective services first; abuse and neglect both reported.
- Document exactly, keep child safe - admit if needed; welfare overrides confidentiality.

Report, record, keep safe.

I-54 Airway foreign bodies in children, aspiration

A toddler with sudden choking then a symptom-free interval and unilateral wheeze has an inhaled foreign body (often a radiolucent nut, usually right-sided) until rigid bronchoscopy proves otherwise - a normal X-ray never excludes it.

Definition
- Inhaled object in larynx, trachea or bronchus.

Causes
- Toddlers 1-3 y: no molars, eat while running; nuts classic.
- Right lung commonest: its bronchus is wider and more vertical.

Right bronchus: wider, straighter, steeper.

Patho-physiology
- Partial block = check valve → air trapped on expiration.
- Complete block → atelectasis; nuts radiolucent.

Partial = trap air; complete = collapse.

Clinical picture
- Choking → symptom-free interval → cough, unilateral wheeze, recurrent pneumonia.
- Laryngotracheal: stridor, cyanosis.

Choke, calm, then cough/wheeze/pneumonia.

Diagnosis
- Holzknecht sign = pendulum movement of the mediastinum = airway foreign body.
- Expiratory and inspiratory chest radiograph or fluoroscopy; normal X-ray never excludes.

Holzknecht = mediastinum swinging like a pendulum.

Treatment
- Choking infant under 1 year: 5 back blows then 5 chest thrusts (not abdominal).
- Over 1 y: abdominal thrusts.
- Definitive: rigid bronchoscopy; symptomatic swallowed coin: emergency endoscopy.

Under 1: back + chest. Over 1: back + tummy.

I-55 Pediatric imaging techniques

In children the imaging ladder is ALARA: justify, optimise, and go non-ionising first - ultrasound, then MRI - keeping CT for polytrauma, significant head injury and emergencies; know the paediatric specifics like the thymus shadow and cranial US through the fontanelle.

Definition
- Imaging chosen by ALARA (as low as reasonably achievable): justification, optimisation, non-ionising first.
  ALARA: As Low As Reasonably Achievable.

Causes
- Children radiosensitive: dividing cells, longer remaining life, organs close to beam.
  Small bodies, long futures.

Patho-physiology
- Blood echogenic, CSF anechoic → cranial US via anterior fontanelle.
- Preterm IVH arises in the germinal matrix.
  Fontanelle = window; germinal matrix = preterm bleed site.

Clinical picture
- Thymus shadow visible till 3 years of age; absent → DiGeorge or SCID.
- Mediastinal shadow narrowing trachea → mass, not thymus.
  Sail sign until 3; no sail = DiGeorge/SCID.

Diagnosis
- Ultrasound first, MRI next; cranial US: brain haemorrhage and hydrocephalus.
- Urinary: US, MCUG, DMSA 4-6 months after infection shows scars, MAG3.
  DMSA = scars, wait 4-6 months.

Treatment
- Polytrauma → acute CT; head trauma: non-contrast CT if GCS ≤ 14 or skull fracture.
- Suspected stroke with normal CT → MRI / MR angiography.
- US for pyloric stenosis, intussusception; upper GI contrast for bilious vomiting.
  Green vomit → upper GI contrast for malrotation.

Section II:

Prevention, infection, neonatology & oncology

(55 topics)

II-1 Standard and recommended immunizations for children and adolescents

Vaccinate on schedule and do not invent contraindications: the only universal one is anaphylaxis to a previous dose or component; live vaccines are avoided in severe immunodeficiency and pregnancy, MMR waits until after 12 months for maternal antibodies to wane, and preterms go by chronological age.

Definition
- Vaccines prime memory B and T cells → fast high-affinity IgG on real infection.
  Rehearsal now, fast response later.

Causes
- Live: MMR, varicella, rotavirus, BCG; avoid in SCID, pregnancy.
- Hungary: BCG to all newborns; varicella mandatory since 2019.
  Live: MMR, Varicella, Rotavirus, BCG - not in SCID or pregnancy.

Patho-physiology
- Maternal antibodies neutralise MMR earlier → given after 12 months.
- Conjugate vaccines work in infants because the protein carrier recruits T-helper cells.
  Conjugate = protein buys T-cell help.

Clinical picture
- MMR fever and rash appear 7-12 days after vaccination.
- Varicella vaccine rash with few lesions = vaccine-strain replication.
  MMR rash a week later, not a day.

Diagnosis
- Only universal contraindication: anaphylaxis to a previous dose or component.
- DTaP: anaphylaxis → no future doses; encephalopathy → hold pertussis.
- Not contraindications: mild illness; preterms use chronological age.
  Only anaphylaxis stops all; preterms go by birthday.

Treatment
- At 2, 4, 6 months: DTaP, IPV, PCV15, rotavirus; hepatitis B at birth, 2, 6 months.
- MMR and varicella at 1 year and 4 years; annual influenza from 6 months.
- Age 11: Tdap, meningococcal, HPV through 26.
  2-4-6 infant, 1 and 4 MMR, 11 teen jabs.

II-2 Normal psychomotor development in childhood (different age groups). Child growth percentiles. Anthropometry.

Development runs cephalocaudal and proximal-to-distal across four domains - judge the trajectory, not a single point: loss of skills, early hand preference, persisting primitive reflexes, not walking by 18 months, or crossing two centile lines are the red flags.

Definition
- Development in 4 domains: gross motor, fine motor/vision, speech/hearing, social.
  4 domains: Gross, Fine/vision, Speech/hearing, Social.

Causes
- Any loss of skills is an urgent red flag.
- Hand preference before 12 months is a red flag.
  Early hand preference = the other hand is weak.

Patho-physiology
- Myelination → cephalocaudal and proximal→distal progression.
- Primitive reflexes normal till 6-12 months; persistence suggests upper motor neuron lesion (e.g. CP).
  Head to toe, core to fingers.

Clinical picture
- Social smile 6 weeks; lifts head 2 months; rolls over 6 months.
- Sits alone, stranger anxiety 9 months; pulls to stand 9-12 months.
- Walks 12-18 months (limit 18 months); kicks ball 2 years.
  Smile 6 wk, roll 6 mo, sit 9 mo, walk 12-18 mo.

Diagnosis
- Weight, length, head circumference on centile charts; judge trajectory.
- Growth failure: weight below 5th percentile or crossing two major lines.
- Failure to thrive: weight → height → head circumference.
  FTT order: Weight → Height → Head.

Treatment
- Birth weight regained by 2 weeks, doubles by 4-5 months, triples by 12 months.
- Length rises 50% by 1 year; growth velocity is the most sensitive marker.
  Weight: back at 2 wk, ×2 at 4-5 mo, ×3 at 1 yr.

II-3 Anthropometry of a newborn. Intrauterin growth restriction (IUGR). Characteristics of a premature baby.

Classify every newborn by weight (LBW <2500 g, VLBW <1500, ELBW <1000) and by weight for gestation (SGA <10th centile): asymmetric late placental FGR spares the head and brings hypoglycaemia, hypothermia and polycythaemia, while prematurity brings RDS, IVH, PDA, NEC and ROP.

Definition
- Term birth weight 2500-4000 g, head circumference ~35 cm; preterm under 37 weeks.
- SGA = birth weight below the 10th centile for gestation.
  Term: 2.5-4 kg, head 35 cm; SGA below 10th.

Causes
- Leading cause of low birth weight in developed countries: smoking.
- Asymmetric FGR usually placental insufficiency; fetal: trisomies, TORCH.
  Smoking: top preventable LBW cause.

Patho-physiology
- Asymmetric (late, placental): head spared; symmetric (1st trimester infection): global growth lag.
- Chronic hypoxia → EPO → polycythaemia (haematocrit over 65%) → hyperviscosity, jaundice.
  Brain-sparing: big head, thin body.

Clinical picture
- IUGR: hypoglycaemia, hypothermia, polycythaemia.
- Preterm: thin red skin, lanugo, soft ears, hypotonia.
- RDS (surfactant deficiency), IVH, PDA, NEC, ROP.
  Preterm problems: RDS, IVH, PDA, NEC, ROP.

Diagnosis
- Low birth weight under 2500 g; VLBW under 1500; ELBW under 1000 g.
- Gestation by dating ultrasound, New Ballard score.
- FGR: Doppler of umbilical artery; absent/reversed end-diastolic flow → urgent delivery.
  LBW 2500, VLBW 1500, ELBW 1000.

Treatment
- Before birth: maternal glucocorticoids (24-34 wk) + MgSO4; after birth: artificial surfactant.
- NEC: NPO, NG tube, IV fluids, antibiotics; surgery for perforation.
- ROP: laser photocoagulation.
  Steroids 24-34 wk; laser the ROP.

II-4 Nutrition of the healthy infant: breastfeeding and weaning. Breastmilk composition. Energy requirements.

Breast is best and works on supply-and-demand (prolactin makes, oxytocin lets down): exclusive for 6 months, solids around 6 months (never before 17 weeks), no cow's milk or honey before 12 months, vitamin D and K for all - and very few maternal conditions truly contraindicate breastfeeding.

Definition
- Breastfeed on demand, exclusively for 6 months, continue to 2 years.
  On demand, 6 months only, 2 years total.

Causes
- Contraindications: infant galactosaemia; maternal HIV, untreated tuberculosis, varicella.
- Hepatitis B and mastitis are not.
  Galactosaemia, HIV, TB, varicella: no breast. Hep B, mastitis: fine.

Patho-physiology
- Suckling → prolactin (synthesis) + oxytocin (let-down); supply follows demand.
- Breast milk protein is one third of cow's; colostrum contains IgA.
  Prolactin produces, oxytocin outs.

Clinical picture
- Adequacy: ≥ 5 wet diapers per day at 5 days; loses up to 10% of birth weight.
- Birth weight regained by 2 weeks.
- Breast milk jaundice: β-glucuronidase.
  Day 5: 5 wet nappies.

Diagnosis
- Cow's milk first given after 12 months → else iron deficiency, renal solute load.
- Energy ~100-110 kcal/kg/day early; fluid ~150 ml/kg/day.
- Kwashiorkor: oedema, swollen abdomen; marasmus: wasting.
  100 kcal and 150 ml per kg per day.

Treatment
- Solids not before 17 weeks, ~6 months; no honey under 12 months (infant botulism).
- Breastfeeding jaundice: feed more often (q2h); milk jaundice: continue.
- All: vitamin D 400 IU daily + vitamin K 1 mg IM at birth.
  No honey before one.

II-5 Psychosomatic diseases. Anorexia nervosa Bulimia nervosa.

Psychosomatic symptoms are real, stress-driven (HPA and brain-gut axis) and should be managed by excluding red flags without endless tests; in eating disorders, the danger is physiological - know the admission criteria for anorexia, the hypokalaemic alkalosis of bulimia and the hypophosphataemia of refeeding.

Definition
- Psychosomatic = real symptoms driven by stress; commonest: functional abdominal pain.
- Anorexia: restriction, low weight; bulimia: binge + compensation weekly for 3 months.
  Bulimia: weekly for 3 months.

Causes
- Stress: school, bullying, family conflict, abuse; separation anxiety → school avoidance.
- Anorexia peaks in adolescent girls.
  Tummy ache on school mornings = separation anxiety.

Patho-physiology
- Stress → HPA + brain-gut axis → visceral hypersensitivity.
- Anorexia: loss of pulsatile GnRH → low LH/FSH/estrogen → amenorrhea.
- Vomiting → HCl loss + aldosterone.
  Starve → GnRH off → periods off.

Clinical picture
- AN: bradycardia, hypotension, lanugo, amenorrhea, mitral valve prolapse.
- BN: parotid hypertrophy, enamel erosion, Russell sign (knuckle calluses).
- Conversion: la belle indifférence.
  Russell = knuckle scars from fingers in throat.

Diagnosis
- Exclude red flags; avoid endless tests.
- BN: hypokalaemic hypochloraemic metabolic alkalosis.
- Admit AN: HR under 40, SBP under 80, temp under 35.5 °C, %mBMI under 70%, syncope.
  AN admission: HR under 40, SBP under 80.

Treatment
- Family-based therapy first-line for adolescent AN; bulimia: psychotherapy ± SSRIs (fluoxetine).
- Refeeding: key electrolyte disturbance hypophosphataemia.
- Somatic symptoms: monthly visits with a single physician; screen self-harm.
  Refeeding: watch phosphate.

II-6 Newborn screening tests. Preventive care

Newborn screening exists because the placenta hides metabolic disease until feeds start: screen at 48-72 h, treat a positive screen as an emergency recall (not a diagnosis), and remember the non-blood screens (hearing, pulse oximetry, hips, red reflex).

Definition
- Screening for diseases silent at birth, harmful, treatable (Wilson-Jungner).
  Silent, Serious, Saveable: the 3 S of screening.

Causes
- Hungary screens 26 congenital metabolic diseases: PKU, hypothyroidism, galactosaemia, biotinidase + tandem-MS.
- Aminoacidopathies, fatty-acid oxidation defects (MCAD), organic acidurias.
  26 in Hungary: think 'two-six, heel-prick fix'.

Patho-physiology
- Placenta clears toxic metabolites in utero → feeding unmasks the defect.
- PKU: deficiency of phenylalanine hydroxylase.
- Galactosaemia: jaundice, cataract, E. coli sepsis after milk feeds.
  Placenta = free dialysis; milk = the unmasking test.

Clinical picture
- MCAD: fasting → hypoketotic hypoglycaemia, sudden death.
- PKU: fair skin, musty odour.
  MCAD: 'Missed meal, no ketones, collapse'. PKU: pale and musty.

Diagnosis
- Heel prick at 48-72 hours → tandem mass spectrometry on dried blood spot, TSH, enzyme assays.
- Positive screen ≠ diagnosis → urgent recall + confirmation.
- Also hearing, pulse oximetry, hips, red reflex; vision every well-child visit (amblyopia).
  Heel prick on day 2-3; screen positive = recall, not diagnosis.

Treatment
- PKU low-phe diet; galactosaemia stop lactose; MCAD avoid fasting.
- All newborns receive vitamin K to prevent haemorrhagic disease; vitamin D 400 IU.
- Hepatitis B mother: immune globulin + vaccine within 12 hours.
  Remove the fuel: phe, lactose, fasting.

II-7 Sudden infant death syndrome. Infantile and paediatric mortality data.

SIDS is a diagnosis of exclusion explained by the triple-risk model (vulnerable infant + critical 2-4-month window + stressor such as prone sleep); prevention is 'back to sleep' in the parents' room on a firm, bare cot mattress, smoke-free.

Definition
- SIDS is sudden death under 1 year, a diagnosis of exclusion after autopsy, scene investigation, history.
  SIDS = 'nothing found' after everything checked.

Causes
- Smoking, bed-sharing (sofa, alcohol), prone/side sleep, soft bedding, overheating, prematurity.
- Low birth weight: less than 2500 grams (5.5 pounds).
  Smoke, Sofa, Stomach, Soft, Sweaty.

Patho-physiology
- Triple-risk model: vulnerable infant + critical developmental period + external stressor.
- Prone → rebreathes CO2 → hypoxia → vulnerable infant fails to arouse.
  Triple risk: wrong baby, wrong age, wrong night.

Clinical picture
- SIDS peaks at 2-4 months of age, during sleep.
- More often boys, winter, preterm, low birth weight.
  2-4 months, asleep, boy, winter.

Diagnosis
- Unexpected death → full autopsy + skeletal survey, metabolic screen; exclude infection, MCAD, abuse.
- Infant deaths: perinatal conditions (prematurity) and congenital malformations; over 1 year, accidents 36-40%.
  Rule out Bugs, Biochemistry, Battering.

Treatment
- "Back to sleep" (supine), avoid bed sharing, smoking cessation, pacifier during sleep.
- Room-share in own cot, firm mattress; breastfeed, immunise; bereavement support.
- Accident prevention: car seats, pool fencing, locked medicines.
  Back to sleep, own cot, smoke-free.

II-8 Clinical symptoms of infections, laboratory diagnostics, normal parameters.

Signs of infection are the host response: tachycardia out of proportion to fever and prolonged CRT signal early shock, any infant <3 months with ≥38 °C gets a full septic screen, and early CRP can falsely reassure because it peaks at 36-48 h.

Definition
- Infection signs are mostly the host response: fever, tachycardia, tachypnoea, poor perfusion.
  Fever, fast heart, fast breath, cold peripheries = host response.

Causes
- Any infant under 3 months with temperature ≥38 °C is high risk (NICE red); mottled skin, non-blanching rash.
  Under 3 months + 38 °C = red flag.

Patho-physiology
- IL-1, IL-6, TNF induce PGE2 in hypothalamus → raised set-point → fever.
- Vasoconstriction → mottling, prolonged CRT → early shock.
  Cytokines → PGE2 → thermostat up.

Clinical picture
- Grunting, RR >60, bulging fontanelle, neck stiffness, petechiae, lethargy.
- Tachycardia out of proportion to fever → shock.
  HR too high for the temperature = shock until proven otherwise.

Diagnosis
- CRP peaks at 36-48 hours (early CRP may reassure falsely); procalcitonin rises within 2-4 hours, more bacterial-specific.
- Normal CSF WBC <5/μL; bacterial meningitis → low glucose.
  PCT is quick (hours); CRP is slow (36-48 h).

Treatment
- Blood cultures before antibiotics; number of cultures and volume of blood decide sensitivity.
- Infant <3 months ≥ 38 °C → full septic screen.
- Shocked child: cultures → antibiotics within the hour; never delay for LP.
  Culture first, volume counts, antibiotics within the hour.

II-9 Antimicrobial therapy

Pick the antibiotic by age-specific bugs, site penetration and severity, culture first and review at 48–72 h; neonates need amoxicillin/ampicillin because Listeria ignores cephalosporins, and know the age-specific toxicities (grey baby, tooth staining, ceftriaxone–calcium).

Definition
- Choose by bug (age, site), penetration, severity, local resistance; culture first.
- Review antibiotic therapy at 48–72 hours to narrow or stop.

Bug, Body site, Badness, Bias (resistance). Review at 48–72 h.

Causes
- Neonate: GBS, E. coli, Listeria. Older: pneumococcus, meningococcus, Hib.
- Listeria is resistant to cephalosporins, so neonatal cover needs amoxicillin.

Neonate bugs GEL: GBS, E. coli, Listeria.

Patho-physiology
- β -lactams inhibit penicillin-binding protein transpeptidases; β -lactamase hydrolyzes the β -lactam ring.
- Aminoglycosides show concentration-dependent killing; β -lactams show time-dependent killing.

Aminoglycosides: Peak. β-lactams: Time.

Clinical picture
- Chloramphenicol → gray baby syndrome; tetracyclines → permanent tooth staining.
- Ceftriaxone in neonates displaces bilirubin and precipitates with calcium.
- Aminoglycosides: acute tubular necrosis, ototoxicity; fluoroquinolones: tendon rupture, cartilage damage.

Gray baby, grey teeth, gristle damage.

Diagnosis
- Blood culture before first dose → review at 48–72 hours.
- Narrow to culture, IV → oral when afebrile, shortest effective duration.

Narrow, Oral, Short.

Treatment
- Neonatal sepsis: ampicillin + gentamicin; suspected meningitis: ampicillin + cefotaxime. Beyond neonate: ceftriaxone.
- CAP: oral amoxicillin; MSSA: nafcillin, oxacillin, dicloxacillin; newborn gonorrhea: ceftriaxone.
- No antibiotics for colds, bronchiolitis, viral wheeze.

Neonate: amp + gent. Older: ceftriaxone.

II-10 Borreliosis (Lyme-disease) and parotitis epidemica (mumps)

Lyme disease: erythema migrans >5 cm after a tick bite is a clinical diagnosis (no serology); doxycycline for most (amoxicillin in young children per European practice), IV ceftriaxone for neuro or high-grade AV block. Mumps: viraemia seeds glands and CNS, which explains every complication; treat supportively and prevent with MMR ×2.

Definition
- Lyme: spirochete Borrelia burgdorferi, Ixodes tick (Europe: Ixodes ricinus). Mumps = paramyxovirus parotitis.

Ixodes = 'I x Lyme'; mumps = Paramyxo-parotid.

Causes
- Lyme: Northeast and upper Midwest US; needs >24–36 h attachment. Mumps: incubation 16–18 days.

Tick on >24–36 h = risk; mumps incubates 16–18 days.

Patho-physiology
- Spirochetes spread centrifugally → expanding erythema with central clearing.
- Mumps: viraemia seeds glands + CNS → explains every complication.

Bullseye spreads out; mumps seeds everywhere.

Clinical picture
- Lyme stage 2: facial palsy, meningitis, carditis (AV block); stage 3: encephalopathy, arthritis.
- Mumps: painful parotid swelling; orchitis in post-pubertal males; meningitis, pancreatitis, deafness.

Lyme 2: 'face, brain, heart'. Mumps: post-pubertal orchitis.

Diagnosis
- Erythema migrans is clinical: tick bite + growing erythema more than 5 cm; no serology.
- Early disseminated/late: ELISA then Western if confirmation needed. Mumps: raised amylase, RT-PCR/IgM.

EM >5 cm + tick bite = treat; no blood test.

Treatment
- Erythema migrans: young child amoxicillin; older doxycycline (US any age, 10 days).
- Neuro/high-grade block: IV ceftriaxone. Prophylaxis if tick attached ≥36 hours, within 72 hours.
- Mumps: supportive; MMR ×2.

Small child: amox; big child: doxy; brain/heart block: ceftriaxone.

II-11 Clinical manifestations of Epstein-Barr virus infection.

Infectious mononucleosis is a teenager with fever, exudative pharyngitis, posterior cervical nodes and splenomegaly; the 'atypical lymphocytes' are reactive CD8 T cells, Monospot fails under 4–5 years (use VCA IgM), treatment is supportive, and the spleen dictates 3–4 weeks off contact sport.

Definition
- Infectious mononucleosis ("kissing disease") is caused by EBV (HHV-4): fever, pharyngitis, nodes, splenomegaly.

Mono tetrad: Fever, Fauces, Fat nodes, Fat spleen.

Causes
- Most common in teenagers; spread by saliva or respiratory secretions; young children mild or silent.
- EBV: Burkitt lymphoma, nasopharyngeal carcinoma.

Kissing teens; Burkitt in Africa, NPC in Asia.

Patho-physiology
- EBV first infects oropharyngeal epithelium, then B cells via CD21.
- Reactive CD8 T cells = Downey type II atypical lymphocytes.
- Polyclonal B activation → heterophile antibodies.

CD21 = 'EBV turns 21, enters B-cell bar'.

Clinical picture
- Fatigue, fever, posterior cervical lymphadenopathy, pharyngitis, hepatosplenomegaly; grey-white tonsillar exudate.
- Raised LFTs (hepatitis) and lymphocytosis.
- Amoxicillin in EBV causes a maculopapular rash - not true penicillin allergy.

Posterior nodes = EBV; anterior = strep.

Diagnosis
- Atypical lymphocytes >10% on smear; Monospot often negative under 4–5 years → VCA IgM.
- Cytopenias or blasts → exclude leukaemia.

Under 5: Monospot lies, VCA IgM tells.

Treatment
- Supportive: rest, fluids, paracetamol (acetaminophen); aciclovir useless.
- Steroids only for airway obstruction, haemolysis or thrombocytopenia.
- No contact sport for 3–4 weeks because of splenic rupture.

No contact sport for 3–4 weeks: protect the spleen.

II-12 Measles, rubella, scarlet fever

Measles = 3 Cs + Koplik spots then a cephalocaudal rash; pneumonia kills and immune amnesia follows. Rubella matters for the fetus (cataract, deafness, PDA). Scarlet fever is a GAS toxin rash: confirm on culture and give 10 days of penicillin to prevent rheumatic fever.

Definition
- Measles (Paramyxoviridae), rubella (togavirus), scarlet fever (group A strep exotoxin).

Measles Paramyxo, Rubella Toga, Scarlet Strep.

Causes
- Measles most contagious (RO 12–18), aerosol; scarlet fever 2–10 years; rubella in first trimester → congenital rubella.

Measles RO 12–18: the most contagious.

Patho-physiology
- Measles kills memory cells → immune amnesia; rubella crosses placenta → organogenesis damage.
- GAS pyrogenic exotoxins = superantigens → capillary dilation.

Measles erases memory; rubella disrupts organogenesis.

Clinical picture
- Measles: 3 Cs + Koplik spots → rash day 3–4, ears downward.
- Rubella: postauricular and suboccipital nodes; CRS: cataract, sensorineural deafness, PDA (80% foetal injury).
- Scarlet: sandpaper rash, strawberry tongue, circumoral pallor, desquamation.

Measles 3 Cs: Cough, Coryza, Conjunctivitis.

Diagnosis
- Measles pneumonia is the commonest death; SSPE years later.
- Scarlet: throat swab; Rapid Strep has low sensitivity, culture confirms; ASO later.

Measles kills via lungs; SSPE haunts years later.

Treatment
- Measles: supportive + vitamin A; contacts MMR ≤ 72 h or Ig ≤ 6 days.
- Scarlet fever: phenoxymethylpenicillin 10 days (US also amoxicillin) → prevents rheumatic fever.
- Prevent: MMR ×2; avoid if pregnant or immunocompromised.

Measles: vitamin A. Scarlet: penicillin 10 days.

II-13 Helminthic and protozoan infections

Think by route and organ: tissue-migrating worms give eosinophilia (Th2/IL-5), Giardia flattens villi (fatty stools), Entamoeba invades (bloody diarrhoea, liver abscess); threadworm is the common European worm (tape test, mebendazole for the household, repeat at 2 weeks), and fever after endemic travel is malaria until films say otherwise.

Definition
- Worms (helminths) and single-cell parasites (protozoa) in children.

Helminth = worm; protozoa = one cell.

Causes
- Enterobius (commonest Europe), Ascaris, Toxocara (puppies), Giardia, Entamoeba, Toxoplasma, malaria.
- Giardia is fecal-oral (daycare, camping streams); Cryptosporidium in HIV/AIDS.

Puppies give Toxocara; streams give Giardia.

Patho-physiology
- Tissue-migrating worms → Th2 / IL-5 → eosinophilia.
- Giardia flattens villi → fat/lactose malabsorption; Entamoeba trophozoites invade colon.

Worms in tissue = eosinophils.

Clinical picture
- Threadworm: nocturnal perianal itch. Toxocara: leukocoria (ocular larva migrans), mimics retinoblastoma.
- Giardia: bloating, foul-smelling floating, fatty stools.
- Entamoeba: bloody diarrhea, right lobe liver abscess.
- Congenital toxo: chorioretinitis, hydrocephalus, diffuse intracranial calcifications. Malaria: fever, anemia, splenomegaly.

Threadworm itches at night.

Diagnosis
- Sticky tape (scotch tape) test; Giardia: trophozoites or cysts in stool.
- Entamoeba: trophozoites with endocytosed RBCs; biopsy with flask-shaped lesions.
- Fever after endemic travel = malaria → urgent thick/thin films.

Fever after travel = malaria films now.

Treatment
- Threadworm: mebendazole, repeat after 2 weeks, treat the whole household.
- Giardia/Entamoeba: metronidazole (or tinidazole); Toxocara: albendazole.
- Congenital toxo: pyrimethamine + sulfadiazine + folate (leucovorin) for 12 months.

Mebendazole twice, whole family.

II-14 Clinical manifestations of Herpes simplex virus, Parvovirus, Coxsackie virus infection.

HSV hides in sensory ganglia and recurs; any suspicion of HSV encephalitis or neonatal HSV means IV aciclovir now. Parvovirus B19 targets erythroid progenitors via P antigen: slapped cheek in children, aplastic crisis in haemolytic anaemias, hydrops in the fetus. Coxsackie A = hand-foot-mouth; B = heart and pleura.

Definition
- HSV (vesicles, latency), parvovirus B19 (single-stranded DNA, fifth disease), Coxsackie (enterovirus).

B19 = fifth disease; Coxsackie = enterovirus.

Causes
- HSV: direct contact; neonatal from genital tract lesions.
- Enteroviruses faecal-oral, summer/autumn.

Enteroviruses love summer.

Patho-physiology
- HSV stays latent in sensory ganglia (HSV-1: trigeminal ganglion) → recurrence.
- B19 infects erythroid progenitors via P antigen → aplastic crisis in sickle cell or spherocytosis.

B19 + short-lived RBCs = aplastic crisis.

Clinical picture
- HSV: herpes labialis, dendritic corneal ulcers; eczema herpeticum = punched-out erosions + fever.
- B19: slapped cheek, lacy rash; fetal hydrops fetalis.
- Coxsackie A: hand, foot and mouth disease; B: myocarditis, Bornholm disease.

Coxsackie A = hAnd foot mouth; B = Bornholm/heart.

Diagnosis
- HSV encephalitis: temporal lobe, CSF HSV PCR, MRI; Tzanck smear also detects HSV.
- Aplastic crisis: falling Hb + reticulocytopenia.
- Pregnant, B19 antibody positive: reassurance; >20 weeks weekly ultrasounds.

HSV loves the temporal lobe.

Treatment
- Suspected HSV encephalitis → IV aciclovir immediately; eczema herpeticum: aciclovir.
- Neonatal HSV: high-dose IV aciclovir 14–21 days; genital HSV in labor: cesarean delivery.
- Aplastic crisis: transfusion; fetal anemia → intrauterine RBC transfusion.

Suspect HSV encephalitis = aciclovir now.

II-15 Varicella infection, herpes zoster

Chickenpox is crops of lesions in all stages (repeated viraemia), infectious from 1–2 days before the rash until all are crusted; give paracetamol not ibuprofen or aspirin, aciclovir for neonates and the immunocompromised, and treat fever returning after day 3–4 as invasive GAS. Zoster is dermatomal reactivation from sensory ganglia.

Definition
- Varicella (chickenpox) = primary VZV; herpes zoster = its reactivation, unilateral dermatomal rash.

Chickenpox first, shingles comeback.

Causes
- Airborne; incubation 10–21 days; infectious 1–2 days before rash until all lesions crusted.

Infectious until every spot is crusted.

Patho-physiology
- Repeated viraemia waves → crops: lesions in different stages of healing together.
- VZV becomes latent in dorsal root (sensory) ganglia → falling T-cell immunity → zoster.

Crops = pox in every stage at once.

Clinical picture
- Itchy centripetal rash (trunk first), headache, fever. Zoster: painful, does not cross midline.
- Fever returning after day 3–4 → invasive group A streptococcus. Hutchinson sign → eye.

Fever back after day 3–4 = think GAS.

Diagnosis
- Clinical (Tzanck smear). Maternal varicella before 20 weeks → congenital varicella syndrome.
- Maternal rash -5 to +2 days of delivery → severe neonatal varicella.

-5 to +2 days around birth = danger window.

Treatment
- Paracetamol, emollients; avoid ibuprofen, never aspirin (Reye).
- Aciclovir for neonates, immunocompromised, severe disease (IV if disseminated); VZIG for high-risk contacts.
- Live attenuated vaccine: 2 doses (1 and 4 y); post-zoster neuralgia: TCAs, gabapentin.

No aspirin = no Reye.

II-16 Skin infections

Skin infections are read by depth: impetigo (superficial, well child), erysipelas (sharp edge), cellulitis (ill-defined), necrotising fasciitis (pain out of proportion, mild skin) — and SSSS is a toxin splitting desmoglein-1, so blisters are sterile and mucosae are spared, unlike TEN.

Definition
- Bacterial, viral, fungal skin infections; depth decides the picture.

Depth decides: impetigo top, erysipelas upper dermis, fasciitis deep.

Causes
- S. aureus and GAS via breaks (eczema, bites, varicella); tinea capitis in young African American children.

Broken skin opens the door.

Patho-physiology
- S. aureus exfoliative toxin cleaves desmoglein-1 in the granular layer → bullous impetigo, SSSS.
- Necrotising fasciitis thromboses perforators → deep death, mild skin.

Desmoglein-1 = SSSS; deep death, mild skin = fasciitis.

Clinical picture
- Impetigo: honey-coloured (golden) crusts, well. Erysipelas: upper dermis, sharp raised edge; cellulitis ill-defined.
- SSSS: Nikolsky +, spares mucous membranes (desmoglein-3) -unlike TEN.
- Necrotising fasciitis: pain out of proportion, bullae, crepitus, shock.

Pain out of proportion = fasciitis until proven otherwise.

Diagnosis
- Proptosis/painful eye movement → orbital cellulitis → CT.
- SSSS blister fluid sterile → culture the focus. Tinea: Wood lamp for Microsporum; scabies: papules and burrows.

Proptosis or painful eye movement = CT orbit.

Treatment
- Impetigo: topical (mupirocin), or oral flucloxacillin (US cephalexin) if widespread/bullous.
- SSSS/orbital: admit, IV antibiotics. Abscess: incision and drainage. Tinea capitis: oral antifungal.
- Necrotising fasciitis: urgent surgery + clindamycin, consider IVIG.

Small: mupirocin; big: flucloxacillin.

II-17 Staphylococcus aureus infections. Toxic shock syndrome

S. aureus walls itself off (coagulase → abscess) so pus needs draining; TSS is a TSST-1 superantigen storm in which blood cultures are usually negative (unlike streptococcal TSS), and you treat with fluids, source removal and a β-lactam plus clindamycin to switch off toxin synthesis.

Definition
- S. aureus: pyogenic invasive disease; TSS = superantigen cytokine storm.

Staph = pus; TSS = storm.

Causes
- Nasal carriage ~1/3. TSS: tampons, retained gauze, burns, wounds, nasal packing, varicella lesions.
- Most common cause of osteomyelitis.

Tampons, packing, burns: hidden toxin factories.

Patho-physiology
- Coagulase builds a fibrin wall → abscesses, furuncles, carbuncles.
- TSST-1 superantigen binds MHC II and TCR V β → T-cell activation (up to 20%).
- MRSA alters penicillin-binding proteins; S. aureus makes β -lactamase.

Coagulase = clot wall = abscess.

Clinical picture
- TSS: high fever, erythroderma, hypotension, multi-organ failure.
- Desquamation of palms/soles 1–2 weeks later.
- Food poisoning: vomiting 1–6 hours after eating (preformed enterotoxin).

TSS: fever, red, low BP, then peel.

Diagnosis
- Bacteraemia → search bone, joint, heart focus + repeat cultures.
- Staph TSS blood culture usually negative; strep TSS positive (toxic shock-like syndrome: GAS).

Staph TSS cultures negative; strep TSS positive.

Treatment
- TSS: fluids, vasopressors, source removal; β -lactam + clindamycin (suppresses toxin synthesis); IVIG if severe.
- MSSA: anti-staph penicillins (nafcillin, oxacillin, dicloxacillin), cefazolin; drain pus.
- MRSA → vancomycin, linezolid; daptomycin for bacteremia; ceftaroline binds altered PBPs.

Clinda stops the toxin factory.

II-18 Neonatal and childhood tetanus. Botulism

Two clostridial toxins, opposite paralyses: tetanospasmin blocks inhibitory GABA/glycine release (trismus, spasms, conscious child), botulinum toxin blocks ACh release (descending flaccid paralysis, constipation, alert afebrile infant). Tetanus gives no immunity, so vaccinate after recovery.

Definition
- Toxin diseases of C. tetani (spastic paralysis) and C. botulinum (flaccid paralysis).

Tetanus Tight; Botulism Floppy.

Causes
- Tetanus: spores in soil/rusty nails, deep wounds, unclean cord with poorly immunized mother.
- Infant botulism: ingested spores from honey or juice (or inhaled dust) in immature gut.

No honey under 1 year.

Patho-physiology
- Tetanospasmin → retrograde axonal transport → blocks GABA/glycine release → spasms.
- Botulinum cleaves SNARE → no acetylcholine release → flaccid + autonomic block.

Tetanus blocks the brakes; botulinum cuts the accelerator.

Clinical picture
- Tetanus: jaw first - trismus (lock jaw), risus sardonicus, spasms, opisthotonus; conscious. Neonate stops sucking.
- Botulism: constipation, early diplopia, ptosis, descending symmetric flaccid paralysis, floppy head; afebrile, alert.

Botulism: descending; jaw lock = tetanus.

Diagnosis
- Tetanus: clinical assessment only; toxin effects, consciousness and sensation preserved.

Tetanus: diagnose with your eyes.

Treatment
- Tetanus: human tetanus immune globulin (HTIG), debridement, metronidazole, benzodiazepines, ICU.
- Prevent: maternal toxoid vaccination; vaccinate after recovery (no immunity after tetanus).
- Infant botulism: supportive + human-derived botulism immune globulin; adults: horse-derived antitoxin; no aminoglycosides.

Aminoglycosides worsen botulism blockade: avoid.

II-19 Inflammatory joint and bone diseases

Haematogenous bacteria settle in the slow-flow metaphysis; under 18 months transphyseal vessels let osteomyelitis spill into the joint. X-rays stay normal for 10–14 days (use MRI), septic arthritis is a surgical emergency, and most children need only a short IV course then oral to ~3 weeks.

Definition
- Osteomyelitis (bone) and septic arthritis (joint) - bacterial, septic arthritis a surgical emergency.

Septic joint = surgical emergency.

Causes
- S. aureus most common; Kingella kingae 6 months–4 years; GBS/E. coli neonates.
- Salmonella in sickle cell; Pseudomonas after nail puncture.

Sickle = Salmonella; sneaker nail = Pseudomonas.

Patho-physiology
- Children: hematogenous spread; bacteria settle in the metaphysis (slow flow in sinusoids) → pus → sequestrum, involucrum.
- Under 18 months, transphyseal vessels spread infection to the joint.

Metaphysis: slow flow, bugs stop and grow.

Clinical picture
- Fever, hot swollen joint, won't bear weight; pseudoparalysis in infants.
- Septic hip held flexed, abducted and externally rotated.

Septic hip: Flexed, ABducted, External rotation (FABER).

Diagnosis
- Blood culture, CRP, ESR; ultrasound effusion; X-ray normal for 10–14 days.
- Joint aspiration: Gram stain, culture, PCR.
- MRI best for osteomyelitis (high ESR, normal X-ray → MRI).

X-ray late, MRI early.

Treatment
- MSSA: nafcillin, oxacillin, dicloxacillin, IV cefazolin; clindamycin or vancomycin if MRSA.
- Short IV (2–4 days) → oral; about 3 weeks total.
- Septic arthritis: joint washout; drain subperiosteal abscess, remove sequestrum.

Short IV, long oral: ~3 weeks total.

II-20 Congenital bone and joint disorders

Screen hips with Ortolani (relocates) and Barlow (dislocates), image with ultrasound before ~4.5 months and X-ray after, and treat early with a Pavlik harness; clubfoot is Ponseti casting; OI (type I collagen) must be considered before labelling fractures as abuse; achondroplasia is FGFR3 gain-of-function with rhizomelia and foramen magnum stenosis.

Definition
- Congenital disorders: DDH, clubfoot, osteogenesis imperfecta, achondroplasia.

DDH, Clubfoot, OI, Achondroplasia.

Causes
- DDH: breech, family history, female sex, first-born, oligohydramnios, tight swaddling.
- OI: COL1A1/COL1A2 (usually AD); achondroplasia: FGFR3 gain-of-function, older fathers.

DDH 4 Fs: Female, First-born, Family, Feet-first (breech).

Patho-physiology
- Femoral head displaced → acetabulum fails to mould → DDH.
- OI: defective type I collagen → fragile bone, blue sclerae.
- FGFR3 overactive → poor endochondral growth → rhizomelic limbs.

Overactive FGFR3 = brakes on growth plate.

Clinical picture
- DDH: Barlow/Ortolani positive; later limited abduction, leg-length discrepancy.
- Clubfoot: rigid cavus, adductus, varus, equinus; OI: low-trauma fractures.
- Achondroplasia: macrocephaly, trident hands, foramen magnum stenosis.

Clubfoot CAVE: Cavus, Adductus, Varus, Equinus.

Diagnosis
- Ortolani relocates a dislocated hip; Barlow dislocates a located one.
- Hip ultrasound before about 4.5 months; X-ray afterwards.
- Consider OI in non-accidental injury. Scoliosis: forward bend test → X-ray.

Ortolani = Out to In; Barlow = Bye-bye out.

Treatment
- DDH: Pavlik harness under 6 months → closed → open reduction.
- Clubfoot: Ponseti casting → Achilles tenotomy → bracing. OI: IV bisphosphonates.
- Scoliosis Cobb 10–30°: monitor; 30–40°: brace; >40–50°: surgery.

Cobb: <30 watch, 30–40 brace, >40–50 operate.

II-21 Juvenile idiopathic arthritis

JIA is a diagnosis of exclusion (onset <16 y, ≥6 weeks): rule out leukaemia and septic arthritis first, then think by subtype — ANA+ oligo girls go blind silently from uveitis, and systemic JIA kills through MAS.

Definition
- JIA: arthritis of unknown cause, onset before 16, lasting at least 6 weeks.

JIA = under 16, over 6 weeks: '16 and 6'.

Causes
- Subtypes: oligoarticular (~50%, young ANA+ girls), polyarticular RF-/RF+, systemic, enthesitis-related (HLA-B27), psoriatic.

Oligo = 'Only a few joints, Often little girls'.

Patho-physiology
- T cells + TNF- α, IL-1, IL-6 → synovial hyperplasia (pannus) → erosion.
- Systemic JIA: IL-1 and IL-6 driven; ANA and RF negative.
- MAS: failed cytotoxic killing → cytokine storm; ESR falls as fibrinogen is consumed.

MAS clue: ferritin rockets, ESR drops.

Clinical picture
- Swollen joint, morning stiffness; limb overgrowth.
- Oligoarticular ANA-positive: silent anterior uveitis (iritis), painless white eye.
- Systemic: quotidian fever, salmon-pink macular rash, hepatosplenomegaly, serositis.

sJIA: fever spikes once daily, rash comes with the fever.

Diagnosis
- Clinical; ANA → uveitis risk; RF/anti-CCP → erosive course.
- Exclude leukaemia (night pain, cytopenia, LDH) and septic arthritis.
- ANA+ oligoarticular: slit lamp every 3 months. MAS: huge ferritin, falling ESR.

Night pain + cytopenia = leukaemia until proven otherwise.

Treatment
- NSAIDs → intra-articular triamcinolone → methotrexate once weekly + folic acid.
- Biologics: anti-TNF, abatacept, tocilizumab; sJIA: anakinra/tocilizumab first-line.
- MAS (most serious complication): IV methylprednisolone, anakinra, ciclosporin.

Ladder: NSAID → inject → MTX weekly → biologic.

II-22 Causes, diagnostic and treatment of acute viral hepatitis

In acute viral hepatitis the transaminases tell you the liver is inflamed, but the INR (uncorrected by vitamin K), glucose and mental state tell you whether the child is dying — PALF is INR ≥1.5 with encephalopathy or ≥2.0 without.

Definition
- Acute inflammation of the liver from hepatotropic viruses A-E.

Hepatotropic alphabet: A-E.

Causes
- A, E: faecal-oral, self-limiting; B, C, D: blood/vertical, can become chronic.
- Others: EBV, CMV, HSV, paracetamol, autoimmune, Wilson.

Vowels (A, E) from the bowels; consonants chronic.

Patho-physiology
- Cytotoxic T cells kill hepatocytes → ALT/AST in thousands; zone I first.
- Impaired bilirubin handling → jaundice, dark urine, pale stools, itch.
- Perinatal HBV becomes chronic in about 90% (vs ~5% adults).

The younger the host, the more chronic HBV becomes.

Clinical picture
- Prodrome: fever, malaise, anorexia, nausea, vomiting → then jaundice; HAV in young children mostly anicteric.

Little kids with HAV: sick but not yellow.

Diagnosis
- IgM anti-HAV; HBsAg + IgM anti-HBc = acute B; anti-HCV + HCV RNA.
- HBeAg predicts infectivity; severity by INR (not corrected by vitamin K), glucose, mental state.
- PALF: INR ≥ 1.5 with encephalopathy, or ≥ 2.0 without.

IgM = 'I'M here now' (acute).

Treatment
- Supportive: fluids, avoid hepatotoxins; admit if INR raised, hypoglycaemic, encephalopathic.
- Chronic only: HBV entecavir/tenofovir; HCV DAAs.
- Newborn of HBsAg+ mother: HBV vaccine + HBIG within 12 hours.

Baby of HBsAg+ mum: vaccine + HBIG in 12 h.

II-23 Tuberculosis

A child with TB is a sentinel for an infectious adult: think contact tracing. Under-2s disseminate (miliary, meningitis), so young contacts get preventive therapy even with negative tests, and TB meningitis needs 12 months plus steroids.

Definition
- Mycobacterium tuberculosis infection; childhood TB = recent infection from an adult contact.

Child with TB = find the coughing adult.

Causes
- Droplet nuclei from adult with cavitary pulmonary TB.
- Risk: age under 2, HIV, malnutrition, measles.

Patho-physiology
- Hilar nodes → Ghon complex = peripheral lung focus + hilar nodes.
- Cord factor → TNF- α → granulomas.
- Under 2 years: blood spread → miliary TB and meningitis.

Under 2: miliary and meningitis.

Clinical picture
- Primary TB mostly asymptomatic; cough >2 weeks, fever, night sweats, weight loss.
- Scrofula, Pott disease, erythema nodosum.

Pott = spine, scrofula = neck nodes.

Diagnosis
- Contact + TST/IGRA + CXR; IGRA has fewer BCG false positives.
- PPD positive: ≥ 5 mm high risk, ≥ 10 intermediate, ≥ 15 low risk.
- Early-morning gastric aspirates or induced sputum → Xpert MTB/RIF, culture. Meningitis: lymphocytes, high protein, very low glucose.

TST cut-offs 5–10–15: high, intermediate, low risk.

Treatment
- RIPE: 2 months HRZ(E) → 4 months HR (rifampin, isoniazid); pyridoxine with isoniazid.
- TB meningitis: 12 months + corticosteroids.
- Contacts under 5 → preventive therapy even if tests negative.

RIPE 2 + RI 4 = 6 months.

II-24 Disorders of the humoral immune system

Antibody deficiency = recurrent encapsulated bacterial sinopulmonary infections starting after maternal IgG wanes (~6 months); measure IgG/IgA/IgM plus vaccine responses, and treat with lifelong immunoglobulin replacement — never live vaccines in XLA.

Definition
- Primary antibody (B-cell) deficiencies: XLA, CVID, selective IgA deficiency, hyper-IgM.

Causes
- XLA: BTK mutation, boys; CVID: no B-cell differentiation into plasma cells.
- Hyper-IgM: mutated CD40L on Th cells. Secondary: nephrotic syndrome, rituximab, malignancy.

Bruton = Boys, BTK, no B cells.

Patho-physiology
- XLA: no B-cell maturation → no opsonising IgG → encapsulated bacteria escape phagocytosis.
- Maternal IgG wanes → onset after ~6 months; hyper-IgM: defective class switching.

Maternal IgG fades at 6 months → infections begin.

Clinical picture
- Recurrent sinopulmonary infections → bronchiectasis; enteroviral meningoencephalitis.
- XLA: absent tonsils and nodes; CVID: autoimmunity, lymphoma.
- IgA deficiency (most common): mostly asymptomatic; transfusion anaphylaxis.

XLA: no tonsils to see.

Diagnosis
- Quantitative IgG, IgA, IgM; IgA deficiency: normal IgG and IgM.
- Vaccine responses (tetanus, Hib, pneumococcal) before and 4–6 weeks after booster.
- Flow cytometry: B cells (CD19) absent in XLA; genetics (BTK).

Levels, function, cells, genes.

Treatment
- Immunoglobulin replacement (IVIG) every 3–4 weeks or subcutaneous weekly.
- Prompt antibiotics; no live vaccines in XLA.
- IgA deficiency: usually none; washed blood products.

XLA: no live vaccines.

II-25 Cellular immunodeficiencies

T-cell defects are combined defects: a lymphopenic infant with thrush, Pneumocystis, chronic viral diarrhoea and failure to thrive has SCID until proven otherwise — isolate, no live vaccines, irradiated CMV-negative blood, and transplant early (before 3.5 months).

Definition
- T-cell defects - almost always combined (antibody fails too): SCID, DiGeorge, Wiskott-Aldrich.

No T help, no B work.

Causes
- SCID: X-linked common γ -chain (IL2RG) commonest, ADA deficiency, RAG mutations.
- DiGeorge: 22q11.2 deletion; WAS: X-linked WAS protein; secondary: HIV, steroids.

22q11: CATCH-22.

Patho-physiology
- No T cells → no control of viruses, fungi, intracellular opportunists; no B-cell help.
- ADA deficiency: dATP toxic to lymphocytes; RAG → VDJ recombination defects.
- DiGeorge: abnormal 3rd/4th pharyngeal pouches → thymic + parathyroid hypoplasia.

Pouches 3 and 4: thymus and parathyroids.

Clinical picture
- First months: thrush, Pneumocystis, chronic viral diarrhoea, failure to thrive.
- DiGeorge: hypocalcaemia, conotruncal heart defects, cleft palate.
- WAS triad: eczema, thrombocytopenia with small platelets, infections.

WAS: eczema, tiny platelets, infections.

Diagnosis
- Lymphopenia, absent thymic shadow; absent T cells on flow cytometry.
- Newborn screening: TRECs. DiGeorge: calcium, PTH, echo, FISH for 22q11.2.

Low TRECs = no new T cells.

Treatment
- Isolation; no live vaccines; only irradiated, CMV-negative blood.
- TMP-SMX (co-trimoxazole) PJP prophylaxis, antifungals, immunoglobulin replacement.
- Cure: stem cell transplant (>90% if <3.5 months); ADA: pegademase, gene therapy.

Transplant early: >90% if under 3.5 months.

II-26 Role of the complement system and diseases of the complement system

Map the complement defect to the disease: early classical → lupus, C3 → pyogenic encapsulated bacteria, C5–C9/properdin → recurrent Neisseria, factor H → aHUS, C1-inhibitor → bradykinin angioedema that ignores adrenaline and antihistamines.

Definition
- Complement: three pathways (classical begins C1, alternative C3) converging on C3 → opsonise, recruit, lyse.

All roads lead to C3.

Causes
- Inherited deficiency of early classical (C1q, C2, C4), C3, terminal (C5-C9), properdin, factor H/I.
- C1-inhibitor deficiency (AD) → hereditary angioedema.

Patho-physiology
- Early classical deficiency → SLE-like; C3 → encapsulated bacteria.
- C5-C9/properdin deficiency → no MAC → recurrent Neisseria infection.
- No C1-inhibitor → bradykinin excess → vascular leak.

MAC missing → meningococcus.

Clinical picture
- Lupus-like disease; severe pyogenic infections; recurrent meningococcal disease.
- Factor H defect → atypical HUS.
- HAE: non-itchy swelling of face, gut, larynx; no urticaria.

HAE: swelling without hives.

Diagnosis
- CH50 (classical), AH50 (alternative), C3, C4 levels.
- Second or unusual-serogroup meningococcal episode → complement screen.
- HAE screening test: low C4 → C1-inhibitor level and function.

HAE: check C4 first.

Treatment
- Vaccinate: meningococcus ACWY + B, pneumococcus, Hib; early antibiotics for fever.
- aHUS: eculizumab (anti-C5) - requires meningococcal vaccination.
- HAE: C1-inhibitor concentrate or icatibant - adrenaline/antihistamines/steroids don't work.

Eculizumab = man-made C5 deficiency.

II-27 HIV infections

Paediatric HIV is mostly vertical and intrapartum, so it is largely preventable (maternal ART + neonatal zidovudine + no breastfeeding → <1%); under 18 months diagnose by PCR, not antibody, and treat everyone at diagnosis.

Definition
- HIV infects CD4 T cells; in children mainly vertical, most intrapartum.

Causes
- Transmission: in utero, intrapartum, breastfeeding; adolescents sexual or drug use.

Patho-physiology
- gp120 binds CD4 + CCR5 (macrophages, early) or CXCR4 (T cells, late); gp41 mediates fusion.
- → CD4 depletion and immune activation.
- Developing thymus, higher viral set points → faster progression than adults.

gp120 grabs, gp41 fuses.

Clinical picture
- Failure to thrive, lymphadenopathy, hepatosplenomegaly, parotid swelling, chronic diarrhoea.
- Persistent candidiasis, LIP, encephalopathy; infant 3-6 months with severe hypoxia → PJP.

Parotid swelling + LIP = paediatric HIV.

Diagnosis
- Under 18 months: HIV DNA/RNA PCR because maternal IgG persists.
- Over 18 months: antibody/antigen tests as in adults.

Under 18 months: test the virus, not antibody.

Treatment
- Prevent: maternal ART, neonatal zidovudine, avoid breastfeeding → transmission <1%.
- ART at diagnosis, regardless of CD4: 2 NRTIs + dolutegravir.
- Co-trimoxazole from 4–6 weeks against PJP; avoid BCG; rifabutin replaces rifampin with protease inhibitors.

II-28 Opportunistic infections

The immune defect predicts the bug: no neutrophils → Pseudomonas, S. aureus, Candida, Aspergillus; no T cells → Pneumocystis, CMV, Cryptococcus, MAC; no antibody/spleen → encapsulated bacteria. Febrile neutropenia = antibiotics within 1 hour.

Definition
- Infections by organisms harmless to healthy hosts; the immune defect predicts the organism.

Defect predicts organism.

Causes
- HIV/AIDS, high-dose immunosuppression (transplant), steroids; chemotherapy neutropenia, CGD.
- Antibody, complement, asplenia (XLA, CVID, sickle cell).

Patho-physiology
- No neutrophils → Pseudomonas, S. aureus, Candida, Aspergillus.
- No T cells → Pneumocystis, CMV, Cryptococcus, MAC.
- No antibody/spleen → encapsulated bacteria.

Clinical picture
- Febrile neutropenia: fever may be the only sign of sepsis.
- PJP: fever, dry cough, hypoxia, high LDH; MAC: high fever, watery diarrhea.
- CMV: retinitis, colitis; Cryptococcus: fungal meningitis.

Diagnosis
- Febrile neutropenia = ≥38°C + neutrophils $<0.5 \times 10^9$/L; cultures line + periphery.
- PJP: bronchoalveolar lavage; CMV retinitis: CD4 ≤ 50, cotton wool spots.

Febrile neutropenia: 38 and 0.5.

Treatment
- Piperacillin-tazobactam within 1 hour; add vancomycin for line/instability.
- PJP: TMP-SMX + steroids if PaO2 ≤ 70 mmHg; MAC: clarithromycin + ethambutol.
- Toxoplasmosis: pyrimethamine & sulfadiazine; varicella: IV aciclovir.

Febrile neutropenia: tazocin in the golden hour.

II-29 Poststreptococcal diseases

Two GAS sequelae, two mechanisms: ARF is molecular mimicry after pharyngitis (Jones criteria, mitral valve, lifelong benzathine prophylaxis); PSGN is immune-complex disease after throat or skin infection (low C3, normal C4, recovers in 8 weeks, not prevented by antibiotics).

Definition
- Non-suppurative sequelae of GAS: acute rheumatic fever and PSGN.

Causes
- ARF: GAS pharyngitis 2-4 weeks before.
- PSGN: pharyngitis 1-2 weeks or skin 3-6 weeks before.

Skin can't give rheumatic fever.

Patho-physiology
- ARF: M protein cross-reacts with self-antigens (molecular mimicry); type II HSR.
- PSGN: immune complexes → complement → low C3 → ↓GFR → salt retention.

ARF = mimicry; PSGN = complexes.

Clinical picture
- ARF (J♥NES): migratory polyarthritis, carditis, nodules, erythema marginatum, chorea.
- Mitral valve most common: acute regurgitation; chronic fish-mouth stenosis.
- PSGN: cola urine, periorbital oedema, hypertension, oliguria.

J♥NES: Joints, Heart, Nodules, Erythema marginatum, Sydenham.

Diagnosis
- Jones: 2 major, or 1 major + 2 minor, + evidence of GAS (ASO/anti-DNase B).
- Histology: Aschoff bodies, Anitschkow cells; echo for subclinical carditis.
- PSGN: low C3, normal C4, normalising within 8 weeks.

Jones: 2 major, or 1 major + 2 minor.

Treatment
- ARF: penicillin V 10 days (anaphylaxis: macrolides); aspirin/naproxen for arthritis.
- Secondary prophylaxis: benzathine penicillin IM every 3-4 weeks.
- PSGN: supportive; strep-throat antibiotics prevent ARF, not PSGN.

Antibiotics stop ARF, not PSGN.

II-30 Pertussis infection

Pertussis kills young infants silently: under 6 months there may be no whoop, just apnoea and cyanosis, so admit, give a macrolide (azithromycin in neonates), treat all close contacts and prevent it with Tdap in pregnancy.

Definition
- Pertussis: toxin-mediated Bordetella pertussis respiratory infection (whooping cough).

Causes
- Aerobic gram-negative coccobacillus, spread by respiratory droplets; attack rate >80%.
- Adolescents/adults often the source.

Patho-physiology
- Filamentous hemagglutinin attaches; tracheal toxin kills ciliated cells → violent coughing.
- Pertussis toxin disables Gi → ↑cAMP → lymphocytosis, hypoglycaemia.
- Lymphocytes >30 × 10^9/L → leukostasis → pulmonary hypertension.

Clinical picture
- Catarrhal (most infectious) → paroxysmal: whoop, post-tussive vomiting → convalescent.
- Infants <6 months: apnoea and cyanosis, often no whoop; pneumothorax risk.

Infants: apnoea, not whoop.

Diagnosis
- PCR of nasopharyngeal swab early; culture on Regan-Lowe or Bordet-Gengou agar.
- FBC: marked lymphocytosis.

Bordet-Gengou for Bordetella.

Treatment
- Early macrolides (azithromycin), also in neonates (erythromycin: pyloric stenosis).
- Admit infants <6 months with apnoea/cyanosis; prophylactic macrolides for all close contacts.
- Prevent: Tdap in pregnancy + DTaP at 2, 4, 6 months.

Tdap in pregnancy protects the newborn.

II-31 Salmonella infection

Separate the two salmonella diseases: non-typhoidal enterocolitis is usually left alone (antibiotics prolong carriage) unless the child is <3 months, immunocompromised or has sickle cell; typhoid is a systemic febrile illness diagnosed by blood culture and treated with ceftriaxone or azithromycin.

Definition
- Non-typhoidal salmonella enterocolitis and typhoid (enteric) fever.

Causes
- NTS: raw chicken/eggs, unpasteurised milk, reptiles (turtles); large infective dose.
- Typhoid: S. Typhi/Paratyphi, humans the only reservoir; travel to endemic areas.

Turtles carry Salmonella.

Patho-physiology
- Gram-negative, non-lactose fermenting, H_2S-positive rod with type III secretion system.
- Typhoid: Peyer patches → macrophages → bacteraemia; necrosis → perforation in week 3.

Typhoid week 3: perforation.

Clinical picture
- NTS: inflammatory diarrhea with leukocytes; sickle cell → osteomyelitis.
- Typhoid: abdominal pain, stepwise fever, relative bradycardia, rose spots.

Sickle cell osteomyelitis: think Salmonella.

Diagnosis
- NTS: stool culture.
- Typhoid: blood culture in week 1 (bone marrow most sensitive); leucopenia typical.

Typhoid week 1: blood.

Treatment
- NTS low-risk: supportive care; antibiotics prolong faecal carriage.
- Antibiotics only if <3 months, immunocompromised, sickle cell, invasive (sickle cell sepsis: ceftriaxone).
- Typhoid: ceftriaxone or azithromycin; carbapenem for XDR; vaccine.

NTS: don't treat unless risky.

II-32 Influenza in children

Influenza in children is mostly supportive (never aspirin — Reye), but high-risk or severe cases get oseltamivir ideally within 48 hours; watch for 'better then worse' bacterial pneumonia, and remember first-time vaccinees under 9 need two doses.

Definition
- Influenza A/B: orthomyxovirus; drift → epidemics, shift (reassortment) → pandemics.

Shift = Sudden pandemic.

Causes
- Respiratory droplets; children main spreaders.
- High risk: under 5, chronic lung/heart/neuro disease, immunosuppression, aspirin therapy.

Patho-physiology
- Drift = point mutations in hemagglutinin or neuraminidase; HA binds sialic acid, NA cleaves it.
- Ciliated cells destroyed → cytokines; denuded airway → secondary bacterial pneumonia.

HA hooks on, NA lets go.

Clinical picture
- Abrupt fever, myalgia, dry cough; infants mimic sepsis, febrile seizures.
- Influenza B: calf myositis; associated with Guillain-Barré; better then worse → pneumonia.

Flu B: 'B' for calf Bite (myositis).

Diagnosis
- Clinical in season; PCR on nasopharyngeal swab for hospitalised/high-risk.

Treatment
- Supportive: paracetamol/ibuprofen; no aspirin (Reye syndrome).
- Oseltamivir (neuraminidase inhibitor) for high-risk/severe, ideally within 48 hours.
- Annual vaccine from 6 months; first time under 9 years: two doses 4 weeks apart.

Under 9 first time: two doses.

II-33 Routine management of the healthy newborn infant. Disorders of the newborn adaptation.

Most newborns just need drying, delayed cord clamping, skin-to-skin, breastfeeding, vitamin K and screening; the art is spotting failed transition — TTN after caesarean, PPHN by pre-/post-ductal gap, hypoglycaemia in at-risk babies, and jaundice before 24 hours.

Definition
- Care of the healthy newborn and disorders of the transition to extrauterine life.

Causes
- Adaptation disorders: TTN, PPHN, hypothermia, hypoglycaemia, jaundice.
- Hypoglycaemia risk: infants of diabetic mothers, small, preterm, cold, asphyxiated.

IDM: high insulin → low glucose.

Patho-physiology
- Labour catecholamines → ENaC clears lung fluid → aeration; TTN = delayed clearance.
- Brown fat (uncoupling protein 1) gives non-shivering heat.

No labour, wet lungs.

Clinical picture
- TTN: term baby after C-section with grunting; hypoglycaemia mnemonic JOLTS.
- Jaundice within 24 h is always pathological; cephalohematoma does not cross sutures.

Jaundice day 1 = pathological.

Diagnosis
- Apgar 1, 5, 10 minutes; examination within 72 h.
- Screening: pulse oximetry, hearing, heel-prick blood spot.
- Pre-/post-ductal saturation gap → PPHN.

Treatment
- Dry and stimulate; delay cord clamping ≥60 s; skin-to-skin; breastfeed.
- Vitamin K 1 mg IM; HBV vaccine + HBIG within 12 hours if mother HBsAg-positive.
- Hypoglycaemia: feed, dextrose gel, IV dextrose; jaundice: phototherapy; TTN: supportive.

II-34 Respiratory Distress Syndrome (RDS)

RDS is surfactant deficiency in the preterm lung: prevent it with antenatal steroids, support from birth with CPAP, and give surfactant once FiO_2 exceeds 0.30 — while covering for GBS, which can look identical.

Definition
- RDS: surfactant deficiency in the preterm lung (mature levels around week 35).

Causes
- Prematurity (most important), maternal diabetes (fetal insulin), caesarean delivery.

Patho-physiology
- Type II pneumocytes make surfactant; none → alveoli collapse (Laplace, P = 2T/r) → hypoxaemia.
- Prematurity → inadequate surfactant → alveoli fail to open → poor oxygenation.

Laplace: small r, big collapse pressure.

Clinical picture
- Within 4 h of birth: tachypnoea, grunting, nasal flaring, recession, cyanosis.

Diagnosis
- CXR: ground-glass with air bronchograms, atelectasis, low lung volumes (BAD).
- Mature lung: L:S ratio ≥ 2.

BAD: bronchograms, atelectasis, diffuse ground-glass.

Treatment
- Prevent: maternal antenatal glucocorticoids.
- CPAP from birth → artificial surfactant if FiO_2 >0.30.
- Caffeine for apnoea; antibiotics until GBS excluded; complications: NEC, ROP, BPD.

II-35 Meconium aspiration syndrome

Meconium aspiration is a post-term, hypoxia-driven disease: ball-valve air trapping (pneumothorax), surfactant inactivation and PPHN — manage non-vigorous babies with prompt ventilation rather than routine tracheal suction, and escalate surfactant → iNO → ECMO.

Definition
- MAS: respiratory distress from meconium aspiration in term and post-term infants.

Causes
- Post-term (postmaturity), growth restriction, pre-eclampsia, oligohydramnios, abnormal CTG.
- Meconium in a preterm → think Listeria.

Preterm meconium = Listeria.

Patho-physiology
- Fetal hypoxia → increased peristalsis and anal sphincter relaxation → meconium passage → aspiration.
- Ball-valve air trapping → pneumothorax; meconium inactivates surfactant; PPHN.

Ball valve → pneumothorax.

Clinical picture
- Meconium-stained skin/nails/cord; distress from birth; barrel chest, crackles.
- Air leak (pneumothorax) in 10-20%.

Diagnosis
- CXR: coarse patchy infiltrates + hyperinflation, flattened diaphragms.
- Pre-/post-ductal saturation gap → PPHN.

MAS: big lungs; RDS: small lungs.

Treatment
- Non-vigorous baby: no routine tracheal suctioning; start ventilation within the first minute.
- NICU: oxygen, CPAP/ventilation, surfactant, inhaled nitric oxide for PPHN.
- ECMO if refractory; antibiotics until sepsis excluded.

Don't suck, ventilate.

II-36 Primary Pulmonary Hypertension of the Newborn (PPHN)

PPHN is a term baby whose pulmonary vascular resistance never falls, so blood shunts right-to-left at the duct and foramen ovale: think of it when cyanosis is out of proportion to the lungs, prove it with a pre/post-ductal gap and echo, and break the hypoxia-acidosis-vasoconstriction circle with gentle care, oxygen without hyperoxia and iNO.

Definition
- PPHN: pulmonary vascular resistance fails to fall after birth.
- Right-to-left shunt at the PDA and foramen ovale.

PPHN = the baby 'forgot' to switch off fetal circulation.

Causes
- Commonest: meconium aspiration; also sepsis, asphyxia, RDS.
- Maternal SSRIs or NSAIDs in pregnancy; diaphragmatic hernia.

Meconium is the Most common.

Patho-physiology
- Shunted blood bypasses alveoli → FiO_2 barely helps.
- Hypoxia + acidosis → more vasoconstriction → vicious circle.

Vicious circle: hypoxia → constriction → more shunt → more hypoxia.

Clinical picture
- Term baby, cyanosis out of proportion to lung disease, labile saturations.
- Loud single S2, tricuspid regurgitation murmur.

Blue out of proportion + labile sats = think PPHN.

Diagnosis
- Pre-post-ductal gap (>3%); pre-ductal = right hand.
- Echocardiography confirms the shunt and excludes cyanotic heart disease (TAPVD).

Right hand = pre-ductal (before the duct).

Treatment
- Minimal handling, sedation, normothermia; oxygen, avoid hyperoxia; surfactant.
- Inhaled nitric oxide 20 ppm if OI >20–25 (relaxes vessels via cGMP).
- Support BP (milrinone, adrenaline, vasopressin); ECMO if refractory.

Calm, warm, iNO, pressors, then ECMO.

II-37 Perinatal infections

Sort perinatal infections by route (transplacental, intrapartum, postnatal) and you know the bugs, the timing and the prevention: HBV vaccine plus HBIG within 12 hours, intrapartum penicillin for GBS, aciclovir for HSV, no breastfeeding in HIV.

Definition
- Infections passed mother → baby transplacentally, intrapartum or postnatally.
  Three routes: Before, During, After birth.

Causes
- Transplacental: TORCH, parvovirus B19, VZV, Zika.
- Intrapartum: GBS, E. coli, HSV, HBV, HIV, gonococcus, chlamydia.
- Postnatal: NICU lines/hands (CoNS, Staph aureus, Candida), breast milk, contacts.
  Placenta = TORCH; canal = GBS/HSV; NICU = CoNS.

Patho-physiology
- Early pregnancy → organogenesis hit → malformation; later → active disease at birth.
- Parvovirus B19 → fetal anaemia and hydrops; HBV in infants → up to 90% carriers.
  Early = malformed; late = inflamed.

Clinical picture
- Neonatal HSV: seizures, vesicles, hepatitis.
- Gonococcal conjunctivitis 2–5 days; chlamydial 5–14 days; silver nitrate: chemical conjunctivitis.
  Conjunctivitis: chemical day 1, Gono 2–5, Chlamydia 5–14.

Diagnosis
- Antenatal screening: HIV, hepatitis B, syphilis; maternal GBS status.
  Screen mum: HIV, HBV, syphilis, GBS.

Treatment
- HBV: vaccine + HBIG within 12 hours (at latest 24 h); breastfeeding allowed.
- HSV → IV aciclovir; gonococcus → ceftriaxone; chlamydia → azithromycin; GBS → penicillin G prophylaxis.
- HIV → ART, no breastfeeding; peripartum chickenpox → VZIG; no fetal scalp electrodes in HIV/HCV.
  HBV baby: jab + HBIG within 12 h.

II-38 TORCH infections

TORCH infections cross the placenta and hit dividing fetal cells, so they share IUGR, CNS and eye damage; the exam lies in the discriminators: periventricular calcification and deafness for CMV, diffuse calcification and chorioretinitis for toxo, the cataract-deafness-PDA triad for rubella, snuffles for syphilis.

Definition
- Congenital Toxoplasma, Other (syphilis, VZV, parvovirus, Zika), Rubella, CMV, HSV.
  TORCH: Toxo, Other, Rubella, CMV, HSV.

Causes
- Maternal infection crossing the placenta (transplacental); CMV is the commonest; highest risk 1st trimester.
  CMV = Commonest congenital infection.

Patho-physiology
- Cross placenta → attack dividing fetal cells → IUGR, CNS and eye damage.
- Parvovirus B19 → hydrops fetalis; VZV → congenital varicella syndrome (limb hypoplasia).
  Parvo kills RBC precursors → hydrops.

Clinical picture
- CMV: blueberry muffin rash, hepatosplenomegaly, jaundice, periventricular calcification, hearing loss.
- Toxo: chorioretinitis, hydrocephalus, diffuse calcification; rubella: cataract, deafness, PDA.
- Syphilis: snuffles, desquamating rash, saber shins.
  CMV circles the ventricles; toxo scatters everywhere.

Diagnosis
- CMV: urine/saliva PCR within 21 days; 80–90% asymptomatic at birth → hearing and eye follow-up.
- Toxoplasmosis: antibody testing.
  CMV PCR: 3 weeks window.

Treatment
- Symptomatic CMV: valganciclovir 6 months; toxo: pyrimethamine + sulfadiazine + folate.
- Syphilis: penicillin G; VZV: VZIG; rubella prevented by MMR.
- HSV lesions in labour: C-section; acyclovir from 36 weeks.
  Toxo triple: pyrimethamine, sulfadiazine, folate.

II-39 Neonatal and late onset neonatal sepsis

Neonatal sepsis is split at 72 hours: early-onset is vertical (GBS, E. coli, Listeria) and late-onset is nosocomial (CoNS from lines); signs are subtle, so any unwell neonate gets cultures and ampicillin plus gentamicin within an hour, swapping to cefotaxime if meningitis.

Definition
- Early-onset (<72 hours) = vertical; late-onset (>72 h) = nosocomial/community.
  72 hours splits mother's bugs from the ward's bugs.

Causes
- Early: GBS, E. coli, Listeria; risks: prolonged ROM, chorioamnionitis, prematurity.
- Late: coagulase-negative staphylococci (lines), Staph aureus, Gram-negatives, Candida.
  Early = GEL: GBS, E. coli, Listeria.

Patho-physiology
- Small neutrophil reserve, low complement/Ig + lines breaching skin → rapid spread.
  Empty neutrophil bank + open skin door.

Clinical picture
- Subtle: apnoea, temperature instability, grunting, poor feeding, mottling.
  Any 'off' baby is septic until proven otherwise.

Diagnosis
- Blood culture, serial CRP, lumbar puncture.
- GBS identified by maternal vaginal screening.
  Swab mum for GBS.

Treatment
- Prevention: intrapartum penicillin G at least 4 hours before delivery; not late-onset GBS.
- Without meningitis: ampicillin + gentamicin within 1 hour; with meningitis: ampicillin + cefotaxime.
- Prior invasive GBS baby → prophylaxis in every subsequent pregnancy.
  Penicillin ≥4 h before birth; antibiotics within 1 h.

II-40 Central nervous system disorders of the neonates

Neonatal brain injury splits by gestation: preterms bleed (germinal matrix IVH in the first 72 hours) and get watershed PVL leading to spastic diplegia, while term babies suffer HIE, stroke and birth trauma; seizures are subtle, so check glucose and calcium first and give phenobarbital.

Definition
- Preterm: IVH and PVL; term: HIE, stroke, infection, kernicterus, birth trauma.
  Preterm: bleeds and white matter; term: HIE.

Causes
- IVH: preterm <32 weeks, within 72 hours; HIE = commonest seizure cause.
- Subgaleal haemorrhage after vacuum delivery; Erb palsy: lateral traction at delivery.
  Vacuum → subgaleal; traction → Erb.

Patho-physiology
- IVH: fragile germinal matrix bleeds → hydrocephalus.
- Watershed white matter ischaemia → PVL → spastic diplegia.
  PVL hits the leg fibres → spastic diplegia.

Clinical picture
- Neonatal seizures are subtle; IVH: low tone, seizures, rapidly increasing head circumference.
- Scalp swelling crossing sutures + shock → subgaleal haemorrhage; Erb palsy: waiter's tip.
  Erb = waiter's tip.

Diagnosis
- Cranial ultrasound; Papile I–IV; grade IV = periventricular haemorrhagic infarction.
- Seizures: check glucose and calcium first, then sodium, magnesium, infection; EEG.
  Seizure? Sugar and calcium first.

Treatment
- Seizures: phenobarbital first line.
- IVH prevention: antenatal steroids, magnesium, delayed cord clamping; avoid pressure/CO2 swings.
- Erb/Klumpke palsy: spontaneous resolution within 3 months (80%).
  Pheno first; Erb heals 80% in 3 months.

II-41 Skin and umbilical disorders of the neonates

Most newborn skin findings are benign and need only reassurance; the skill is spotting the few that are not (SSSS, HSV, omphalitis, a problem haemangioma) and reading the umbilicus: what leaks out tells you the remnant, and a cord still attached after 3 weeks with neutrophilia means leukocyte adhesion deficiency.

Definition
- Benign newborn rashes, vascular lesions, skin infections and umbilical problems.
  Mostly benign; spot the infection.

Causes
- Benign: erythema toxicum, milia, miliaria, nevus simplex, dermal melanocytosis.
- Infective: bullous impetigo/SSSS, candida, HSV, omphalitis.
  Erythema toxicum, milia, miliaria: all harmless.

Patho-physiology
- SSSS: exfoliative toxin cleaves desmoglein-1 → mucosa spared.
- Omphalitis → umbilical vein → sepsis, portal vein thrombosis.
  Desmoglein-1 = skin only; mucosa spared.

Clinical picture
- Erythema toxicum: sterile pustules full of eosinophils; miliaria: sweat-duct blockage with heat.
- Strawberry hemangioma grows rapidly then regresses; umbilical hernia soft, reducible.
- Urine at navel = patent urachus; faeces = vitelline duct; granuloma: soft, moist, pink.
  Urine = urachus; faeces = vitelline.

Diagnosis
- Cord separates at 5–15 days; delayed + neutrophilia → leukocyte adhesion deficiency (CD18).
- Single umbilical artery → congenital and chromosomal anomalies.
  Late cord + high neutrophils = LAD (CD18).

Treatment
- Benign rashes → reassurance; hemangioma → propranolol if eye, airway, ulcer.
- Granuloma → silver nitrate; umbilical hernia: monitor, resolves by age 5.
- SSSS → IV flucloxacillin; omphalitis → IV antibiotics; HSV → aciclovir.
  Haemangioma near eye/airway → propranolol.

II-42 Necrotising enterocolitis (NEC). Meconium ileus

NEC is the preterm on feeds whose belly distends, stools turn bloody and platelets fall; the X-ray shows pneumatosis and portal gas, and management is bowel rest, decompression and broad antibiotics with surgery for perforation. Meconium ileus is the term-baby obstruction that means cystic fibrosis until proven otherwise.

Definition
- NEC: ischaemic-inflammatory bowel necrosis in preterms after feeds start.
- Meconium ileus: terminal ileum blocked by sticky meconium = cystic fibrosis (~15%).
  Meconium ileus = think CF.

Causes
- NEC: prematurity (biggest risk factor), formula feeding, abnormal flora, poor mucosal perfusion; breast milk protective.
  Breast milk protects; prematurity predisposes.

Patho-physiology
- Immature barrier + abnormal flora + poor perfusion (e.g. hypoxaemia) → bacteria invade wall → pneumatosis.
  Gas in the wall = bacteria in the wall.

Clinical picture
- NEC: abdominal distension, nutritional intolerance, bloody stool; instability, low platelets, acidosis.
- Meconium ileus: distension, bilious vomiting, no meconium.
  NEC triad: distension, intolerance, bloody stool.

Diagnosis
- NEC X-ray: pneumatosis intestinalis, portal venous gas; Rigler sign = perforation.
- Baby cannot be moved: horizontal X-ray from side position, lying on back.
- Meconium ileus: soap-bubble, microcolon; later sweat test.
  Rigler = both bowel wall sides seen = perforation.

Treatment
- NEC: NPO (bowel rest), NG decompression, IV fluids, TPN, IV antibiotics (gram-negative, anaerobes).
- Surgery for perforation, peritonitis, or worsening despite medical therapy.
- Meconium ileus: Gastrografin enema → surgery if it fails.
  NEC: NPO, NG, TPN, antibiotics.

II-43 Congenital diaphragmatic hernia. Oesophageal atresia

CDH and oesophageal atresia are both diagnosed in the delivery room: a scaphoid abdomen with heart sounds on the right means intubate, never bag-mask; drooling and choking at the first feed mean pass an NG tube and watch it coil. CDH is a physiological emergency (pulmonary hypoplasia and hypertension), not a surgical one.

Definition
- CDH: usually left-sided posterolateral Bochdalek hernia.
- OA: proximal blind pouch, usually with distal TEF (85%).
  Bochdalek = Back and Left.

Causes
- CDH: abdominal viscera herniate into chest; OA/TEF: associated VACTERL anomalies (mesodermal defects).
  VACTERL: vertebral, anal, cardiac, TE, renal, limb.

Patho-physiology
- CDH: pulmonary hypoplasia + pulmonary hypertension determine survival.
- OA: fetus can't swallow → polyhydramnios; saliva pools → choking, aspiration.
  OA: can't swallow → polyhydramnios.

Clinical picture
- CDH: respiratory distress, scaphoid abdomen, heart sounds on right.
- OA: excess drooling, choking at first feeding; distal fistula: abdominal distension.
- Cough since birth, always during feeding → H-type TEF.
  CDH: scaphoid belly, heart on the right.

Diagnosis
- OA: firm NG tube coils in pouch on X-ray; gastric gas = distal fistula.
- Pure OA: gasless abdomen; VACTERL screen, echo.
  Coiled tube + gas = OA with distal TEF.

Treatment
- CDH: intubate, no bag-mask, NG tube, gentle ventilation, treat pulmonary hypertension.
- CDH surgery after stabilisation: physiological, not surgical, emergency.
- OA/TEF: surgery after Replogle tube; look for VACTERL.
  CDH: never bag-mask.

II-44 Duodenal stenosis, bowel atresia, omphalocele, gastroschisis

Bilious vomiting in a newborn is obstruction below the ampulla until proven otherwise: a flat abdomen with a double bubble is duodenal atresia (think Down syndrome, exclude malrotation), many distended loops is jejunoileal atresia. For wall defects: a membrane-covered midline sac (omphalocele) brings syndromes; bare bowel right of the cord (gastroschisis) brings young mothers and is otherwise isolated.

Definition
- Congenital gut obstructions (duodenal, jejunoileal, anal atresia) and abdominal wall defects.
  Blocked gut vs gut outside.

Causes
- Duodenal: failed recanalisation, trisomy 21; jejunoileal: in-utero vascular accident.
- Omphalocele: Beckwith-Wiedemann, trisomies 13, 18; gastroschisis: young mothers.
  Double bubble = Down; omphalocele = syndromes.

Patho-physiology
- Obstruction below ampulla → bilious vomiting from first feed, polyhydramnios.
  Below ampulla → green vomit.

Clinical picture
- Duodenal: bilious vomiting, flat abdomen; jejunoileal: distension, many loops.
- Omphalocele: midline, covered by sac; gastroschisis: right of cord, no membrane.
- Omphalocele has other major malformations (cardiac, genitourinary, neural tube).
  Gastroschisis = Gut bare, right side.

Diagnosis
- Duodenal atresia: double bubble on abdominal X-ray; exclude malrotation.
  Double bubble = duodenal atresia.

Treatment
- Duodenal: decompression, fluids → surgical repair (duodenoduodenostomy).
- Wrap bowel in sterile saline dressings, NG tube; primary closure, else staged silo repair.
- High anal atresia: colostomy → PSARP; low → anoplasty.
  High anal atresia → colostomy first.

II-45 Follow-up of the premature baby Bronchopulmonary dysplasia (BPD). Retinopathy of prematurity (ROP).

Following up a preterm means BPD (still needing oxygen or support at 36 weeks PMA), ROP screening (<31 weeks or <1501 g), corrected age for growth and development but chronological age for vaccines, plus iron, vitamin D and RSV prophylaxis.

Definition
- BPD = oxygen or respiratory support at 36 weeks PMA.
- ROP = abnormal retinal vessel growth in preterm babies.
  BPD = still on O2 at 36 weeks.

Causes
- Prematurity plus oxygen toxicity (RIB), ventilator trauma, infection.
  Oxygen helps but harms the preterm.

Patho-physiology
- BPD: arrested alveolar/vascular growth + injury → simplified alveoli, pulmonary hypertension.
- ROP: high oxygen → hypoxic retina → VEGF surge → neovascularisation.
  ROP: oxygen → hypoxia → VEGF → new vessels.

Clinical picture
- BPD: month of oxygen need, ground-glass opacities on X-ray (scarring).
- Follow growth, neurodevelopment (PVL → spastic diplegia), hearing, vision.
  Follow-up: grow, develop, hear, see.

Diagnosis
- Corrected age for growth/development (~2 years); chronological for vaccines.
- ROP screen: <31 weeks or <1501 g; BPD with rising O2 → echo.
  Corrected for milestones; chronological for jabs.

Treatment
- Supplement iron, vitamin D; RSV prevention: nirsevimab or palivizumab.
- BPD prevention: maternal glucocorticoids, surfactant, caffeine; oxygen, nutrition.
- ROP: laser photocoagulation or anti-VEGF.
  ROP: laser or anti-VEGF.

II-46 Lymphomas

Hodgkin is the adolescent with painless rubbery neck nodes, B symptoms and Reed-Sternberg cells (CD15/CD30), cured in more than 90%; childhood NHL is high grade and fast, meaning tumour lysis (Burkitt) and mediastinal airway compromise (lymphoblastic). Biopsy before steroids and never lie a mediastinal mass flat under GA.

Definition
- Hodgkin: adolescent lymphoma with Reed-Sternberg cells; childhood NHL is high grade.

RS cell = owl's eye = Hodgkin.

Causes
- NHL: Burkitt t(8;14) MYC, lymphoblastic, DLBCL, ALCL.
- Hodgkin: nodular sclerosis commonest (70%).

Burkitt: 8 meets 14, MYC.

Patho-physiology
- Hodgkin: reactive inflammatory cells, contiguous spread; cytokines → B symptoms.
- Burkitt lysis → TLS: high K, phosphate, urate, low calcium.

TLS: high K, PO4, urate; low Ca.

Clinical picture
- Painless rubbery cervical/supraclavicular nodes; B symptoms: fever, drenching night sweats, weight loss >10%.
- Burkitt: abdominal mass; lymphoblastic: anterior mediastinal mass.
- Suspicious node: >2 cm, hard, supraclavicular, persistent.

Burkitt = belly; lymphoblastic = chest.

Diagnosis
- Excision biopsy: RS cells CD15, CD30; CXR, PET-CT; staging predicts prognosis.
- No steroids before biopsy.

Biopsy before steroids.

Treatment
- Hodgkin: response-adapted chemotherapy, cure >90%.
- TLS: hyperhydrate, rasburicase (contraindicated in G6PD deficiency).
- Mediastinal mass → avoid general anaesthesia and lying flat.

Mediastinal mass: don't lie them flat.

II-47 Leukaemia

ALL is the commonest childhood cancer (peak 2-5 years): blasts crowding the marrow give pallor, bruising, infection and bone pain. Diagnose with marrow, flow cytometry and genetics, risk-stratify by age, WBC, genetics and above all MRD, and remember the CNS is a sanctuary that needs intrathecal therapy.

Definition
- Commonest childhood cancer; ALL ~80%, peak 2-5 years.

ALL peaks 2-5 years.

Causes
- Risk factor: Down syndrome.

Down syndrome → leukaemia risk.

Patho-physiology
- Blasts fill marrow → anaemia, bleeding, infection; infiltration → bone pain, organomegaly; CNS a sanctuary.

CNS = sanctuary site.

Clinical picture
- Pallor, bleeding, infection, bone pain, HSM, lymphadenopathy; mediastinal mass in T-ALL.

T-ALL = Thymus mass.

Diagnosis
- Bone marrow examination + flow cytometry, genetics; LP.
- Good: age 1-10, WBC <50, hyperdiploidy/ETV6-RUNX1; minimal residual disease strongest.

MRD is the strongest predictor.

Treatment
- Induction, consolidation + intrathecal methotrexate, intensification, maintenance (~2 y); ~90% cure.
- Febrile neutropenia → IV antibiotics within 1 hour; co-trimoxazole vs Pneumocystis.
- High-risk/relapse: allogeneic SCT; graft versus leukaemia = donor cells attack leukaemic cells.

GvL: donor cells attack leukaemia.

II-48 Disorders of platelets, thrombocytopenia, thrombocytosis

Platelet bleeding is skin and mucosa, not joints. A well child with isolated thrombocytopenia weeks after a virus is ITP: no marrow, observe if only skin signs; but anything else abnormal (other cell lines, organomegaly, bone pain) means marrow to exclude leukaemia.

Definition
- Low, dysfunctional or high platelets; bleed into skin and mucosa, not joints.

Platelets: skin/mucosa; factors: joints.

Causes
- Destruction (ITP, NAIT, TTP/HUS, DIC, hypersplenism); underproduction (leukaemia, aplasia).
- Dysfunction, normal count: Glanzmann, Bernard-Soulier, vWD, aspirin.

Normal count, still bleeding → function defect.

Patho-physiology
- ITP: IgG against GPIIb/IIIa → platelets consumed by splenic macrophages.
- TTP: ADAMTS13 deficiency → uncleaved vWF → microthrombi; Glanzmann: aggregation defect; Bernard-Soulier: adhesion.

Glanzmann = Gather fails; Bernard-Soulier = Binding fails.

Clinical picture
- Petechiae, purpura, epistaxis; feared complication intracranial bleeding.
- ITP: well child, purpura 3 weeks after viral illness, no organomegaly.

ITP: well child, post-viral, isolated purpura.

Diagnosis
- ITP: isolated low platelets, large platelets on film, increased megakaryocytes.
- Other cell line low, HSM or bone pain → bone marrow; TTP: raised LDH.

Other line low or HSM → marrow.

Treatment
- ITP: cutaneous only → observation (75-80% resolve in 6 months); mucosal bleeding → corticosteroids.
- Significant bleeding → IVIG; chronic ITP (>12 months): romiplostim/eltrombopag.
- TTP: plasma exchange; thrombocytosis is reactive → no treatment, no aspirin.

ITP: watch skin, treat mucosa.

II-49 Disseminated intravascular coagulation, haemolytic uremic syndrome (HUS)

DIC, HUS and TTP all consume platelets and fragment red cells; the clotting screen splits them: prolonged PT/APTT, low fibrinogen and high D-dimer mean DIC, while HUS and TTP have normal clotting. Typical HUS follows bloody diarrhoea from STEC and is supportive only; atypical HUS gets eculizumab; TTP gets plasma exchange.

Definition
- DIC: consumptive coagulation. HUS triad: microangiopathic haemolytic anaemia, thrombocytopenia, acute kidney injury.
- TTP: deficiency or inhibition of ADAMTS13 (vWF metalloprotease).

HUS triad: MAHA, low platelets, AKI.

Causes
- DIC: sepsis, shock, trauma, malignancy.
- Typical HUS follows bloody diarrhoea from Shiga toxin-producing E. coli (O157:H7).
- Atypical HUS: alternative complement pathway dysregulation.

Typical = toxin; atypical = complement.

Patho-physiology
- DIC: tissue factor → microthrombi + consumption + fibrinolysis → clots and bleeds.
- Shiga toxin binds Gb3 receptors on glomerular endothelium → microthrombi → RBC fragmentation, AKI.

Gb3 = toxin's door on the glomerulus.

Clinical picture
- DIC: bleeding plus thrombosis.
- HUS: anaemia, low platelets, AKI after bloody diarrhoea.

DIC = clot and bleed at once.

Diagnosis
- DIC labs: low platelets, prolonged PT and APTT, low fibrinogen, high D-dimer.
- Clotting times are normal in HUS but prolonged in DIC; TTP has raised LDH.

Normal PT/APTT = HUS, not DIC.

Treatment
- DIC: treat the cause; platelets, FFP, cryo only if bleeding.
- Typical HUS: supportive care, dialysis; avoid antibiotics and antimotility drugs.
- Atypical HUS → eculizumab; TTP → plasma exchange, glucocorticoids, rituximab.

Atypical HUS: eculizumab blocks C5.

II-50 Porphyria

Porphyrias are haem-synthesis blocks: early precursors (ALA, PBG) are neurotoxic and give acute attacks (AIP: abdominal pain, neuropathy, psychiatric change, dark urine, no photosensitivity), while late porphyrins give photosensitivity (PCT, EPP). Screen an attack with urinary PBG and treat with haem arginate and glucose; avoid P450 inducers.

Definition
- Inherited haem-synthesis enzyme defects; precursors before the block accumulate.

Block → backlog before the block.

Causes
- Acute intermittent porphyria (AIP): autosomal dominant, deficiency of porphobilinogen deaminase (HMBS).
- Porphyria cutanea tarda: UROD deficiency, most common porphyria.
- Commonest in children: erythropoietic protoporphyria, ferrochelatase deficiency. Triggers: P-450 inducers, barbiturates contraindicated.

Barbiturates trigger AIP.

Patho-physiology
- Early haem precursors (ALA, PBG) are neurotoxic → acute attacks.
- Late porphyrins cause photosensitivity.

Early = nerves; late = light.

Clinical picture
- AIP mnemonic Wine In PAIN: wine-coloured urine, inducers, psychological disturbance, abdominal pain, neuropathy; no photosensitivity.
- Hyponatraemia in an attack is due to SIADH.
- PCT: blistering cutaneous photosensitivity and hyperpigmentation.

Wine In PAIN.

Diagnosis
- Screening test in an acute attack: urinary porphobilinogen; also raised ALA.

Attack → check urine PBG.

Treatment
- Acute attack: IV haem arginate (haemin) and glucose, which inhibit ALA synthesis.
- PCT: low-dose hydroxychloroquine or phlebotomy.

Haem + glucose shut ALA synthase.

II-51 Histiocytosis

Two different histiocytoses: LCH is a clonal (BRAF V600E) neoplasm of CD1a+/CD207+ Langerhans cells that punches holes in bone and can cause central DI; HLH is a cytokine storm from failed perforin killing — think fever, pancytopenia, splenomegaly and sky-high ferritin.

Definition
- LCH: proliferation of Langerhans cells; HLH: systemic overactivation of macrophages and cytotoxic T cells.

LCH = Langerhans cells grow; HLH = Hyper-Lymphohistiocytic Hyperinflammation.

Causes
- LCH: BRAF V600E, peak 1-3 years; Hand-Schüller-Christian often in children over 3.
- HLH: primary genetic or secondary to EBV, systemic JIA.

BRAF in LCH; EBV and sJIA (macrophage activation syndrome) trigger secondary HLH.

Patho-physiology
- LCH cells are positive for CD1a and CD207 (langerin); infiltrate bone, skin, pituitary, organs.
- HLH: failed perforin cytotoxicity → cytokine storm.

LCH cells: CD1a + CD207 (Birbeck granules). HLH: no perforin = no off-switch.

Clinical picture
- LCH: lytic (punched-out) skull lesion, painful lytic long-bone lesion with normocalcaemia, skin rash.
- Pituitary involvement causes central diabetes insipidus.
- HLH: fever, pancytopenia, hepatosplenomegaly.

LCH triad (Hand-Schüller-Christian): skull lesions, DI, exophthalmos.

Diagnosis
- LCH: biopsy + staging; risk organs are liver, spleen and bone marrow.
- HLH shows very high ferritin, high triglycerides, low fibrinogen.

HLH labs: ferritin sky-high, TG high, fibrinogen low.

Treatment
- Multisystem LCH needs systemic chemotherapy; single bone lesion may need only curettage or observation.
- HLH: immunosuppression (steroids ± chemotherapy); primary (familial) HLH needs HSCT.

Single bone: scrape or wait. Familial HLH: transplant.

II-52 Central nervous system tumours in children

Brain tumours are the commonest solid tumours of childhood and about half sit in the posterior fossa — so the classic story is morning headache and vomiting from obstructive hydrocephalus plus ataxia; get an urgent MRI of brain and spine and never do an LP first.

Definition
- CNS tumours are the commonest solid tumours of childhood.
- In children, primary CNS tumours are usually infratentorial; about half are posterior fossa.

Kids: tumours in the back (posterior fossa); adults: tumours on top.

Causes
- Pilocytic astrocytoma: commonest primary CNS tumour; good prognosis.
- Medulloblastoma (primitive neuroectodermal, cerebellar granule cells); ependymoma in the 4th ventricle.
- Turcot syndrome: FAP + malignant CNS tumours. Pineal tumours → Parinaud syndrome.

Pilocytic = Pleasant prognosis; Medullo = Malignant midline cerebellum.

Patho-physiology
- Posterior fossa mass → obstructive hydrocephalus → raised ICP.
- Medulloblastoma grows rapidly, spreads via CSF (drop metastasis); pilocytic = cyst + mural nodule.

Medullo 'drops' down the spine via CSF.

Clinical picture
- Raised ICP headache worst on waking (nocturnal CO_2 retention → vasodilatation); vomiting.
- Ependymoma: relief in the knee-chest position. Parinaud: upward gaze palsy, convergence-retraction nystagmus.

Morning headache + vomiting = ICP until proven otherwise.

Diagnosis
- Urgent MRI of brain and spine, and no lumbar puncture with raised ICP.
- Ependymoma: perivascular pseudorosettes; medulloblastoma: MYC amplification.

Raised ICP: scan, never tap.

Treatment
- Maximal safe resection; dexamethasone, CSF diversion for hydrocephalus.
- Cranial radiotherapy is avoided under about 3 years; chemotherapy; lifelong late-effect follow-up.

Under 3: spare the brain from radiation.

II-53 Abdominal malignancies - Wilms tumour, neuroblastoma

Well toddler with a smooth flank mass that stays on one side = Wilms; sick, pale toddler with a hard mass crossing the midline, raccoon eyes or dancing eyes = neuroblastoma (check urinary VMA/HVA). Any abdominal mass gets same-day review and ultrasound.

Definition
- Neuroblastoma: arises from sympathetic neural crest cells, most often the adrenal medulla.
- Wilms tumour: malignant kidney tumour of metanephric mesenchyme (blastema).

Neuro = nerve crest (adrenal); Wilms = kidney blastema.

Causes
- Neuroblastoma: N-myc (MYCN) amplification is the strongest adverse marker.
- Wilms: 3-4 years; WAGR = Wilms, aniridia, genitourinary anomalies, developmental delay.

WAGR = Wilms, Aniridia, GU anomalies, Retardation.

Patho-physiology
- Neuroblastoma: small blue round cells, Homer-Wright rosettes; markers bombesin and NSE.
- Wilms spreads most often to the lungs.

Clinical picture
- Neuroblastoma: abdominal mass, periorbital ecchymoses (orbital bone metastases), opsoclonus-myoclonus-ataxia.
- Wilms: smooth flank mass that does not cross the midline in a well child aged 3-4 years.

Sick child + crossing mass = neuroblastoma; well child + flank mass = Wilms.

Diagnosis
- Any abdominal mass → same-day review + ultrasound (Wilms: abdominal ultrasound).
- Neuroblastoma: urinary VMA and HVA raised; MIBG, marrow, biopsy.

VMA/HVA in urine = neuroblastoma.

Treatment
- Neuroblastoma high-risk: chemo, surgery, stem cell rescue, radiotherapy, anti-GD2.
- Wilms: Europe (SIOP) gives chemotherapy before surgery; the US (COG) usually operates first.

SIOP = Shrink first (chemo); COG = Cut first.

II-54 Malignant bone and soft tissue tumours

Osteosarcoma = teenage metaphysis (knee), sunburst/Codman triangle, radio-resistant; Ewing = diaphysis, onion-skin, t(11;22), mimics osteomyelitis, radiosensitive; rhabdomyosarcoma = commonest soft tissue sarcoma (botryoides in girls). All like to metastasise to the lungs.

Definition
- Osteosarcoma: malignant proliferation of osteoblasts; peak 10-20 years.
- Ewing sarcoma: neuroectoderm-derived; rhabdomyosarcoma: commonest soft tissue sarcoma of childhood.

Osteo = teens' growth spurt.

Causes
- Ewing carries t(11;22) (EWSR1-FLI1); boys.
- Osteosarcoma: hereditary retinoblastoma, Li-Fraumeni, prior radiation.
- Embryonal rhabdomyosarcoma (sarcoma botryoides): girls.

Ewing = 11;22 (sum 33). Osteo = RB + Li-Fraumeni.

Patho-physiology
- Osteosarcoma arises in the metaphysis of long bones, often knee; cells make osteoid.
- Ewing: small round blue cells in the diaphysis (femur).
- Commonest metastasis site: lungs.

Osteo = metaphysis, Ewing = diaphysis (shaft).

Clinical picture
- Osteosarcoma: painful enlarging mass, pathological fracture, bone pain with swelling.
- Ewing may mimic osteomyelitis. Sarcoma botryoides: grape-like vaginal mass with bleeding.
- Soft tissue lump >2 cm, deep, enlarging or painful → urgent ultrasound.

Night bone pain + swelling = image it.

Diagnosis
- Osteosarcoma X-ray: sunburst and Codman triangle; Ewing: onion-skin periosteal reaction.
- Rhabdomyoblasts stain for desmin and myogenin; soft tissue sarcoma → core needle biopsy.

Sunburst = osteo; onion skin = Ewing.

Treatment
- Osteosarcoma: en bloc resection and chemotherapy; radio-resistant.
- Ewing is radiosensitive: chemo + surgery and/or radiotherapy.
- Soft tissue sarcoma: wide excision, chemotherapy, radiation.

Ewing radiates well; Osteo resists.

II-55 Early and late onset adverse effects of chemotherapy

Chemotherapy hits fast-dividing tissue (marrow nadir 7–14 days → febrile neutropenia = antibiotics within 1 hour) and each drug has a signature organ toxicity: anthracycline-heart, cisplatin-ears/Mg, cyclophosphamide-bladder, vincristine-nerves/gut, bleomycin-lungs, etoposide-secondary AML.

Definition
- Chemotherapeutic drugs are effective against rapidly dividing cells → early and late toxicities.

Causes
- High-turnover tissues (marrow, mucosa, hair) plus drug-specific organ toxicity.
- Etoposide (topoisomerase II inhibitor) is linked to secondary AML.

Patho-physiology
- Marrow nadir 7–14 d → neutropenia → infection risk.
- Tumour lysis syndrome: high potassium, high phosphate, high urate, low calcium.

TLS: everything up except calcium.

Clinical picture
- Anthracyclines → dilated cardiomyopathy; cisplatin → high-frequency sensorineural hearing loss, renal magnesium wasting.
- Cyclophosphamide/ifosfamide → haemorrhagic cystitis (acrolein); vincristine → peripheral neuropathy, constipation, paralytic ileus.
- Bleomycin → lung fibrosis, mucositis; vinblastine → myelosuppression.

Doxo = heart, Cis = ears/kidney, Cyclo = bladder, Vincristine = nerves, Bleo = lungs.

Diagnosis
- Febrile neutropenia = emergency → cultures.
- Late effects: cardiac, gonadal, endocrine, neurocognitive and second cancers.

Treatment
- Febrile neutropenia: IV broad-spectrum antibiotics within 1 hour.
- Protect with dexrazoxane (anthracyclines) and mesna (cyclophosphamide); rasburicase contraindicated in G6PD deficiency.
- Vincristine is fatal if given intrathecally; lifelong follow-up.

Mesna = bladder; dexrazoxane = heart.

Section III: Organ systems (55 topics)

III-1 Diabetes mellitus. Management of diabetic ketoacidosis

New type 1 diabetes is a same-day referral, not an outpatient test. In DKA: fluids first, then insulin 0.05–0.1 U/kg/h with no bolus, replace potassium (total body K^+ is always depleted), add dextrose as glucose falls — and watch for cerebral oedema (headache, bradycardia), the leading cause of death.

Definition
- Type 1 DM: autoimmune beta-cell destruction → absolute insulin deficiency; peaks 4–6 y, puberty.
- Four symptoms: polyuria, polydipsia, weight loss despite good appetite, fatigue.

4 Ws: Wee, Wa(ter), Weight loss, Weary.

Causes
- Autoimmune: antibodies to GAD, IA-2, ZnT8, insulin.
- Commonest associated autoimmune diseases: coeliac disease and Hashimoto thyroiditis.

Type 1 brings friends: coeliac + Hashimoto.

Patho-physiology
- No insulin → hyperglycaemia → osmotic diuresis → dehydration.
- Ketone production → acidosis → Kussmaul breathing; total body K^+ depleted.

Clinical picture
- Polyuria, polydipsia, weight loss despite good appetite, fatigue; secondary enuresis.
- DKA: vomiting, abdominal pain, Kussmaul breathing, ketotic breath, drowsiness.

Kussmaul = deep breaths to blow off acid.

Diagnosis
- Typical symptoms + high glucose (18.2 mmol/L) → paediatric diabetes centre immediately.
- DKA: glucose >11 mmol/L, venous pH, plus ketones.

Treatment
- DKA: fluids, then insulin infusion 0.05–0.1 U/kg/h with no bolus; potassium, dextrose.
- Leading cause of death: cerebral oedema (headache, bradycardia) → mannitol or 3% saline.
- Most modern: insulin pump with an integrated glucose sensor.

No bolus, go slow: protect the brain.

III-2 Metabolic syndrome. Type 2 diabetes mellitus

Obese pubertal teen with acanthosis nigricans, negative islet antibodies and high C-peptide = type 2 diabetes: metformin plus lifestyle, insulin if ketotic or HbA1c ≥8.5%. A slim child with a three-generation family history is MODY — and genotype changes treatment.

Definition
- Metabolic syndrome: central obesity + ≥ 2 of high TG, low HDL, HTN, high fasting glucose.
- T2DM: obese pubertal teen; islet autoantibodies negative, high C-peptide.

T2 teen: fat, antibody-free, C-peptide high.

Causes
- Central obesity, puberty, family history of T2DM, high-risk ethnic group.
- MODY: autosomal dominant, 3 generations, not obese.

MODY: 3 generations, no obesity.

Patho-physiology
- Visceral fat → insulin resistance → hyperinsulinaemia + free fatty acids → the cluster.
- Insulin resistance → acanthosis nigricans, fatty liver, PCOS.

Clinical picture
- Thick, dark, velvety skin in neck and axillae: acanthosis nigricans.
- Often asymptomatic on screening; polyuria, polydipsia, occasionally DKA; raised ALT, sleep apnoea, SCFE.

Diagnosis
- Type 1 vs type 2: islet autoantibodies and C-peptide in every new case.
- GCK-MODY: mild stable fasting hyperglycaemia. Diabetes <6 months → genetic testing.

Diabetes <6 months old = genes, not autoimmunity.

Treatment
- First-line drug without ketosis: metformin plus lifestyle; insulin if ketosis or HbA1c ≥ 8.5%.
- HNF1A-MODY very sensitive to sulfonylureas; GCK-MODY needs no treatment.
- Screen complications from diagnosis.

GCK: leave it; HNF1A: sulfonylurea.

III-3 Diabetes insipidus

Polyuria with high plasma osmolality and dilute urine (<300 mOsm/kg) is diabetes insipidus; the desmopressin step separates central (urine concentrates >50%) from nephrogenic (no response). Central DI always needs an MRI looking for craniopharyngioma, germinoma or LCH.

Definition
- Central DI = deficient vasopressin (ADH); nephrogenic DI = renal resistance to it.
- Dilute polyuria.

Central = no signal; nephrogenic = deaf kidney.

Causes
- Central: craniopharyngioma, germinoma, LCH, trauma, neurosurgery, meningitis.
- Nephrogenic: X-linked AVPR2, lithium, hypercalcaemia, hypokalaemia.

Nephrogenic: Lithium, high Ca, low K.

Patho-physiology
- No ADH or V2 resistance → no aquaporin-2 in collecting duct → dilute urine.
- Free water loss → ↑ plasma osm → thirst; infants → hypernatraemic dehydration.

Clinical picture
- Polyuria >2 L/m²/day, nocturia, secondary enuresis.
- Intense thirst, prefers cold water; infant FTT, vomiting, fever.

Diagnosis
- DI confirmed by high plasma osmolality with urine osmolality below 300 mOsm/kg.
- After water deprivation, desmopressin concentrates urine by >50% in central DI, not nephrogenic.
- Central DI needs MRI: craniopharyngioma, germinoma, Langerhans cell histiocytosis.

Desmopressin works = central; no response = nephrogenic.

Treatment
- Central → desmopressin; main risk is hyponatraemia.
- Nephrogenic → low-solute diet and thiazide diuretics; stop lithium.
- Correct hypernatraemia ≤ 0.5 mmol/L per hour.

III-4 Disorders of neuro- and adenohypophysis

Hypopituitarism is congenital (midline defects, e.g. septo-optic dysplasia — neonatal hypoglycaemia, micropenis, prolonged jaundice) or acquired (craniopharyngioma, irradiation). GH deficiency = slow growth with delayed bone age and normal proportions; and always replace cortisol before thyroxine.

Definition
- Hypo- or hyperfunction of anterior (GH, ACTH, TSH, LH, FSH, prolactin) or posterior (vasopressin) pituitary.

Anterior 'FLAT PiG': FSH, LH, ACTH, TSH, Prolactin, GH.

Causes
- Congenital: midline defects - septo-optic dysplasia, cleft palate, single central incisor.
- Acquired: craniopharyngioma, cranial irradiation, trauma, histiocytosis, infection.

Midline face/brain defect = check pituitary.

Patho-physiology
- GH deficiency → low growth velocity, delayed bone age, normal proportions.
- Craniopharyngioma compresses the optic chiasm → bitemporal hemianopia; DI.

Clinical picture
- Newborn hypopituitarism: hypoglycaemia, micropenis, prolonged jaundice.
- Prolactinoma: galactorrhoea, amenorrhoea. Weight gain with falling height velocity suggests Cushing syndrome.
- Posterior: DI, or SIADH.

Fat + slowing height = Cushing; fat + tall = simple obesity.

Diagnosis
- Low IGF-1 → failure of two GH stimulation tests → pituitary MRI.
- Craniopharyngioma: suprasellar, cystic and calcified.

Treatment
- GHD → recombinant GH until final height.
- Always replace cortisol before thyroxine.
- Prolactinoma: cabergoline; craniopharyngioma → surgery ± radiotherapy; SIADH → fluid restriction.

Steroids before thyroid.

III-5 Congenital adrenal hyperplasia (CAH)

CAH is autosomal recessive 21-hydroxylase deficiency in ~95%: girls present with virilised genitalia at birth, boys look normal and crash at 1–3 weeks with salt-wasting (low Na, high K, low glucose, shock). Raised 17-OHP clinches it; IV hydrocortisone, saline and dextrose save lives.

Definition
- CAH: AR defect of cortisol synthesis; ~95% 21-hydroxylase deficiency, commonest adrenocortical enzyme defect.

21 = 95%.

Causes
- Autosomal recessive CYP21A2 mutation.
- Rare: 11β-hydroxylase deficiency (virilisation and hypertension), 17α-hydroxylase.

11β: virilised + hypertensive.

Patho-physiology
- Low cortisol → no feedback → ACTH up → adrenal hyperplasia.
- Salt-wasting: insufficient glucocorticoid and mineralocorticoid, excessive androgen.

Clinical picture
- Girls: virilised genitalia at birth.
- Boys: salt-wasting crisis at 1–3 weeks - hyponatraemia, hyperkalaemia, low glucose, shock.
- Simple virilising: precocious pseudopuberty.

Girls: genitals at birth. Boys: crisis week 1–3.

Diagnosis
- Diagnostic marker: raised 17-hydroxyprogesterone (newborn screening); high ACTH and renin.
- Karyotype, pelvic ultrasound, CYP21A2 genetics.

Treatment
- Crisis: IV hydrocortisone, 0.9% saline 10–20 mL/kg, IV dextrose.
- Long-term: hydrocortisone, fludrocortisone and salt in infancy.
- Stress dosing in illness; emergency card.

Sick day = double dose; carry the card.

III-6 Thyroid diseases

Congenital hypothyroidism (mostly dysgenesis/ectopic gland) is caught by heel-prick TSH on day 3–5 and must be on levothyroxine 10–15 µg/kg/day within 2 weeks to protect the brain. In older children Hashimoto is the commonest acquired cause; Graves is TRAb-driven and treated with methimazole — sore throat or fever means an urgent blood count.

Definition
- Congenital or acquired hypothyroidism; hyperthyroidism mostly Graves (Basedow's disease, TRAb-driven).

Causes
- Commonest congenital cause: thyroid dysgenesis (~85%), most often an ectopic gland.
- Maternal iodine deficiency → temporary primary hypothyroidism in the newborn.
- Acquired: Hashimoto thyroiditis, commonest in children.

Patho-physiology
- No hormone → slowed metabolism, growth and brain development → delayed bone age.
- TSH-receptor stimulating antibodies → gland overdrive → thyrotoxicosis.

Clinical picture
- Neonate: prolonged jaundice, constipation, hypotonia, macroglossia, umbilical hernia.
- Hashimoto: painless goitre, growth failure.
- Basedow's: weight loss, tachycardia, Graefe sign.

Basedow 3: weight loss, tachycardia, Graefe.

Diagnosis
- Heel-prick TSH day 3–5.
- Hashimoto: high TSH, low free T4, anti-TPO. Graves: suppressed TSH, high T4/T3, TRAb.

Treatment
- Congenital: levothyroxine 10–15 µg/kg/day within the first 2 weeks.
- Graves: methimazole + propranolol; propylthiouracil is avoided because of severe hepatotoxicity.
- Sore throat/fever → urgent blood count.

Sore throat on methimazole = check neutrophils.

III-7 Disorders of calcium and phosphorus homeostasis. Rickets.

Nutritional rickets is a vitamin D problem of unsupplemented, dark-skinned breastfed infants: low phosphate, very high ALP and high PTH, with craniotabes, rosary, wide wrists and bowed legs. Normal PTH with low phosphate points to XLH (FGF23). Prevent with 400 IU vitamin D daily.

Definition
- Rickets = failed growth-plate mineralisation; mostly vitamin D deficiency in unsupplemented breastfed infants.

Causes
- Commonest: vitamin D deficiency - unsupplemented breastfeeding, dark skin, little sunlight.
- Phosphopenic: X-linked hypophosphataemic rickets (PHEX, high FGF23), Fanconi syndrome.

XLH: PHEX broken, FGF23 high, phosphate lost.

Patho-physiology
- Low vitamin D → less Ca/P absorption → PTH rises → phosphate wasting.
- Low Ca × P → growth plate cannot mineralise → deformity.

Clinical picture
- Soft, ping-pong-ball skull bones = craniotabes; rachitic rosary, Harrison sulcus, wide wrists, bowed legs.
- Infants <6 mo: hypocalcaemic seizures, tetany, laryngospasm.

Rosary + Harrison + wide wrists + bowlegs.

Diagnosis
- Nutritional rickets bloods: phosphate low, ALP very high, PTH high.
- XLH: low phosphate with normal calcium and normal PTH.
- X-ray: cupping, fraying and splaying of the metaphyses.

XLH: phosphate low, everything else normal.

Treatment
- Prevent: all infants receive 400 IU of vitamin D daily.
- Nutritional: high-dose vitamin D ≥ 12 weeks + calcium; XLH: phosphate + calcitriol or burosumab.
- Hypocalcaemia that won't correct with IV calcium - check magnesium.

Calcium won't rise? Check magnesium.

III-8 Systemic lupus erythematosus (SLE)

Childhood SLE is an adolescent-girl disease that is more severe than adult lupus — nephritis in most. ANA ≥1:80 is the entry ticket, anti-dsDNA tracks activity, anti-Sm is most specific, low C3/C4 means activity, and a high CRP in a febrile lupus patient means infection until proven otherwise. Everyone gets hydroxychloroquine.

Definition
- Multisystem autoimmune disease: immune complexes against nuclear antigens.
- cSLE: adolescent girls, more severe than adult disease.

Causes
- Female predominance after puberty; more common in non-white populations.
- Anti-histone antibodies suggest drug-induced lupus (hydralazine, procainamide, isoniazid).

Drug lupus: 'HIP' - Hydralazine, Isoniazid, Procainamide.

Patho-physiology
- Immune complexes deposit → organ damage; in active SLE complement C3 and C4 fall.
- Maternal anti-Ro/La antibodies cause congenital heart block.

Clinical picture
- Fever, malar rash sparing nasolabial folds, photosensitivity, painless oral ulcers.
- Non-erosive arthritis, serositis, nephritis 50–80%, seizures/psychosis.
- Coombs-positive haemolysis, leucopenia, thrombocytopenia.

Diagnosis
- Entry criterion: ANA at a titre of at least 1:80.
- Anti-dsDNA tracks activity and nephritis; anti-Sm most specific.
- High CRP in a febrile lupus patient suggests infection, not flare.

ANA = screen; Sm = specific; dsDNA = disease activity.

Treatment
- Every SLE patient takes hydroxychloroquine + sun protection.
- Moderate → azathioprine, methotrexate or mycophenolate.
- Severe → IV methylprednisolone → mycophenolate or IV cyclophosphamide; rituximab if refractory.

III-9 Dermatomyositis, scleroderma

JDM is a complement-mediated capillary vasculopathy of skin and muscle: proximal weakness with Gowers sign, heliotrope rash and Gottron papules, calcinosis far more than in adults, and no routine cancer screening. Steroids plus methotrexate first. Paediatric scleroderma is usually localised (linear) and treated with methotrexate.

Definition
- JDM: autoimmune small-vessel vasculopathy of skin and muscle; commonest childhood myopathy.
- Commonest scleroderma in children: localised (linear); systemic sclerosis rare.

Causes
- JDM peaks 5-10 y, girls more often.

Patho-physiology
- Complement deposition → capillary loss in muscle and skin → weakness, rash, ulcers.
- Rarely linked to malignancy (unlike adults) → no routine cancer screening.

Adult DM: look for cancer; child DM: don't.

Clinical picture
- Weakness of the proximal muscles of the limbs and neck flexors; Gowers sign.
- Purple eyelids = heliotrope rash; red scaly papules over knuckles = Gottron papules.
- Calcinosis much commoner than in adult dermatomyositis.

Gottron on knuckles, heliotrope on lids.

Diagnosis
- CK, LDH, AST/ALT, aldolase + MRI (T2/STIR) muscle oedema.
- Biopsy: perifascicular atrophy. SSc antibodies: anti-Scl-70, anticentromere.

Treatment
- First-line JDM: steroids plus methotrexate; IVIG if severe.
- Localised scleroderma: methotrexate ± initial steroids; physiotherapy.
- SSc: CCB for Raynaud, ACE inhibitor for renal crisis.

III-10 Vasculitis. Henoch-Schönlein purpura

HSP (IgA vasculitis) is the commonest childhood vasculitis: palpable purpura on buttocks and legs with a normal platelet count, plus joints, gut and kidneys. Treatment is supportive; steroids ease severe gut pain but do not prevent nephritis — so check urine and BP for 6-12 months.

Definition
- Commonest vasculitis of childhood: Henoch-Schönlein purpura (IgA vasculitis), small vessel, 3-10 years.

Causes
- Follows upper respiratory infection (often streptococcal); autumn and winter.
- Takayasu is large-vessel, Kawasaki medium-vessel; granulomatosis with polyangiitis: c-ANCA (PR3).

Takayasu big, Kawasaki medium, HSP small.

Patho-physiology
- IgA small-vessel vasculitis → palpable purpura with a normal platelet count.
- Gut → colic, bleeding, ileo-ileal intussusception; kidney → haematuria, proteinuria.

Purpura + normal platelets = vasculitis.

Clinical picture
- Palpable purpura on buttocks and legs (always).
- Arthritis of ankles/knees (~75%); colicky abdominal pain (~50-75%).
- Nephritis (~20-50%) usually within 4-6 weeks.

Diagnosis
- Clinical: purpura (mandatory) + abdominal pain, IgA histology, arthritis or renal involvement.
- FBC/clotting normal; urine protein-creatinine ratio, BP; US if severe pain.

Treatment
- Supportive: rest, paracetamol (acetaminophen), fluids; NSAIDs only if kidneys normal.
- Steroids treat severe abdominal pain but do not prevent nephritis.
- Monitor urine and blood pressure for 6-12 months.

III-11 Urinary tract infections, localisation, course, treatment, complications, VUR

In a febrile infant, think UTI until a clean urine says otherwise: infants present non-specifically, under-3-months go in for IV cephalosporin, and upper-tract infection is what scars kidneys, so look for VUR/obstruction behind recurrent pyelonephritis.

Definition
- UTI: cystitis (bladder mucosa) vs pyelonephritis (renal parenchyma).

Bladder = burning; kidney = fever.

Causes
- Most frequent pathogen: E. coli (P fimbriae); Klebsiella, Proteus, Enterococcus.
- Risks: VUR, posterior urethral valves, constipation, dysfunctional voiding.

E. coli: P for Pyelo-fimbriae.

Patho-physiology
- Bowel flora ascends; primary VUR is caused by a short intramural ureteric tunnel.
- Pyelonephritis → fever, high CRP → renal scarring → HTN, CKD.

Short tunnel = leaky valve.

Clinical picture
- Infants: non-specific - fever, vomiting, poor feeding, jaundice, sepsis.
- Older: dysuria, frequency (cystitis) vs fever, loin pain (pyelonephritis).

Febrile infant, no focus: check the urine.

Diagnosis
- Clean/catheter sample: dipstick nitrite + leucocyte esterase, culture.
- Pyelonephritis sediment: leucocytes, white blood cell casts and bacteria.
- VUR: MCU, sonocystography, dynamic kidney scintigraphy; DMSA scan 4-6 months later shows scarring.

Casts = kidney; DMSA = scars.

Treatment
- Infants under 3 months: admit, IV antibiotics (cefotaxime/ceftriaxone).
- Pyelonephritis: oral cefalexin or co-amoxiclav 7-10 days; cystitis 3 days.
- VUR: fix bowel/bladder, prophylaxis; surgery for breakthrough infections.

Under 3 months = IV; pyelo 7-10, cystitis 3.

III-12 Nephrolithiasis in childhood

A child with a stone has a metabolic disease until proven otherwise: image with ultrasound, analyse the stone, and do a full blood/urine metabolic screen in every child - hypercalciuria is the commonest culprit.

Definition
- Urinary tract stones in children; mostly metabolic.

Child stone = look for metabolism.

Causes
- Commonest metabolic abnormality: hypercalciuria; also hypocitraturia, hyperoxaluria, cystinuria, distal RTA.
- Infection (Proteus), anatomical stasis, drugs (furosemide in preterms, topiramate).

Calcium tops the list.

Patho-physiology
- Excess solute + little water + low citrate (main inhibitor) → supersaturation → crystals.
- Struvite forms in alkaline urine via urease-producing organisms such as Proteus.

Citrate is the stone's enemy.

Clinical picture
- Adolescents: renal colic flank → groin, nausea, haematuria.
- Young children: atypical - vague pain, irritability, UTI, painless haematuria.
- Hexagonal crystals in urine indicate cystinuria.

Hexagon = cystine.

Diagnosis
- First-line imaging: ultrasound; KUB for calcium stones; low-dose CT only if unclear.
- Metabolic work-up in every child: blood + urine calcium, oxalate, citrate, cystine; stone analysis.

US first, CT last.

Treatment
- NSAIDs + paracetamol, hydration; obstructed infected kidney → urgent drainage.
- <5 mm usually pass (tamsulosin); else ESWL, ureteroscopy, PCNL for staghorn.
- Prevent: fluids, low salt, thiazide (reduces urinary calcium), potassium citrate, tiopronin for cystine.

Obstructed + infected = drain now.

III-13 Glomerular diseases

Timing and complement sort nephritic children: haematuria 1-2 days after an URTI with normal C3 is IgA; 1-3 weeks after pharyngitis (3-6 after impetigo) with low C3 that recovers by 6-8 weeks is PSGN.

Definition
- Glomerulonephritis: glomerular inflammation → haematuria ± proteinuria; nephritic syndrome.

Nephritic = blood; nephrotic = protein.

Causes
- Most frequent type: IgA nephropathy; PSGN after group A strep.
- Others: lupus, MPGN, HSP nephritis, ANCA, anti-GBM.
- Haematuria with sensorineural deafness suggests Alport syndrome.

Alport: can't pee, can't see, can't hear.

Patho-physiology
- Immune complexes break GBM → red cells + casts, proteinuria.
- ↓ GFR + ongoing Na^+ reabsorption → oedema, hypertension, oliguria.

Leaky filter + salt retention.

Clinical picture
- Cola urine, periorbital oedema, hypertension, oliguria.
- IgA: macroscopic haematuria 1-2 days after URTI. PSGN: 1-3 weeks after pharyngitis, 3-6 weeks after impetigo.

IgA = days; PSGN = weeks.

Diagnosis
- Urine: red blood cells with red blood cell casts, most commonly with proteinuria.
- IgA: normal C3. PSGN: C3 low, normalises by 6-8 weeks; ASO/anti-DNase B.

Low C3 = PSGN; normal C3 = IgA.

Treatment
- PSGN: supportive - fluid/salt restriction, furosemide, antihypertensives, penicillin.
- IgA: ACE inhibitor; steroids for heavier disease.
- RPGN: pulse methylprednisolone + immunosuppression; plasma exchange if anti-GBM.

PSGN supportive; RPGN pulse.

III-14 Nephrotic syndrome

A 2-6-year-old boy with morning puffy eyes, frothy urine and no haematuria/hypertension has minimal change disease: start prednisolone without biopsy, and remember the killers are infection (pneumococcal peritonitis), thrombosis and hypovolaemia.

Definition
- Heavy proteinuria (UPCR ≥ 200 mg/mmol), albumin <30 g/L, oedema, + hyperlipidaemia.

Protein out, albumin down, fat up, swelling.

Causes
- Commonest cause in young children: minimal change disease, age 2-6, boys, ~80-90%.
- Atypical if <1 or >12 y, HTN, macroscopic haematuria, low C3, renal impairment.

Small boy, 2-6 = minimal change.

Patho-physiology
- Circulating factor → podocyte effacement → selective albumin leak → low oncotic pressure → oedema.
- Thrombosis: urinary loss of antithrombin III; lost IgG → infection.

Lose AT-III = clot; lose IgG = infect.

Clinical picture
- Periorbital oedema in the morning → legs, scrotum, ascites, effusions.
- Peritonitis classically caused by Streptococcus pneumoniae; also renal vein thrombosis, hypovolaemia/AKI.

Morning puffy eyes.

Diagnosis
- Urine protein-creatinine ratio, serum albumin, lipids.
- No remission after 4 weeks of daily prednisolone = steroid-resistant; needs biopsy and genetic testing.

4 weeks no remission = biopsy.

Treatment
- First episode: oral prednisolone daily for 4 weeks, then alternate days for 4 weeks.
- Albumin 20% for symptomatic hypovolaemia; pneumococcal vaccine; no added salt.
- Relapsing/dependent → CNI, mycophenolate, rituximab; resistant → CNI + ACE inhibitor.

4 weeks daily, 4 weeks alternate.

III-15 Acute and chronic kidney failure. Kidney transplantation.

In paediatric AKI first ask pre-, intra- or post-renal (FENa <1% = pre-renal; think PUV in a boy, HUS after bloody diarrhoea), then hunt the killers - hyperkalaemia and fluid overload; CAKUT drives most childhood CKD, best ended by pre-emptive living-donor transplant.

Definition
- KDIGO AKI: creatinine rise ≥ 26.5 µmol/L (0.3 mg/dL) in 48 h or ≥ 1.5× in 7 days.
- CKD: abnormal structure/function >3 months; G5 <15 mL/min/1.73 m².

AKI = 48 h/7 days; CKD = 3 months.

Causes
- Pre-renal commonest; post-renal: PUV.
- Intrinsic AKI after bloody diarrhoea: HUS; commonest CKD cause: CAKUT.

Bloody diarrhoea then AKI = HUS.

Patho-physiology
- Low perfusion → Na and water kept → FENa <1%; BUN rises relatively more.
- No K^+ excretion → hyperkalaemia → arrhythmia.
- CKD: ↓ erythropoietin → anaemia; PTH ↑, Ca^{2+} ↓, phosphorus ↑.

Pre-renal: kidney is thirsty, keeps salt.

Clinical picture
- Oliguria, fluid overload, hypertension, uraemia.
- CKD: anaemia, bone disease, HTN, growth failure.
- HUS triad: microangiopathic haemolytic anaemia, thrombocytopenia, acute kidney injury.

HUS: anaemia, low platelets, AKI.

Diagnosis
- Creatinine, urine sodium, ultrasound; hyperkalaemia ECG: peaked T, wide QRS.

Peaked T = potassium.

Treatment
- Treat cause, stop nephrotoxins, balance fluids (insensible 300-400 mL/m²/day + UO).
- Hyperkalaemia: IV calcium gluconate → salbutamol/insulin-glucose → resin → dialyse.
- CKD: ESA, binders; pre-emptive living-donor transplantation; atypical HUS: eculizumab.

Calcium protects, insulin shifts, resin removes.

III-16 Developmental abnormalities of the kidneys and the urinary tract

CAKUT is the leading cause of childhood CKD: antenatal hydronephrosis with a normal ureter is usually PUJ obstruction, a boy with a distended bladder and poor stream has PUV until proven otherwise (catheterise now), and VUR matters because pyelonephritis scars kidneys.

Definition
- CAKUT: congenital kidney/urinary tract anomalies - commonest cause of childhood CKD.

CAKUT = commonest CKD cause.

Causes
- Failed ureteric bud-mesenchyme induction → agenesis, dysplasia; abnormal ascent → horseshoe kidney.
- Duplex kidney: upper-pole ureter obstructs, lower refluxes; PUJ, PUV obstruct.

Duplex: upper obstructs, lower refluxes (Weigert-Meyer).

Patho-physiology
- Bilateral agenesis → oligohydramnios → pulmonary hypoplasia (Potter sequence).
- PUV → back-pressure → renal dysplasia, CKD; short intravesical ureter → VUR.
- VUR → pyelonephritis → focal scars → hypertension, CKD.

No urine, no fluid, no lungs.

Clinical picture
- Antenatal hydronephrosis: commonest cause (normal ureter) PUJ obstruction.
- PUV (boys): commonest lower tract obstruction; poor stream, palpable bladder, UTI.
- VUR: recurrent pyelonephritis; siblings ~30%. Horseshoe kidney: under inferior mesenteric artery.

Normal ureter + big pelvis = PUJ.

Diagnosis
- US after first days (early scans underestimate dilatation).
- VUR diagnosed and graded (I-V) with MCUG; MAG3 drainage; scars on DMSA.

MCUG grades, MAG3 drains, DMSA scars.

Treatment
- PUJ → pyeloplasty if function falls.
- PUV → urgent bladder catheter → endoscopic valve ablation.
- VUR: treat UTIs, prophylaxis if high grade; reimplantation for breakthrough infection.

PUV: catheter first, ablate next.

III-17 Renal tubulopathy

Map the tubulopathy to the segment and the diuretic it mimics: proximal = Fanconi (cystinosis), loop = Bartter (furosemide: hypercalciuria), DCT = Gitelman (thiazide: hypocalciuria, low Mg), collecting duct = distal RTA (urine pH >5.5) and nephrogenic DI.

Definition
- Inherited tubular transporter defects, mapped to nephron segments.

Know the segment, know the syndrome.

Causes
- Proximal: Fanconi (commonest cause in children: cystinosis); loop: Bartter; DCT: Gitelman.
- Collecting duct: distal RTA, type 4 RTA, nephrogenic DI (X-linked AVPR2).

Bartter = loop (B before G, loop before DCT).

Patho-physiology
- Bartter/Gitelman: salt wasting → renin, aldosterone up → hypokalaemic alkalosis.
- Distal RTA: no H+ secretion → normal-gap acidosis; urine pH stays >5.5; hypercalciuria, hypocitraturia → nephrocalcinosis.

Distal RTA: urine pH >5.5 despite acidosis.

Clinical picture
- Fanconi: glycosuria with normal blood glucose (proximal tubulopathy), phosphaturia → rickets, failure to thrive.
- Bartter mimics furosemide and causes hypercalciuria, nephrocalcinosis, polyhydramnios.
- Gitelman mimics thiazides and causes hypomagnesaemia, hypocalciuria, cramps, tetany.

Bartter = loop diuretic; Gitelman = thiazide.

Diagnosis
- Blood gas, K, Mg, urine pH and calcium; cystinosis → corneal crystals.
- Nephrogenic DI: dilute urine, hypernatraemia, no response to desmopressin.

DDAVP fails = nephrogenic.

Treatment
- Cystinosis: cysteamine + replace water, bicarbonate, phosphate, K.
- Bartter: indometacin + K; Gitelman: Mg + K.
- Distal RTA: alkali (citrate); nephrogenic DI: free water, thiazide.

Thiazide treats DI (paradox).

III-18 Chromosomal abnormalities

Know the dysmorphology and the 'screening list' of each aneuploidy, and remember the counselling twist: translocation Down syndrome (~4%) is the one where you karyotype the parents because recurrence risk can be high.

Definition
- Numerical or structural chromosome errors: trisomies, microdeletions, sex chromosome aneuploidies.

Number or structure.

Causes
- Down syndrome: ~95% non-disjunction (maternal age), ~4% translocation (karyotype the parents), ~1% mosaic.
- Trisomy 18/13; 22q11 deletion; Williams 7q11.23; Turner 45,X; Klinefelter 47,XXY.

Translocation Down = test the parents.

Patho-physiology
- Down: extra copies of APP gene → early Alzheimer; endocardial cushion defect → AVSD.
- 22q11 deletion: thymic hypoplasia → infections; parathyroid hypoplasia → hypocalcaemia.
- Turner short stature: SHOX haploinsufficiency.

CATCH-22.

Clinical picture
- Down: hypotonia, duodenal atresia (trisomy 21), Hirschsprung, leukaemia, hypothyroid.
- Edwards: clenched fists, overlapping fingers, rocker-bottom feet; Patau: holoprosencephaly, cleft lip, polydactyly.
- 22q11: conotruncal heart defects; Turner: webbed neck, coarctation; Klinefelter: tall, small firm testes, infertility.

Edwards = 18 = Election age; Patau = 13 = Puberty.

Diagnosis
- Prenatal: combined first-trimester test, NIPT → CVS/amniocentesis with QF-PCR/karyotype.
- Translocation Down → karyotype parents; microdeletions → microarray, FISH.

Screen then confirm invasively.

Treatment
- Down: screen heart, thyroid, hearing, eyes, coeliac; multidisciplinary care.
- Turner: growth hormone + oestrogen; check for Y material (gonadoblastoma).
- Klinefelter: testosterone at puberty.

Turner with Y = remove gonads.

III-19 Mendelian inherited diseases (autosomal dominant and recessive), common disorders

Read the pedigree before the gene: vertical with male-to-male transmission and 50% per child is AD (often de novo with older fathers); horizontal with carrier parents, consanguinity and 25% per pregnancy is AR - and an unaffected sib of an AR child is a carrier with 2/3 probability.

Definition
- Single-gene disease: AD needs one mutant allele; AR needs both.

Dominant = one; recessive = two.

Causes
- AD: NF1, Marfan (fibrillin-1), achondroplasia (FGFR3 gain-of-function), TSC, ADPKD, FH.
- AR: CF, PKU, SMA (SMN1 deletion), sickle, CAH, galactosaemia, Wilson.

Structural = dominant; enzyme = recessive.

Patho-physiology
- AD: haploinsufficiency, dominant negative, gain of function; de novo mutations common, linked to advanced paternal age.
- AR: most enzyme deficiencies are AR because 50% activity is sufficient, so carriers are healthy.

50% enzyme is enough.

Clinical picture
- AD pedigree: vertical, each child has 50% risk, male-to-male transmission.
- AR pedigree: horizontal, two carrier parents → 25% risk per pregnancy; consanguinity.
- Healthy sibling of an AR child: 2/3 carrier; meconium ileus → cystic fibrosis.

Unaffected sib of AR = 2/3 carrier.

Diagnosis
- Mutation analysis, gene panels, exome/genome sequencing; carrier screening.
- Prenatal (CVS, amniocentesis); newborn screening for PKU, CAH, galactosaemia, CF, SMA.

Screen what you can treat.

Treatment
- Non-directive genetic counselling with risk calculation.
- Cascade testing of relatives; disease-specific therapy (e.g. nusinersen for SMA).

Non-directive counselling.

III-20 X-linked inherited diseases, common disorders

X-linked recessive: affected boys, carrier mothers, never father-to-son, and every daughter of an affected man is a carrier; Duchenne is the archetype - Gowers sign, calf pseudohypertrophy, CK >10 000, steroids, and no suxamethonium.

Definition
- XLR: affected boys, carrier mothers, no male-to-male transmission.
- XLD: both sexes; affected father → all daughters affected.

No male-to-male in X-linked.

Causes
- A carrier mother has a 50% risk of each son being affected.
- Haemophilia father: all daughters obligate carriers; ~1/3 of Duchenne new mutations.

Carrier mum: half her sons.

Patho-physiology
- Males hemizygous; lyonisation protects most carriers.
- Duchenne: out-of-frame dystrophin deletion; Becker: in-frame one.
- Fragile X: CGG repeat in FMR1; XLA: BTK.

Out of frame = Duchenne; in frame = Becker.

Clinical picture
- DMD age 2-5: Gowers sign, calf pseudohypertrophy, falls.
- Haemophilia: haemarthrosis; fragile X: ID, long face; XLA: infections from ~6 months.
- WAS: eczema, thrombocytopenia with small platelets, infections; XLD: hypophosphataemic rickets, Rett syndrome.

WAS: eczema + tiny platelets.

Diagnosis
- First test in suspected Duchenne: serum creatine kinase (>10 000 U/L) → genetic testing.
- Haemophilia: isolated long APTT, normal PT and platelets.

DMD CK >10 000.

Treatment
- DMD: corticosteroids (prednisolone [prednisone] or deflazacort), physio, cardiac/respiratory surveillance.
- Avoid suxamethonium; haemophilia: factor prophylaxis, emicizumab, no IM/NSAIDs.
- Live vaccines must be avoided in XLA.

XLA: no live vaccines.

III-21 Multifactorial and polygenic genetic disorders

Multifactorial threshold traits give empiric, not Mendelian, risks (~2-5% for first-degree relatives of an isolated malformation), and risk climbs with severity, more affected relatives and when the rarer sex is affected (Carter effect) - but first exclude a syndrome, and give folate.

Definition
- Disease appears when combined genetic and environmental liability crosses a threshold.

Liability threshold.

Causes
- Boys: pyloric stenosis (about 4-5 times commoner in boys), cleft lip, Hirschsprung, club foot.
- Girls: DDH, isolated cleft palate, NTDs; also CHD, diabetes, asthma.

Pyloric boys; DDH girls.

Patho-physiology
- Relatives share genes → curve shifts toward threshold → empiric recurrence.
- Carter effect: when the less commonly affected sex is affected, relatives have a higher recurrence risk.

Rare sex affected = more genes.

Clinical picture
- First-degree recurrence of an isolated malformation is 2-5%; falls sharply in second-degree relatives.
- Risk ↑ with severity, ≥ 2 affected relatives, affected rarer sex.

2-5% first-degree.

Diagnosis
- Other anomalies/dysmorphism/delay → look for syndrome (22q11, trisomy 13) first.
- NTD screening: raised maternal serum AFP + anomaly ultrasound.

Raised AFP = open NTD.

Treatment
- Counsel with empiric risks; NTD recurrence ~3-5% without folate.
- Folic acid 400 µg (all) or 5 mg daily (high-risk woman) preconception to 12 weeks.
- US: 4 mg after previous NTD pregnancy.

400 µg all; 5 mg high risk.

III-22 Congenital metabolic disorders of amino acid metabolism

An aminoacidopathy baby is well at birth (the placenta dialyses) and crashes after protein feeds looking septic; in a crisis stop protein, give 10% dextrose, check ammonia/gas/lactate - and let the smell, lens and liver give away the specific diagnosis.

Definition
- Mostly autosomal recessive enzyme defects of amino acid metabolism.

Enzymes = recessive.

Causes
- Classic PKU: deficiency of phenylalanine hydroxylase; MSUD: branched-chain ketoacid dehydrogenase.
- Tyrosinaemia 1: fumarylacetoacetate hydrolase; homocystinuria: cystathionine β -synthase.

PKU = PAH.

Patho-physiology
- Placenta clears metabolites → well at birth → sick after protein feeds.
- PKU: low tyrosine (precursor of melanin) → fair hair; leucine → cerebral oedema.

Well at birth, sick after milk.

Clinical picture
- Sepsis-like: poor feeding, vomiting, lethargy, seizures, coma.
- Untreated PKU: musty odour; sweet-smelling urine + neonatal encephalopathy: maple syrup urine disease.
- Tyrosinaemia 1: liver failure + Fanconi; homocystinuria: marfanoid, lens dislocates downwards (Marfan: upwards).

Homocystinuria: lens down; Marfan: up.

Diagnosis
- PKU: newborn heel-prick phenylalanine.
- Crisis: glucose, gas, lactate, ammonia, amino acids, urine organic acids.

Crisis bloods: glucose, gas, lactate, ammonia.

Treatment
- Acute metabolic crisis: stop protein and give high-energy IV glucose (10%).
- Ammonia scavengers/haemofiltration, treat sepsis; PKU: lifelong low-Phe diet ± sapropterin.
- Tyrosinaemia → nitisinone; homocystinuria → pyridoxine (B6); MSUD → BCAA-restricted diet.

Stop protein, start glucose.

III-23 Congenital metabolic disorders of glucose and glycogen metabolism

Pair the low sugar with the liver: hypoglycaemia + big liver + lactic acidosis, high urate and lipids = von Gierke (never fast); hypoketotic hypoglycaemia on fasting = MCAD; jaundice, cataract and E. coli sepsis after milk = galactosaemia; symptoms at weaning = fructose intolerance.

Definition
- Inherited defects of glycogen, galactose or fructose metabolism.

Glycogen, galactose, fructose.

Causes
- Von Gierke (GSD I) is deficiency of glucose-6-phosphatase (AR).
- Pompe (GSD II), McArdle (GSD V), galactosaemia (GALT), fructose intolerance (aldolase B); MCAD: AR.

Von Gierke = G6Pase.

Patho-physiology
- G6Pase block → raised lactate, uric acid and triglycerides.
- MCAD deficiency causes hypoketotic hypoglycaemia during fasting.

MCAD: no ketones.

Clinical picture
- Low blood sugar in an infant with normal temperature and hepatomegaly: von Gierke disease.
- Pompe: hypertrophic cardiomyopathy and hypotonia; McArdle: exercise cramps, myoglobinuria, second wind.
- Neonatal jaundice, cataract, E. coli sepsis after milk: galactosaemia; hereditary fructose intolerance at weaning.

Milk = galactose; weaning fruit = fructose.

Diagnosis
- GSD I: fasting hypoglycaemia 2-4 h post-feed, lactic acidosis, high urate.
- Galactosaemia: red-cell GALT, urine reducing substances; McArdle: raised CK.

Hypo 2-4 h after feed = GSD I.

Treatment
- GSD I: never fast - frequent/overnight feeds, uncooked cornstarch, allopurinol.
- Pompe: enzyme replacement; galactosaemia: stop lactose now, soy formula.
- MCAD: avoid fasting; fructose intolerance: exclude fructose, sucrose, sorbitol.

Cornstarch keeps GSD I going overnight.

III-24 Lysosomal storage diseases

A child normal at birth who then loses skills, especially with organomegaly, coarse features or a cherry-red spot, has a storage disease until proven otherwise; check leucocyte enzyme activity - and remember ERT cannot treat the brain, so Hurler needs HSCT before ~2 years.

Definition
- Lysosomal enzyme deficiency → substrate stored in the cells that turn over most.

Full lysosomes, swollen organs.

Causes
- Mostly recessive; the two X-linked ones: Fabry and Hunter.
- Gaucher is deficiency of β -glucocerebrosidase; Niemann-Pick, Tay-Sachs (hex A), Hurler (α -L-iduronidase), Pompe.

Fabry and Hunter hunt X.

Patho-physiology
- Macrophage → hepatosplenomegaly, cytopenias; neuron → regression, cherry-red spot.
- Connective tissue (GAGs) → coarse face, dysostosis multiplex, valves, corneal clouding.

Macrophage, neuron, matrix.

Clinical picture
- Normal at birth, then loss of skills = key alarm.
- Cherry-red spot without hepatosplenomegaly: Tay-Sachs; with hepatosplenomegaly: Niemann-Pick.
- Gaucher: massive splenomegaly, bone crises; Hurler: corneal clouding (Hunter: none); Fabry: burning hands/feet.

Hunter sees clearly (no clouding).

Diagnosis
- Enzyme activity in leucocytes or dried blood spots.
- Urinary GAGs/oligosaccharides, genetic testing; newborn screening for Pompe, MPS I.

Measure the missing enzyme.

Treatment
- ERT does not treat CNS disease: it does not cross the blood-brain barrier.
- Gaucher, Fabry, Hunter: ERT; substrate reduction.
- HSCT before ~2 y in Hurler preserves cognition; multidisciplinary supportive care.

ERT can't get into the brain.

III-25 Rhinitis, adenoiditis, tonsillitis

Most sore throats are viral; treat GAS (fever, exudate, tender anterior nodes, no cough, age 5-15) with 10 days of penicillin V to prevent rheumatic fever (not PSGN), avoid amoxicillin when EBV is possible, and spot the red flags - quinsy and unilateral foul discharge.

Definition
- Upper airway inflammation: rhinitis, adenoid hypertrophy, tonsillitis.

Nose, adenoids, tonsils.

Causes
- Rhinitis: viral or allergic (IgE).
- Tonsillitis: mostly viral; group A strep commonest bacterial, 5-15 years.

GAS age 5-15.

Patho-physiology
- Adenoid hypertrophy peaks at 3-7 years: blocked nose, Eustachian tubes → snoring, OSA, otitis media.
- GAS → rheumatic fever; PSGN follows pharyngitis by 1-2 weeks.

Adenoids peak 3-7.

Clinical picture
- Allergic: sneezing, itch; unilateral foul discharge in a toddler = foreign body.
- GAS: fever, exudate, tender anterior nodes, no cough.
- Trismus, hot-potato voice and deviated uvula: peritonsillar abscess (quinsy).

One-sided smelly nose = foreign body.

Diagnosis
- GAS: rapid antigen test; negative backed by throat culture (low sensitivity).
- Adenoids: nasendoscopy or lateral neck X-ray; sleep study if OSA uncertain.

Negative RADT? Culture it.

Treatment
- Allergic rhinitis: most effective is an intranasal steroid; viral: no antibiotics.
- GAS: penicillin V for 10 days (prevents rheumatic fever, not PSGN); avoid amoxicillin if EBV possible.
- Adenectomy: recurrent URTIs, otitis media, blocked nasal breathing, OSA; tonsillectomy: 7/1 y, 5/y×2, 3/y×3.

Amoxicillin + EBV = rash.

III-26 Acute upper respiratory tract infections, acute sinusitis, bronchitis

Most upper airway infections are viral and need no antibiotics; the skill is spotting the few that do (GAS pharyngitis, AOM, true bacterial sinusitis) and the airway emergencies (croup vs epiglottitis) where the throat must not be examined.

Definition
- Upper airway infections: pharyngitis, sinusitis, croup, epiglottitis, bronchitis - mostly viral.

Upper airway = mostly viruses; hunt the bacterial and the blocked.

Causes
- Viral: adenovirus, rhinovirus, EBV; croup = parainfluenza.
- Bacterial: GAS; sinusitis and otitis media: S. pneumoniae.

Croup = Parainfluenza (P for 'parrot bark').

Patho-physiology
- Cold swells ostiomeatal complex → sinus can't drain → bacterial colonisation.
- Croup: subglottic oedema in cricoid ring → small radius → high resistance.

Poiseuille: halve the radius, resistance x16.

Clinical picture
- Strep: fever, exudate, tender nodes, no cough.
- Croup: barking cough, stridor; epiglottitis: drooling and tripod posture; bronchiolitis: wheeze, not stridor.
- Pertussis in young infants: apnoea and cyanosis, often no whoop.

Bark = croup, drool = epiglottis, wheeze = bronchioles.

Diagnosis
- Bacterial sinusitis: over 10 days, double worsening, or severe onset; CT only for complications.
- Strep: rapid antigen test or culture. Epiglottitis: do not examine the throat.

10 days, double dip, or severe start.

Treatment
- Antibiotics only: Streptococcus angina, acute otitis media, acute bacterial sinusitis; not colds or bronchiolitis.
- Penicillin V for 10 days to prevent rheumatic fever; croup: single oral dexamethasone.
- Epiglottitis: secure the airway; oseltamivir within 48 hours.

Abx trio: Strep throat, AOM, bacterial sinusitis.

III-27 Infections of the ear

Children get AOM because their Eustachian tube is short, horizontal and floppy; most AOM settles with analgesia, so the exam skill is knowing who gets amoxicillin, when OME needs grommets, and recognising mastoiditis.

Definition
- AOM: middle-ear infection; OME: effusion without infection; otitis externa: canal infection.
  Media = middle ear; externa = outside canal.

Causes
- Commonest bacterial cause of AOM: S. pneumoniae, NTHi, Moraxella; day-care, smoking, Down.
- Otitis externa: Pseudomonas aeruginosa.
  Swimmer's ear = Pseudomonas in the pool.

Patho-physiology
- Children get AOM because their Eustachian tube is short, horizontal and floppy.
- Cold → tube blocks → effusion → bulging drum; perforation → otorrhoea.
- OME: fluid dampens drum → conductive hearing loss.
  Child tube: short, flat, floppy.

Clinical picture
- AOM: otalgia, ear-pulling, fever, bulging drum.
- OME: dull retracted drum, type B tympanogram.
- Typical sign of otitis externa: pain and/or sensitivity of the tragus.
  Tug the tragus: pain = externa.

Diagnosis
- Otoscopy; tympanometry for OME.
- Protruding ear with postauricular swelling after AOM suggests mastoiditis → IV antibiotics, CT.
  Ear sticks out after AOM = mastoid.

Treatment
- AOM: analgesia first; amoxicillin if under 2 years with bilateral disease or otorrhoea.
- OME: wait 3 months; grommets in chronic serous otitis media or Eustachian tube dysfunction.
- Externa: aural toilet, topical antibiotic ± steroid drops.
  Grommets for glue that won't go.

III-28 Pneumonia and complications in infants and children

Childhood pneumonia is a clinical diagnosis built on age-specific tachypnoea and work of breathing; treat with amoxicillin, admit on hypoxaemia (SpO_2 <92%) or grunting, and suspect empyema when fever persists after 48 h of antibiotics.

Definition
- Lung infection with alveolar exudate; complications are effusion, empyema, abscess.
  Fluid in the alveoli, pus in the pleura.

Causes
- Neonates: GBS, E. coli, Listeria; preschool: viruses commonest overall.
- Typical pneumonia: most commonly S. pneumoniae; school age: Mycoplasma; pneumatoceles after influenza → S. aureus.
  Age ladder: GBS → virus → Mycoplasma.

Patho-physiology
- Alveolar exudate → low compliance + shunt → tachypnoea, indrawing, grunting, hypoxaemia.
- Parapneumonic effusion stages: exudative, fibrinopurulent, organising.
  Exudate → fibrin → peel.

Clinical picture
- WHO tachypnoea threshold for age 1-5 years: 40/min (≥ 60 <2 mo, ≥ 50 2-12 mo).
- Crackles, bronchial breathing, dullness; abdominal pain from diaphragmatic pleura.
- Fever persisting after 48 hours of antibiotics suggests empyema or abscess.
  Fever after 48 h of antibiotics = look for pus.

Diagnosis
- Clinical; CXR not routine if managed at home.
- Admit if SpO_2 <92%, grunting, apnoea, poor feeding; US shows effusion septations.
  Treat the child, not the X-ray.

Treatment
- First-line oral antibiotic for childhood CAP: amoxicillin; add macrolide if Mycoplasma.
- Severe: IV ampicillin/penicillin G or ceftriaxone (US); co-amoxiclav/cefuroxime (UK).
- Empyema: drain; thick pus dissolved by fibrinolysis (urokinase/streptokinase) → VATS if fails.
  Pus: drain it, dissolve it, then VATS.

III-29 Congenital lung malformations

Congenital lung lesions are now mostly picked up antenatally; the key emergency is CDH, where pulmonary hypoplasia and pulmonary hypertension kill and bag-mask ventilation must be avoided.

Definition
- Congenital anomalies of diaphragm, lung parenchyma and chest wall.
  Three places: diaphragm, lung, wall.

Causes
- CDH is usually a left posterolateral (Bochdalek) defect: pleuroperitoneal canal fails to close.
- CPAM, sequestration, lobar emphysema; two most frequent chest deformities: pectus excavatum and pectus carinatum.
  Bochdalek = Back and Left.

Patho-physiology
- CDH: gut in chest → pulmonary hypoplasia + pulmonary hypertension drive mortality.
- Lobar emphysema most often affects the left upper lobe: deficient cartilage → ball-valve.
  Ball-valve: air in, can't get out.

Clinical picture
- CDH: cyanosis, scaphoid abdomen, heart sounds displaced to the right.
- CPAM/sequestration: recurrent pneumonia in the same lobe, distress.
  CDH: scaphoid belly, heart pushed right.

Diagnosis
- Often antenatal ultrasound; CPAM = cysts that communicate with the airway.
- Sequestration has a systemic arterial supply, usually from the aorta (Doppler/CT angiography).
- Pectus: Haller index on CT/MRI, echocardiography, spirometry.
  Sequestration = separate from bronchi, fed by aorta.

Treatment
- In suspected CDH avoid bag-and-mask ventilation; intubate, NG tube on suction; NO/ECMO, repair when stable.
- CPAM/sequestration/lobar emphysema: resection if symptomatic.
- Pectus excavatum: Nuss procedure; carinatum: compressive brace.
  CDH: no bag, tube in, NG out.

III-30 Cystic fibrosis. Lung transplantation

CF is an AR CFTR chloride-channel defect: dehydrated secretions explain everything (bronchiectasis, pancreatic insufficiency, meconium ileus, salty sweat); diagnose with IRT screening then sweat chloride ≥60 mmol/L, and treat with clearance, enzymes and CFTR modulators.

Definition
- Autosomal recessive CFTR defect; commonest mutation F508del causes protein misfolding.
  F508del: Folded wrong, Fails to arrive.

Causes
- Carrier ~1 in 25 Europeans; class II F508del, class III gating (G551D).
  1 in 25 carry → 1 in 2500 affected.

Patho-physiology
- No Cl^- secretion + unopposed ENaC → dehydrated mucus → infection → bronchiectasis.
- Blocked pancreatic ducts → exocrine insufficiency; sweat duct cannot reabsorb chloride → salty sweat.
  Lungs dry, sweat salty: same channel, opposite jobs.

Clinical picture
- Meconium ileus (X-ray: soap-bubble; contrast enema: microcolon), steatorrhoea, FTT despite good appetite.
- Main chronic lung pathogen: Pseudomonas aeruginosa (S. aureus early); nasal polyps, male infertility.
- Salty baby, hyponatraemic dehydration in heat; ADEK deficiency.
  Salty, skinny, sticky.

Diagnosis
- Newborn screening measures immunoreactive trypsinogen (IRT) → CFTR panel.
- Diagnostic sweat chloride: ≥60 mmol/L (30-59 intermediate); faecal elastase low.
  Sweat test: 60 or more = CF.

Treatment
- Airway clearance, dornase alfa (cleaves neutrophil DNA), hypertonic saline, Pseudomonas eradication, azithromycin.
- Enzymes every meal, high-calorie diet, ADEK, salt; CFTR modulators (elexacaftor/tezacaftor/ivacaftor).
- End-stage: bilateral lung transplant (avoids seeding from native lung); limited by bronchiolitis obliterans.
  Dornase cuts DNA strands in pus.

III-31 Bronchial asthma

Asthma = inflammation + hyperresponsiveness + reversible obstruction: low-dose ICS is the cornerstone controller, and in an acute attack a silent chest or a normal/rising $PaCO_2$ means exhaustion, not improvement.

Definition
- Asthma = chronic airway inflammation + bronchial hyperresponsiveness + reversible obstruction.
  Inflamed, twitchy, reversible.

Causes
- Atopy (eczema, rhinitis), family history, tobacco smoke, prematurity, obesity.
- Triggers: viruses, exercise, allergens, smoke.
  Viruses trigger most childhood attacks.

Patho-physiology
- Th2/mast cells → IL-4/5/13, histamine, leukotrienes (C4, D4, E4).
- → bronchospasm (β2-agonist) + oedema (steroid) + mucus → remodelling.
  Spasm → β2; swelling → steroid.

Clinical picture
- Recurrent wheeze, night cough, exercise symptoms, chest tightness.
- Life-threatening: silent chest, SpO_2 <92%, PEF <33%; normal or rising PaCO2 signals exhaustion.
  Silent chest and normal CO2 = danger.

Diagnosis
- Under 5 years: clinical diagnosis (trial of ICS); from 5 years FeNO and spirometry.
- FeNO ≥ 35 ppb; reversibility: FEV1 increase of at least 12%; PEF variability.
  12% better after salbutamol = reversible.

Treatment
- Cornerstone controller: inhaled corticosteroid (low dose) → MART/LABA or LTRA → biologics; spacer, action plan.
- Acute: oxygen, salbutamol, ipratropium, prednisolone (US: dexamethasone).
- Severe acute asthma not responding to nebulised bronchodilators: add IV magnesium sulfate → PICU.
  ICS is the foundation; magnesium is the rescue.

III-32 Allergy in infants and children. Definition of atopy, atopic diseases.

Atopy is a Th2-driven tendency to make IgE; separate fast IgE-mediated reactions (minutes–2 h, anaphylaxis risk → IM adrenaline 10 µg/kg) from slow non-IgE reactions (hours–days), and confirm food allergy by elimination and re-challenge rather than by IgE tests alone.

Definition
- Atopy is a genetic tendency to produce IgE driven by a Th2 response.
  Atopy = Th2 → IgE.

Causes
- Atopic march, in order: food allergy - atopic dermatitis - bronchial asthma - allergic rhinitis.
- Loss-of-function mutations in filaggrin break the skin barrier; milk, egg, peanut, nuts, fish.
  Food → skin → chest → nose.

Patho-physiology
- Th2 (IL-4/13) → IgE on mast cells → cross-linking → histamine, leukotrienes.
- IL-5 → eosinophils → late-phase chronic inflammation.
  IL-5 = eosinophils.

Clinical picture
- IgE-mediated (minutes-2 h): urticaria, angioedema, vomiting, wheeze, anaphylaxis.
- Non-IgE (hours-days): eczema flare, reflux, proctocolitis, FPIES.
  Fast = IgE; slow = cell-mediated.

Diagnosis
- Food allergy frequency in childhood: parents report approx. 7-10%; in reality 1-2 %.
- Allergen specific IgE is not totally reliable; elimination diet, then allergen re-challenge after symptoms disappear.
  Remove, resolve, re-challenge.

Treatment
- Avoidance, dietitian, action plan; milk-allergic infant → extensively hydrolysed formula, not goat's milk.
- Anaphylaxis: IM adrenaline 10 µg/kg (0.01 mL/kg of 1 mg/mL), anterolateral thigh; repeat at 5 min.
  Adrenaline: 0.01 mL/kg of 1:1000, thigh.

III-33 Symptoms and treatment of decompensated heart failure

Infant heart failure looks like a feeding problem (sweating, tachypnoea, FTT) with hepatomegaly as the key sign; timing points to the cause — first week = duct-dependent left lesion (give PGE1), 4–8 weeks = large L→R shunt as PVR falls.

Definition
- Heart cannot meet demand → congestion and low output; decompensated = shock/collapse.
  Congested and under-perfused.

Causes
- First week: duct-dependent left lesions (HLHS, coarctation), SVT, sepsis.
- Large VSD: heart failure at 4-8 weeks because pulmonary vascular resistance falls (AVSD, PDA).
- Later: myocarditis, cardiomyopathy, arrhythmia, Kawasaki, anaemia.
  Week 1 = duct; week 6 = shunt.

Patho-physiology
- ↓ Output → sympathetic + RAAS → tachycardia, sweating, salt/water retention.
- ↑ Pulmonary venous pressure → tachypnoea; ↑ systemic venous pressure → hepatomegaly.
  Back-up behind the left = lungs; right = liver.

Clinical picture
- Poor feeding with sweating, tachypnoea and failure to thrive.
- Most reliable sign of heart failure in an infant: hepatomegaly; gallop, cardiomegaly, weak pulses.
  Infant HF: feeds, sweats, fails to grow.

Diagnosis
- Echocardiography defines cause; CXR (cardiomegaly), ECG.
- BNP/NT-proBNP, electrolytes, renal function, gas and lactate.
  Echo for cause; lactate for severity.

Treatment
- Sit up, careful O_2 high-calorie NG feeds (up to 150 kcal/kg/day).
- Furosemide + spironolactone (spares potassium, blocks aldosterone), ACE inhibitor; milrinone if decompensated.
- First-week duct-dependent lesion → prostaglandin E1; then repair defect or transplant.
  Week-1 failure: open the duct with PGE1.

III-34 Cyanotic congenital heart disease

Cyanotic CHD = the 5 Ts; a blue newborn who doesn't pink up with 100% O_2 (failed hyperoxia test) gets prostaglandin E1 — TGA presents in the first days without a murmur, Fallot later with spells.

Definition
- Congenital heart lesions causing central cyanosis - the 5 Ts.
  5 Ts: Tetralogy, TGA, Truncus, Tricuspid, TAPVR.

Causes
- R→L shunt, low pulmonary flow: Fallot, pulmonary/tricuspid atresia, critical PS.
- Commonest cyanotic lesion in first days: transposition of the great arteries; truncus (22q11), TAPVR.
  TGA = day 1 blue; Fallot = later blue.

Patho-physiology
- Cyanosis visible at ~3-5 g/dL deoxygenated haemoglobin.
- Fallot: RVOT obstruction, VSD, overriding aorta, RV hypertrophy; spell → more R→L shunt.
- Chronic hypoxia → EPO → polycythaemia, clubbing; brain abscess.
  Spell = more blood bypasses the lungs.

Clinical picture
- Fallot: murmur softens in a tet spell; boot-shaped heart.
- TGA: profound cyanosis first hours-days, often no murmur; egg on a string.
  Spell: bluer and quieter.

Diagnosis
- Hyperoxia test: PaO2 does not rise with 100% oxygen in cardiac cyanosis.
- Echocardiography; pulse-oximetry screening (right hand + foot) detects critical CHD.
  Cardiac shunt ignores oxygen.

Treatment
- Blue newborn not improving on O_2 → PGE1 (watch apnoea).
- Tet spell: knee-chest (raises SVR), O_2, morphine, fluids, propranolol, phenylephrine; repair 3–6 mo.
- TGA: prostaglandin → balloon atrial septostomy → arterial switch in first 2–3 weeks.
  Knee-chest squeezes arteries, pushes blood to lungs.

III-35 Non-cyanotic congenital heart disease

Acyanotic CHD splits into L→R shunts (VSD commonest, ASD, AVSD, PDA) and obstructions (coarctation, AS, PS); learn the signature finding of each and remember that untreated shunts can reverse to Eisenmenger.

Definition
- Congenital heart disease without cyanosis: left-to-right shunts or obstructive lesions.
  Shunts or blocks, but pink.

Causes
- Commonest defect: VSD; also ASD, AVSD (Down), PDA (preterm).
- Obstruction: coarctation (Turner), aortic stenosis, pulmonary stenosis (Noonan).
  Turner-Coarct, Noonan-PS, Down-AVSD.

Patho-physiology
- Long-standing left-to-right shunt reverses = Eisenmenger syndrome.
- A small VSD gives a louder murmur: high pressure gradient.
  Small hole, loud noise.

Clinical picture
- 4 signs: heart murmur, heart failure, cyanosis, peripheral pulse abnormalities.
- Fixed wide splitting of S2 = ASD; PDA: continuous machinery murmur, bounding pulses.
- Coarctation: upper-limb hypertension, weak, delayed femoral pulses.
  Fixed split = ASD; machinery = PDA.

Diagnosis
- Congenital defects are diagnosed with echocardiography.
- Holosystolic murmur at left lower sternal border = VSD.
- Innocent murmur: soft, systolic, postural, no other signs.
  Innocent: soft, systolic, shifts with posture.

Treatment
- Large VSD: surgery in first months; ASD: catheter device at 3-5 years.
- Critical coarctation: prostaglandin E1; preterm PDA: ibuprofen, indomethacin or paracetamol.
  Open the duct: PGE1; close it: NSAID.

III-36 Cardiomyopathy

Think of cardiomyopathy as 'can't empty' (DCM), 'can't fill' (HCM) and 'can't stretch' (RCM): DCM gets heart-failure drugs, HCM gets β-blockers and no inotropes/vasodilators, RCM goes early to transplant — and always screen first-degree relatives.

Definition
- Heart muscle disease: dilated (commonest), hypertrophic, restrictive.

DCM: commonest; can't squeeze.

Causes
- DCM: viral myocarditis, anthracycline (doxorubicin) chemotherapy, Duchenne.
- HCM: autosomal dominant sarcomere gene mutation; newborn: infant of a diabetic mother.

Doxorubicin dilates; sarcomeres thicken.

Patho-physiology
- DCM: can't empty → low EF → dilation → systolic failure, arrhythmia, thrombi.
- HCM: can't fill → dynamic LVOT obstruction → sudden death.
- RCM: can't stretch → huge atria, early pulmonary hypertension.

Empty, fill, stretch.

Clinical picture
- DCM: tachypnoea, poor feeding, sweating, hepatomegaly, gallop, cool peripheries.
- HCM: exertional syncope/dyspnoea; murmur increases with Valsalva and standing.

Less preload, louder HCM.

Diagnosis
- Echo: dilated low-EF LV or hypertrophy; also ECG, CXR, troponin, BNP, cardiac MRI.
- Screen first-degree relatives.

One case, screen the family.

Treatment
- DCM: ACE-i, ARB, β -blockers (carvedilol), spironolactone improve survival; transplant if end-stage.
- HCM: beta-blockers; avoid inotropes, vasodilators, excess diuretics; high-risk: ICD.
- RCM → early transplant.

HCM: slow and full; avoid inotropes.

III-37 Myocarditis, pericarditis - symptoms and management

Myocarditis hides as 'tachycardia out of proportion to fever' with hepatomegaly; pericarditis declares itself with positional pain, rub and widespread concave ST elevation — and any effusion with Beck triad or pulsus paradoxus needs urgent pericardiocentesis.

Definition
- Myocarditis: heart muscle inflammation; pericarditis: inflamed pericardium ± effusion.

Muscle vs sac.

Causes
- Commonest cause of myocarditis in children: viral infection (enteroviruses, adenovirus, parvovirus B19).
- Pericarditis: viral, purulent (Staphylococcus aureus), TB, autoimmune, post-pericardiotomy.

Myocarditis: think enterovirus.

Patho-physiology
- Viral + immune myocyte loss → ↓ stroke volume → tachycardia, congestion, arrhythmias.
- Tamponade: fluid → ventricles cannot fill → stroke volume falls.

Tamponade: squeezed shut from outside.

Clinical picture
- Myocarditis: tachycardia out of proportion to fever; hepatomegaly = reliable sign of venous congestion.
- Pericarditis: pain relieved by sitting up and leaning forward, friction rub.
- Tamponade: Beck triad, pulsus paradoxus (inspiratory fall in systolic pressure over 10 mmHg).

Pulsus paradoxus: >10 mmHg fall on inspiration.

Diagnosis
- Myocarditis: sinus tachycardia, low-voltage QRS; troponin, BNP, echo, cardiac MRI.
- Pericarditis: widespread concave ST elevation with PR depression; echo shows effusion.

Pericarditis ECG: smiley ST, sad PR.

Treatment
- Myocarditis: small fluid boluses, furosemide, inotropes, ECMO; no competitive sport for 3-6 months.
- Pericarditis: NSAID + colchicine; tamponade → urgent pericardiocentesis.
- Purulent: surgical drainage + IV anti-staphylococcal antibiotics.

Tamponade: needle it now.

III-38 Infective endocarditis - risk factors, prophylaxis, management

IE = vegetation on damaged endothelium seeded by bacteraemia; diagnose with ≥3 blood cultures before antibiotics plus echo, treat with 4–6 weeks of IV bactericidal therapy, and give dental prophylaxis only to the high-risk (prosthetic material, prosthetic valves, previous IE).

Definition
- Infection of the endocardium, usually on a valve or defect.

Bugs on valves.

Causes
- Risk: CHD with prosthetic material, prosthetic valves, previous IE.
- Neonates with normal hearts: central venous catheters.
- Viridans streptococci: subacute, after dental bacteraemia; S. aureus: acute.

Viridans = slow, from teeth; aureus = aggressive.

Patho-physiology
- Turbulent jet damages endothelium → platelet-fibrin vegetation → seeded by bacteraemia.
- Emboli; immune complexes → glomerulonephritis, Osler nodes, Roth spots.

Jet → nick → clot → colonise.

Clinical picture
- Prolonged fever, malaise, new/changing murmur, splenomegaly (FROM JANE).
- Painful finger-pulp nodules are Osler nodes; painless palmar macules are Janeway lesions.

Osler = Ouch; Janeway = painless.

Diagnosis
- Basis of diagnosis: haemoculture and echocardiography.
- Before antibiotics take at least three blood cultures from separate venepunctures.

Three cultures before one antibiotic.

Treatment
- IV bactericidal antibiotics for 4-6 weeks: amoxicillin + flucloxacillin [nafcillin] + gentamicin.
- Prosthetic material → vancomycin + gentamicin; surgery for heart failure, emboli.
- Prophylaxis, high-risk: single oral dose of amoxicillin 30-60 minutes before dental work.

Prophylaxis: one dose, an hour before the dentist.

III-39 Causes of malabsorption. Celiac disease

Separate maldigestion (no enzymes — CF) from mucosal malabsorption (coeliac, CMPA, Giardia); for coeliac, test anti-tTG IgA plus total IgA while eating gluten, and treat with a strict lifelong gluten-free diet monitored by antibodies.

Definition
- Maldigestion (no enzymes) vs mucosal malabsorption.
- Coeliac: permanent immune enteropathy triggered by gluten.

Enzymes missing vs wall damaged.

Causes
- Maldigestion: cystic fibrosis (commonest cause of pancreatic insufficiency in children).
- Mucosal: coeliac, cow's milk allergy, Giardia, Crohn, lactase deficiency.
- Coeliac needs HLA-DQ2/DQ8; at risk: T1DM, Down, Turner, IgA deficiency.

No DQ2/DQ8, no coeliac.

Patho-physiology
- Gliadin → immune villous atrophy, crypt hyperplasia, intraepithelial lymphocytosis → malabsorption; fat loss drags ADEK.

Flat villi, deep crypts, many lymphocytes.

Clinical picture
- Classical 6 m-2 y: diarrhoea, distension, wasted buttocks, FTT.
- Extraintestinal: hepatitis, osteoporosis, arthritis, isolated iron deficiency, dermatitis herpetiformis.

Big belly, flat bottom.

Diagnosis
- First-line: anti-tTG IgA plus total IgA, on a gluten-containing diet.
- IgA at least 0.2 g/l to read tTG and EMA reliably.
- Avoid biopsy: positive symptoms, tTG AND EMA, positive HLA-DQ2/DQ8.

No gluten, no antibodies.

Treatment
- Strict lifelong gluten-free diet; antibodies monitor adherence.
- CMPA → extensively hydrolysed formula; amino-acid formula if severe.

Antibodies tell you if they cheat.

III-40 Diseases of oral cavity. Diseases of the oesophagus.

Mouth disease in children is mostly infection (HSV, coxsackie, Candida); oesophageal disease turns on a few patterns — TEF (coiled NG tube, VACTERL), EoE (≥15 eos/HPF), achalasia (bird's beak) — and button batteries are a 2-hour emergency.

Definition
- Mouth: mostly infections; oesophagus: reflux, EoE, achalasia, TEF, foreign bodies.

Mouth infects; oesophagus obstructs.

Causes
- HSV-1 gingivostomatitis, coxsackie A herpangina, Candida thrush (persistent: immunodeficiency).
- TEF: VACTERL association; commonest type is atresia with distal fistula (85%).

Herpangina = coxsackie at the back.

Patho-physiology
- Atopy → eosinophilic oesophagitis → rings, strictures → food impaction.
- Button battery burns through the oesophageal wall in 2 hours.

Battery burns in 2 hours.

Clinical picture
- Gingivostomatitis: swollen bleeding gums, anterior ulcers; herpangina: soft palate only.
- Physiological reflux resolves by 12-18 months.
- TEF: coughing, choking, cyanosis; achalasia: dysphagia, regurgitation.

Front = herpes; back = herpangina.

Diagnosis
- EoE biopsy: at least 15 eosinophils per high-power field.
- Achalasia: bird's-beak on barium, absent peristalsis on manometry.
- TEF: NG tube coiling on X-ray; pure atresia: gasless abdomen.

Coiled NG = atresia.

Treatment
- Gingivostomatitis: fluids, analgesia, aciclovir within 72 h.
- Severe reflux: fundoplication; achalasia: Heller myotomy.
- TEF: surgery, look for VACTERL; swallowed coin: emergency endoscopy (within 2 hours).

Myotomy for achalasia; wrap for reflux.

III-41 Chronic liver failure in children

Think of chronic liver failure as three lost jobs: can't clear (jaundice, itch), can't build (INR, albumin) and can't flow (portal hypertension). Biliary atresia is the commonest cause, and Wilson disease is the treatable one you must not miss in any child over 4-5 years.

Definition
- Chronic loss of hepatic clearance, synthesis and flow in children.

Liver jobs: CLEAR, BUILD, FLOW.

Causes
- Commonest cause of chronic liver failure in children: biliary atresia.
- Alpha-1 antitrypsin, Alagille; Wilson (AR, ATP7B), autoimmune hepatitis, hepatitis B/C.

Baby liver failure? Think the Bile ducts first: Biliary atresia.

Patho-physiology
- Can't clear (jaundice, itch), can't build (INR, albumin), can't flow (portal hypertension).
- Wilson: copper can't reach bile → liver, basal ganglia, cornea, haemolysis.

Wilson = copper Wanders: liver, basal ganglia, cornea, RBCs.

Clinical picture
- Jaundice, pale stools, pruritus, bruising, ascites, splenomegaly, varices, ADEK deficiency.
- Wilson symptoms start after 4-5 years of age.

Wilson waits: no symptoms before 4-5 years.

Diagnosis
- INR not correcting with IV vitamin K = synthetic liver failure.
- Wilson: serum caeruloplasmin low, 24-hour urinary copper high, KF rings, ATP7B genetics.
- Acute liver failure with Coombs-negative haemolysis suggests Wilson disease.

Vitamin K fails to fix INR = factory broken.

Treatment
- Cholestatic children: MCT formula + ADEK; ursodeoxycholic acid, cholestyramine for itch.
- Spironolactone, lactulose, band ligation; rifaximin lowers ammoniagenic bacteria.
- Wilson: penicillamine/trientine, then zinc; end-stage: liver transplantation.

Wilson: Pull copper out (penicillamine), then Zinc blocks intake.

III-42 Biliary atresia. Liver transplantation in children.

Any term baby who is still jaundiced after 2 weeks needs a split (conjugated) bilirubin. A thriving baby with pale stools has biliary atresia until proven otherwise, and the Kasai works best before about 6 weeks.

Definition
- Progressive fibro-obliteration of extrahepatic bile ducts in infancy.

Atresia = ducts Absent/obliterated, Early infancy.

Causes
- Probably perinatal viral/immune trigger; isolated, or with polysplenia/situs anomalies, cystic.

Syndromic BA: Spleen too many (polysplenia), organs flipped.

Patho-physiology
- Blocked ducts → conjugated bilirubin refluxes → jaundice, dark urine; no stercobilin → acholic stool.
- Retained bile → fibrosis → cirrhosis, portal hypertension; no bile → vitamin K bleeding.

No bile in gut = white poo, dark wee, bleeding.

Clinical picture
- Typical stool feature: clay-coloured (acholic) stool, dark urine.
- Term infant jaundiced beyond 2 weeks; thriving, so missed as breast-milk jaundice.

Yellow baby + white stool = surgical emergency.

Diagnosis
- Jaundice beyond 2 weeks needs a conjugated (split) bilirubin; raised GGT, INR.
- Ultrasound: triangular cord sign, small or absent gallbladder; definitive: intraoperative cholangiogram.
- Diagnose earlier, but no later than 6 weeks of infancy.

Cord sign, Cholangiogram confirms, Catch it by 6 weeks.

Treatment
- Kasai portoenterostomy ASAP; ursodiol, prophylactic antibiotics, ADEK, MCT formula.
- Fever and returning jaundice = ascending cholangitis → IV antibiotics.
- Most common cause of liver transplantation in children; tacrolimus.

Kasai = Kiss bowel to liver hilum.

III-43 Inflammatory bowel disease. Crohn disease and ulcerative colitis in childhood

In children, Crohn disease often presents as growth failure and delayed puberty before any gut symptoms, while UC presents with bloody diarrhoea. Calprotectin decides who needs endoscopy, and in paediatric Crohn the first-line induction is exclusive enteral nutrition, not steroids.

Definition
- Crohn: transmural with skip lesions; UC: continuous from the rectum.

Crohn: Cobblestones, skip, full thickness. UC: Upward from rectum, Continuous.

Causes
- Chronic immune-mediated gut inflammation (IBD) of childhood.

Patho-physiology
- Crohn: transmural, non-caseating granulomas → strictures, fistulae, perianal disease.
- UC: mucosal only, crypt abscesses → bloody diarrhoea; linked to PSC.

Crohn = granulomas; UC = crypt abscesses.

Clinical picture
- Crohn: weight loss; often presents first with growth failure and delayed puberty.
- Perianal fissures: not in the midline, deep, not painful.
- UC: bloody diarrhoea, urgency, tenesmus; high GGT → think sclerosing cholangitis.

Short child, late puberty? Check for Crohn.

Diagnosis
- Bloods, stool culture; faecal calprotectin gatekeeps endoscopy.
- Ileocolonoscopy + upper endoscopy with biopsies + MR enterography.

Calprotectin opens the endoscopy gate.

Treatment
- Crohn: first choice exclusive enteral nutrition 6-8 wk, then thiopurine/methotrexate, anti-TNF.
- UC: 5-ASA (mesalamine) → steroids → thiopurine/anti-TNF; colectomy curative.
- PUCAI ≥65: IV steroids; days 3/5 → infliximab or ciclosporin.

Crohn kids: Drink, don't steroid (EEN first).

III-44 Pancreatitis in childhood

Paediatric pancreatitis is diagnosed by INSPPIRE: 2 of 3 from typical pain, enzymes ≥3× ULN and imaging. Look for the cause (bile, trauma including NAI, drugs, systemic disease). Treat with fluids, opioids and early enteral feeding, and give no routine antibiotics.

Definition
- Acute pancreatic inflammation with autodigestion, may progress to SIRS.

Causes
- Biliary, trauma (handlebar, NAI), drugs: valproate, azathioprine, L-asparaginase.
- Infection (mumps), HUS, HSP, metabolic; recurrent → genetic testing: PRSS1, SPINK1, CFTR.

Recurrent pancreatitis: PRSS1, SPINK1, CFTR.

Patho-physiology
- Premature intra-acinar activation of trypsinogen → autodigestion → cytokines → SIRS, capillary leak.
- Fat saponification binds calcium → hypocalcaemia.

Fat + calcium = soap → low calcium.

Clinical picture
- Epigastric pain radiating to back, eased by leaning forward; vomiting, fever.
- Tachycardia, ileus; rarely Grey Turner or Cullen sign.

Grey Turner = flank (Turn sideways); Cullen = umbilicus.

Diagnosis
- INSPPIRE needs 2 of 3: typical pain, amylase/lipase ≥ 3× ULN, imaging.
- Ultrasound first; check calcium, glucose, triglycerides, LFTs.

INSPPIRE: 2 of 3 (pain, enzymes, imaging).

Treatment
- Fluids ~1.5-2× maintenance, analgesia incl. opioids; enteral feeding within 48-72 hours.
- No routine antibiotics - only infected necrosis or cholangitis.
- Treat cause: stop drug, cholecystectomy same admission; pseudocyst forms after more than 4 weeks.

Pseudocyst needs >4 weeks to mature.

III-45 Cleft lip and cleft palate

In cleft care the timeline matters: lip at about 3 months, palate at 1 year (before first words), alveolar bone graft at 8–11 years. Every cleft palate needs a hearing check, because tensor veli palatini dysfunction causes glue ear.

Definition
- Congenital cleft from failed fusion of facial processes; ~1 in 700 births.

1 in 700: commonest craniofacial anomaly.

Causes
- Maternal smoking, alcohol, phenytoin/valproate, folate deficiency.
- Isolated cleft palate is commoner in girls and more often syndromic, e.g. 22q11 deletion.

Palate alone? Think Girls and 22q11.

Patho-physiology
- Cleft lip: failed fusion of the medial nasal process with the maxillary process.
- Palatal shelves fail → no suction seal, VPI.
- Tensor veli palatini dysfunction → Eustachian tube can't open → otitis media with effusion.

Cleft palate = leaky seal + blocked ear tube.

Clinical picture
- Feeding difficulty, nasal milk reflux; glue ear.
- Hypernasal speech (VPI); Pierre Robin: micrognathia, glossoptosis, U-shaped cleft palate.

Pierre Robin: small jaw, tongue back, U-shaped cleft.

Diagnosis
- Often seen at 20-week anomaly scan (lip more than palate).
- Any cleft palate → hearing test; syndromic features → 22q11 testing.

Palate? Test ears and 22q11.

Treatment
- Cleft team; squeezable bottles, upright feeding, weight monitoring.
- Lip repaired at about 3 months; soft palate closed at 1 year, before first words.
- Grommets, speech therapy, alveolar bone graft 8–11 years.

Rule of 3s and 1: lip 3 months, palate 1 year.

III-46 Ileus, intussusception

Separate mechanical from paralytic ileus, and watch for colicky pain that becomes constant, because that means strangulation. In a 3-month to 3-year-old with paroxysmal colic, pallor and drawn-up legs, ultrasound for the target sign and an air enema come before any redcurrant jelly appears.

Definition
- Ileus: bowel obstruction, mechanical (lumen blocked) or paralytic (peristalsis stops).

Mechanical = plugged pipe; paralytic = stopped pump.

Causes
- Mechanical: adhesions, incarcerated hernia, intussusception, volvulus, atresia.
- Paralytic: post-surgery, sepsis, peritonitis, hypokalaemia, opioids, vincristine.
- Intussusception: usual lead point Peyer's patches; over 3 y: Meckel's, polyp, lymphoma.

Over 3 years intussusception? Hunt a lead point.

Patho-physiology
- Mechanical → tinkling bowel sounds; paralytic → silent, distended abdomen.
- Colicky pain turning constant suggests strangulation: ischaemia, fever, rising lactate.
- Ileocolic intussusception → mucosal ischaemia → redcurrant-jelly stool (late).

Colic turning constant = bowel dying.

Clinical picture
- Intussusception 3 months - 3 years: paroxysmal colic, pallor, legs drawn up, lethargy.
- Sausage mass (RUQ); vomiting, later bilious.

Intussusception: 3 months to 3 years.

Diagnosis
- Ultrasound: target (doughnut) sign.
- Commonest cause of strangulation ileus in children: incarcerated hernia.
- Suspected SBO: unstable → X-ray; stable → CT with IV contrast.

Target/doughnut on US = intussusception.

Treatment
- Obstruction: drip and suck, analgesia, surgical review; complete SBO: laparotomy.
- Intussusception: resuscitate → air (pneumatic) enema reduction; ~10% recur.
- Surgery: strangulation, peritonitis, perforation, shock, failed reduction.

Air pushes it back; ~10% come back.

III-47 Diseases of the colon. Obstipation Hirschsprung disease.

About 95% of childhood constipation is functional (withholding, then a dilated rectum, then overflow soiling) and needs macrogol disimpaction then months of maintenance. Delayed meconium beyond 48 h with an empty rectum and an explosive gush on DRE is Hirschsprung until a suction biopsy says otherwise.

Definition
- About 95% of childhood constipation is functional; Hirschsprung = congenital aganglionosis.

95% functional; Hirschsprung is the exception.

Causes
- Functional: weaning, toilet training, school, anal fissure (tight sphincter → mucosal tear).
- Hirschsprung: failed neural crest migration; Down syndrome association.

Hirschsprung + Down = classic pair.

Patho-physiology
- Withholding → harder stool → pain → dilated rectum → overflow soiling (liquid stool around impacted faeces).
- Aganglionic rectum can't relax → functional obstruction.

Withhold → hard → hurt → hold: vicious cycle.

Clinical picture
- Functional: withholding postures, packed rectum, soiling; infantile dyschezia: infant strains ~10 minutes.
- Hirschsprung: meconium delayed beyond 48 hours, distension, bilious vomiting.
- Rectum empty with explosive release on withdrawal.

Hirschsprung: empty rectum + explosive gush.

Diagnosis
- Rectal suction biopsy: absent ganglion cells, raised acetylcholinesterase.
- Contrast enema: transition zone; manometry: absent rectoanal inhibitory reflex.
- Anal fissure: physical exam (DRE, anoscopy).

No ganglia, no RAIR = Hirschsprung.

Treatment
- Functional: disimpaction with macrogol, then maintenance for months, not days; toileting.
- Hirschsprung: rectal washouts → pull-through operation.
- Feared complication: enterocolitis - fever, distension, explosive foul diarrhoea.

Macrogol for months, not days.

III-48 Acute appendicitis

Appendicitis classically starts as periumbilical visceral (T10) pain, moves to the RIF, and vomiting follows the pain. Under-5s perforate early (short omentum, perforation after 36–48 h). Use a score to stratify, ultrasound first, and never withhold analgesia.

Definition
- Acute inflammation of the appendix from luminal obstruction.

Causes
- Obstruction of the lumen by a faecolith (older) or lymphoid hyperplasia (younger, after viral illness).

Young = lymphoid; Old = stone (faecolith).

Patho-physiology
- Obstruction → pressure, venous failure, bacterial invasion → gangrene → perforation after 36–48 hours.
- Visceral T10 → periumbilical pain → parietal peritoneum → localised RIF pain.
- Children under 5 perforate more often: short omentum.

Umbilicus (T10) → RIF as peritoneum gets involved.

Clinical picture
- Pain first epigastric/periumbilical, later right lower quadrant; vomiting comes after the pain.
- McBurney tenderness, pain on hopping/cough; Rovsing's sign: pressing LIF hurts RIF.

Pain first, vomit second.

Diagnosis
- Paediatric Appendicitis or Alvarado score: low discharge, high surgery, middle imaging.
- First-line imaging is ultrasound: non-compressible tube >6 mm; CT only if inconclusive.

Appendix >6 mm, non-compressible = inflamed.

Treatment
- Analgesia given freely does not mask the diagnosis; IV fluids, antibiotics.
- Laparoscopic appendicectomy.
- Appendix mass → antibiotics first ± drainage, interval appendicectomy.

Mass? Cool it first, cut later.

III-49 Testis and ovarian torsion. Cryptorchidism

Testicular torsion is a clinical diagnosis: take the boy to theatre, not to the scanner. You have about 6 hours for ~90% salvage, and both sides are fixed. Ovarian torsion is untwisted and preserved even when dusky. Undescended testes are referred by 3–6 months and fixed at 6–18 months.

Definition
- Torsion: testis/ovary twists on its pedicle; cryptorchidism: testis not in scrotum.

Causes
- Testis: bell-clapper deformity, usually bilateral; ovary: cyst or teratoma as lead weight.
- Cryptorchidism: preterm (up to 30%) vs term 2–4%.

Bell-clapper: swings like a bell, both sides.

Patho-physiology
- Torsion: veins occlude first → arterial failure → infarction.
- Undescended testis → germ cells degenerate → infertility and testicular germ cell tumour.

Veins choke first, arteries die last.

Clinical picture
- Testis: sudden pain, vomiting, high-riding horizontal lie, cremasteric reflex absent.
- Appendix testis torsion: blue dot sign, reflex preserved.
- Ovary: sudden unilateral lower abdominal pain, vomiting, may mimic appendicitis.

Blue dot = benign appendix testis.

Diagnosis
- Testicular torsion is clinical → theatre, not scan.
- Ovary: US shows enlarged oedematous ovary; preserved Doppler does not exclude.
- Bilateral impalpable testes → karyotype, exclude CAH.

Torsion: theatre, not scan.

Treatment
- Testis: explore within 6 hours (~90% salvage), detort, fix both sides.
- Ovary: laparoscopic detorsion, untwisted and preserved even if dusky.
- Cryptorchidism: refer by 3–6 months, orchidopexy at 6–18 months.

6 hours to save the testis.

III-50 Neural tube defects

Prevent neural tube defects with folic acid (400 µg for everyone, 5 mg if high risk) before conception. At birth, protect the placode (moist sterile dressing, prone, no latex) and close it within 48 h. Then watch for hydrocephalus from the Chiari II malformation and manage the neurogenic bladder.

Definition
- Failed neural tube closure (week 4): cranial day 25, caudal day 28.
- Anencephaly cranially; spina bifida (occulta, meningocele, myelomeningocele) caudally.

Top closes day 25, bottom day 28.

Causes
- Folate deficiency (TMP-SMX in 1st trimester), previous NTD, valproate/carbamazepine, maternal diabetes, obesity.

Folate fights NTDs.

Patho-physiology
- Exposed placode damaged → deficits below the level.
- Most myelomeningoceles: tethered cord → Chiari II → hydrocephalus.

Myelomeningocele → Chiari II → hydrocephalus.

Clinical picture
- Flaccid leg weakness, sensory loss, neurogenic bladder/bowel, club feet, scoliosis.
- Hydrocephalus; stridor/apnoea from brainstem compression; latex allergy.

Spina bifida = no latex.

Diagnosis
- Raised maternal serum alpha-fetoprotein; ultrasound lemon (frontal bones) and banana (cerebellum) signs.
- At birth: level, head circumference, fontanelle, hips, feet, bladder.

Lemon skull, banana cerebellum.

Treatment
- Prevent: folic acid 400 µg daily routinely, 5 mg if high risk.
- Birth: moist sterile dressing, prone, no latex; close within 48 hours.
- VP shunt/ETV; clean intermittent catheterisation, oxybutynin, physio.

400 µg routine; 5 mg high risk.

III-51 Perinatal asphyxia, Hypoxic-ischemic encephalopathy (HIE)

HIE has a latent phase of 1–6 h before secondary energy failure, and that latent phase is the cooling window. Cool babies ≥36 weeks with moderate or severe encephalopathy to 33.5 °C for 72 h, starting within 6 h, and rewarm slowly.

Definition
- Perinatal oxygen deprivation causing encephalopathy (HIE), staged by Sarnat.

Causes
- Abruption, cord prolapse, uterine rupture, shoulder dystocia, maternal hypotension.
- Supports: sentinel event, Apgar ≤ 5 at 10 min, pH <7.0, base deficit ≥ 12–16.

pH below 7 = serious acidosis.

Patho-physiology
- Hypoxia → ATP fails → primary energy failure → oedema, excitotoxicity.
- Latent phase 1–6 h (window) → secondary energy failure 6–72 h → neuronal death.
- Diving reflex spares brain, heart, adrenals; kidneys, gut, liver suffer.

6-hour window: cool before secondary failure.

Clinical picture
- Mild: hyperalert, jittery; moderate: lethargy, hypotonia, seizures.
- Severe: coma, flaccid, apnoea; multiorgan ATN, PPHN, DIC, hypoglycaemia.

Mild jittery, moderate floppy, severe coma.

Diagnosis
- Cord gas, aEEG (normalising by 24–48 h good).
- MRI days 4–10: basal ganglia and thalamus injury = poor prognosis.
- Apgar: Appearance, Pulse, Grimace, Activity, Respiration (10-point scale).

Deep grey (BGT) damage = bad news.

Treatment
- Cooling: moderate or severe HIE, babies ≥ 36 weeks.
- 33.5 °C for 72 hours, started within 6 hours; rewarm 0.5 °C per hour.
- Normocapnia, avoid hyperthermia, restrict fluids; seizures → phenobarbital.

33.5 for 72, start by 6.

III-52 Hydrocephalus. Infantile cerebral palsy

Hydrocephalus in an infant shows as a head crossing centiles, a bulging fontanelle and sunsetting eyes. Treat with a VP shunt or ETV, and later headache, vomiting and drowsiness mean the shunt is blocked. CP is a non-progressive injury to the developing brain with a picture that evolves; loss of skills is never CP.

Definition
- Hydrocephalus: CSF accumulation with ventricular dilatation.
- CP: essential criterion non-progressive, residual brain injury of developing brain.

CP doesn't progress; the picture changes.

Causes
- Hydrocephalus: aqueduct stenosis, Chiari II, tumour, post-IVH/meningitis, congenital toxoplasmosis.
- CP: antenatal (most), prematurity (PVL, IVH) strongest, HIE, kernicterus.

CP biggest risk: being born early.

Patho-physiology
- Obstruction or failed absorption → CSF builds → skull expands if sutures open.
- Preterm IVH arises in the germinal matrix.
- CP lesion static, picture evolves: hypotonia → spasticity, contractures.

Germinal matrix = preterm bleed site.

Clinical picture
- Infant: head circumference crossing centiles, bulging fontanelle, sunsetting eyes.
- CP: spastic commonest; dyskinetic (basal ganglia), ataxic (cerebellum).
- Hand preference before 12 months suggests hemiplegia; loss of skills is not CP.

Hand preference <12 months = hemiplegia.

Diagnosis
- Hydrocephalus/IVH: cranial ultrasound (infant), MRI; bulging fontanelle + vomiting → brain imaging.
- CP: standardised movement assessment + MRI; classify function by GMFCS.

GMFCS grades CP mobility I–V.

Treatment
- Ventriculoperitoneal shunt or endoscopic third ventriculostomy; headache/vomiting/drowsiness = blockage.
- CP: multidisciplinary physio/OT/SLT; baclofen, botulinum toxin, hip surveillance.

Shunt + vomiting + drowsy = blocked.

III-53 Meningitis, brain abscess

Give antibiotics without delay. The LP can wait if there are signs of raised ICP, shock or coagulopathy, but the antibiotics never wait. Under 3 months use cefotaxime + amoxicillin (Listeria), and over 3 months ceftriaxone plus dexamethasone. Brain abscess means ring enhancement and no LP.

Definition
- Bacterial meningeal infection; brain abscess = walled-off pus in brain.

Causes
- Neonates: Streptococcus agalactiae (GBS) and E. coli; Listeria.
- Later: meningococcus, pneumococcus; abscess: ear/sinus spread, cyanotic CHD (right-to-left shunt bypasses lung filter).

Newborn: GBS, E. coli, Listeria.

Patho-physiology
- Cytokines open BBB → protein elevated, glucose decreased, neutrophils.
- Oedema, impaired CSF absorption → raised ICP.

Bacteria: protein Up, glucose Down.

Clinical picture
- Neck stiffness, Kernig, Brudzinski; infants: fever, irritability, bulging fontanelle.
- Non-blanching rash → meningococcal sepsis.
- Abscess: headache, focal signs, seizures; fever in only half.

Non-blanching rash = meningococcus until proven otherwise.

Diagnosis
- LP: neutrophils >1000/µL, protein >1 g/L, CSF:blood glucose <0.4.
- Raised ICP, shock, coagulopathy: delay LP, never delay antibiotics.
- Abscess: ring-enhancing lesion; lumbar puncture contraindicated.

LP can wait, antibiotics can't.

Treatment
- Antibiotics without delay: under 3 months cefotaxime + amoxicillin (covers Listeria); over 3 months ceftriaxone.
- Dexamethasone over 3 months reduces hearing loss; test hearing within 4 weeks; contact prophylaxis.
- Abscess: drain + IV antibiotics 4–8 weeks.

Under 3 months: add amoxicillin for Listeria.

III-54 Encephalitis. Facial palsy

Any febrile child with altered behaviour has HSV encephalitis until proven otherwise, so start IV aciclovir on suspicion. For facial palsy, the forehead separates central (spared) from peripheral, and every palsy needs an ear exam and a blood pressure before you call it Bell palsy.

Definition
- Encephalitis = brain parenchyma inflammation; facial palsy = central (UMN) or peripheral (LMN).

Causes
- Encephalitis: HSV-1 (key treatable cause), enterovirus, VZV; neonates HSV-2; anti-NMDAR, ADEM.
- Facial palsy: Bell (exclusion), otitis, Lyme, Ramsay Hunt (VZV), hypertension, leukaemia.

Bell's = last resort label.

Patho-physiology
- HSV encephalitis typically affects the temporal lobe.
- Upper face has bilateral cortical supply → forehead spared in central lesions.

Forehead spared = Central (brain) lesion.

Clinical picture
- Encephalitis: fever + altered behaviour/consciousness, seizures, focal signs.
- Peripheral: whole half-face, can't close eye, taste loss; central: forehead sparing.
- Bilateral palsy: Lyme, Guillain-Barre, leukaemia; Ramsay Hunt: ear vesicles.

Bilateral facial palsy: Lyme, GBS, Leukaemia.

Diagnosis
- LP: lymphocytic CSF, normal glucose; HSV PCR; MRI, EEG.
- Every facial palsy: examine the ear, measure blood pressure; atypical course → image brain.

Facial palsy: ear + BP every time.

Treatment
- IV aciclovir on suspicion, 14-21 days (21 neonates); crystal-induced AKI risk.
- ADEM methylprednisolone; anti-NMDAR in girls → search for ovarian teratoma.
- Palsy: protect the eye, treat cause; prednisolone within 72 h.

Suspect HSV? Aciclovir now, prove later.

III-55 Neurocutaneous syndromes

The neurocutaneous syndromes share an ectodermal origin, so skin findings flag brain disease: café-au-lait macules and Lisch nodules mean NF1, ash-leaf macules with infantile spasms mean TSC, and a V1 port-wine stain with seizures means Sturge-Weber. Know the targeted drug for each: selumetinib, vigabatrin/everolimus, aspirin.

Definition
- Neurocutaneous syndromes: skin + nervous system lesions (shared ectodermal origin).

Skin + brain: same ectoderm parent.

Causes
- NF1: neurofibromin, chromosome 17; NF2: merlin, chromosome 22 (AD).
- TSC1/TSC2 (AD); Sturge-Weber: somatic GNAQ mutation, not inherited.

NF2 = chromosome 22 (two twos); NF1 = 17.

Patho-physiology
- NF1: neurofibromin loss → Ras unchecked → neural crest tumours.
- TSC: mTOR overactivity → hamartomas; SWS: leptomeningeal angioma → cortical ischaemia.

TSC = mTOR → everolimus blocks it.

Clinical picture
- NF1: ≥ 6 café-au-lait macules, freckling, neurofibromas, optic glioma, Lisch nodules.
- NF2: bilateral vestibular schwannomas; TSC: ash-leaf macules, angiofibromas, rhabdomyomas, infantile spasms.
- SWS: V1 port-wine stain, seizures, hemiparesis, glaucoma.

NF2 = 2 ears (bilateral vestibular schwannomas).

Diagnosis
- NF1: two or more diagnostic features; annual eye checks.
- TSC: Wood lamp for ash-leaf; infantile spasms show hypsarrhythmia (West syndrome).
- SWS imaging: tram-track calcification.

Tram-tracks = Sturge-Weber.

Treatment
- NF1 plexiform neurofibroma: selumetinib (MEK inhibitor).
- TSC spasms: first-line vigabatrin; SEGA: everolimus.
- SWS: seizure control, low-dose aspirin, glaucoma follow-up, pulsed-dye laser.

TSC spasms? Vigabatrin.